T2K 6.914×10^{20}-POT Neutrino 4.011×10^{20}-POT Anti-Neutrino 3-Flavour Joint Analysis

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Abstract

We report the results of the 3-flavour $\nu/\bar{\nu}$ joint analysis, performed by the *VALOR* group, on the combined Run 1+6 dataset, using data taken in both neutrino and antineutrino beam mode. The goal of this analysis is to search for CP violation by setting limits on δ_{CP} and measure the oscillation parameters $\sin^2\theta_{13}$, $\sin^2\theta_{23}$, Δm^2_{32} (Δm^2_{13}) (Δm^2_{32} for normal hierarchy and Δm^2_{13} for inverted hierarchy). Though T2K has limited sensitivity to the mass hierarchy given the relatively short baseline, fits under both hypotheses are performed to show which is preferred by data. Furthermore the search for $\bar{\nu}_e$ appearance and the measurement of $\bar{\nu}_\mu$ disappearance are shown as well.

This analysis adapts the work presented in Refs. [1], [2], [3], [4], [5], [6], [7], [8] [9] and [10] to this dataset. The Run 1-6 dataset corresponds to an integrated J-PARC neutrino beam exposure of 6.914×10^{20} POT in neutrino mode and 4.011×10^{20} POT in antineutrino mode.

This analysis predicts 125.9 ± 6.5 (syst) for neutrino μ-like, 34.3 ± 1.8 (syst) for antineutrino μ-like, 26.6 ± 1.8 (syst) for neutrino e-like, and 3.22 ± 0.24 (syst) for antineutrino e-like single-ring events in Super-K, given the oscillation parameters of the Asimov data set A in table 12, corresponding to $\sin^2\theta_{23} = 0.528$, $\delta_{CP} = -1.601$ and normal hierarchy, and observes respectively 124 for neutrino μ-like, 34 for antineutrino μ-like, 31 for neutrino e-like, 3 for antineutrino e-like.

A joint analysis of all the four selected samples is performed in a framework of 3-flavour oscillations including matter effects in constant-density matter.

For μ-like samples the templates are built as a function of the reconstructed neutrino energy, while for e-like the analysis templates are built as a function of the reconstructed neutrino energy and lepton angle, in order to improve the separation of different flavour/reaction components and increase the sensitivity to δ_{CP}. The observed single-ring events spectra are compared to the expected spectra in order to infer all the 3-flavour oscillation parameters.

As in [10], all the systematic parameters are marginalised (see section 4) in the fit and confidence intervals are quoted for all the oscillation parameters. Since the most important result of this analysis release is the search for CP violation, the Feldman-Cousins method [59] is performed only for the simultaneous fit of δ_{CP} and mass hierarchy.

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1. Introduction

In this technical note, we report the results of a 3-flavour $\nu/\bar{\nu}$ joint analysis, performed by the *VALOR* group, on the combined Run 1-6 dataset, using data taken in both neutrino mode (6.914×10^{20} POT), with the polarity set to forward horn current (FHC), and antineutrino (4.011×10^{20} POT) mode with polarity set reverse horn current (RHC). This analysis builds on our other analyses presented in Refs. [1], [2], [3], [4], [5], [6], [7], [8] [9] and [10]. As in [10], a hybrid frequentist-Bayesian method [60], which consists of marginalising instead of profiling the nuisance parameters, is used.

The consistency of the data with the expectation is given by comparing the predicted reconstructed neutrino energy and outgoing lepton angle spectra of 1 ring events at Super-K to the observed spectra. The method of calculation of these predicted spectra is presented in section 2. This analysis uses flux tuning based on NA61/SHINE 2009 data [17][50]. It also uses an indirect extrapolation from ND280 to Super-K, which is implemented via a fit of flux and cross-section parameters to ND280 measurements of the muon momentum-angle distributions of $\bar{\nu}_\mu$ and ν_μ for different event topologies with the neutrino interaction vertex either in FGD1 or FGD2 (BANFF fit) [29].

A significant difference between this and the previous analyses is that for e-like samples the analysis templates are built as a function of both the reconstructed neutrino energy and lepton angle, in order to improve the separation of the different reaction / flavour components, in particular $\bar{\nu}_\mu \to \bar{\nu}_e$ and $\nu_\mu \to \nu_e$, fundamental in measuring δ_{CP}. Furthermore the ν_e intrinsic beam and NC components can also be better separated from signal. Though the new variable θ increases the number of template bins and so the CPU time, the number of reconstructed neutrino energy bins has not been reduced, in order to continue to have a very good sensitivity to the oscillation pattern. For μ-like samples the templates are built as a function of the reconstructed neutrino energy as in the past analyses.

Updated nominal predictions of the reconstructed neutrino energy and lepton angle spectrum of single ring events in Super-K are presented for Run 1-6 in section 2.

The oscillation analysis is performed using neutrino oscillation probabilities computed in a 3-flavour framework including constant-density matter effects. Details of our oscillation probability calculations, estimates of the numerical accuracy of our calculations and comparisons with calculations used by the alternative T2K oscillation analyses are given in Appendices A and B of Ref. [4].

As in our previous analyses, the oscillation fit is made using the likelihood-ratio method [56]. The mathematical details of this method are given in Sec. 3.2 in Ref. [1]. We consider a total of 102 parameters including energy scale, neutrino flux, neutrino cross-section, efficiency, initial-state nuclear environment and final-state re-scattering systematics plus the oscillation parameters (nuisance parameters). All the nuisance parameters are marginalised [60]. The nuisance parameters and their effects on the extrapolated single-ring event spectra are described in section 3. The analysis method is presented in detail in section 4.

Fit validations are presented in section 5. The following updates have been added: the coverage studies study for the δ_{CP} vs mass hierarchy fit is shown in section 5.2. The main goal of this analysis is to search for CP violation in the context of the PMNS matrix and new confidence intervals, produced with the Feldman-Cousins method [59], are shown in section 6.2. These sections as well as appendix E have been added in this new version. Confidence intervals of the other oscillation parameters are set using the constant $\Delta\chi^2$ method. Sensitivity studies are shown in section 5.1. The results of the measurement using the Run 1-6 dataset is given in section 6 of this note.

Finally, a summary is given in section 7.

2. Predictions of single ring event spectra in Super-K

2.1. Methodology

In this analysis a comparison of the reconstructed energy (E_r) and reconstructed lepton angle (θ) spectrum of single ring events observed at Super-K with the predicted spectrum is used to build the test statistic to quote confidence intervals of the oscillation parameters. For μ-like samples the templates are built as a function of the reconstructed neutrino energy, as in the past analyses, and a single θ bin is used. The predicted number $N_{SK;r,\theta}$ of single ring events in the r-th and θ-th bin is computed for each selected sample as follows:

$$N_{SK;r,\theta} = \sum_{m} \sum_{t} \sum_{r\prime} P_{m;t} \cdot T_{r;r\prime;f^{SK}_{E;r}} \cdot S_{m;t;r\prime,\theta;\vec{f}} \cdot N^{MC}_{SK;m;r\prime,\theta;t} \quad (1)$$

In eq. (1), $N^{MC}_{SK;m;r\prime,\theta;t}$ is the input Super-K Monte Carlo (MC) template containing the number of events in the single ring MC sample with true reaction mode m in the true energy bin t, the reconstructed energy bin $r\prime$ and reconstructed lepton angle bin θ. The construction of the MC templates is discussed in section 2.2. Both flux-tuning weights based on NA61/SHINE hadron-production data (see section 2.3) and flux and cross section tuning weights based on ND280 data (see section 2.4) are applied to the nominal MC templates. $S_{m;t;r\prime,\theta;\vec{f}}$ is an overall, multiplicative, systematic error factor depending on a vector of nuisance (systematic) parameters $\vec{f}$, which is function of the reaction mode m, the true energy bin t, the reconstructed energy bin $r\prime$, and the reconstructed lepton angle bin θ. A complete description of the nuisance parameters considered in this analysis is given in section 3. $T_{r;r\prime;f^{SK}_{E;r}}$ is a transfer function describing the migration of events between the reconstructed energy bins r and $r\prime$ due to uncertainty in the Super-K reconstructed energy scale, expressed here in terms of the nuisance parameter $f^{SK}_{E;r}$. Finally, $P_{m;t}$ is the 3-flavour oscillation probability applied to the true energy bin t of the Super-K MC template which corresponds to mode m; application of the term $P_{m;t}$ is discussed in section 2.5.

2.2. Construction of the nominal Super-K Monte Carlo templates

The nominal Super-K single ring MC analysis templates $N^{MC}_{SK;m;r\prime,\theta;t}$ are constructed by applying either the single e-like ring selection cuts or the single μ-like ring selection cuts to the official Super-K MC samples. The selection is exactly the same for neutrino and antineutrino beam mode. In total 4 samples are obtained:

- neutrino mode single μ-like ring sample
- antineutrino mode single μ-like ring sample
- neutrino mode single e-like ring sample
- antineutrino mode single e-like ring sample

Details of the MC samples are given in section 2.2.1. The calculation of the normalisation of each MC sample is described in section 2.2.2. The official single μ-like ring selection cuts are listed in section 2.2.3. while the official single e-like ring selection cuts are listed in section 2.2.4. The list of MC templates used in this analysis is shown in section 2.2.5, while the true and reconstructed energy as well as reconstructed lepton angle binning of the templates is discussed in section 2.2.6.

2.2.1. Input Monte Carlo samples

This analysis uses the official 14a Super-K MC samples. These were generated using the 13a JNUBEAM flux [37, 38], NEUT 5.3.2 [41], skdetsim v13p90 and the 14b Super-K software [34] with nominal tunings. The CCQE model is reweighted from the Spectral Function to a Relative Fermi Gas model with a relativistic Random Phase Approximation on an event by event basis. The NC1γ events are scaled up by a factor of two in accordance with improved calculations of single photon emission in NC interactions [35]. Lastly, a simplified reweighting, as a function of the outgoing pion energy, is used to scale the CCCoherent events from the Rein Sehgal model to the Berger-Sehgal model prediction [39, 40]. Six MC samples were generated for both neutrino and antineutrino beam mode running (horn current $\pm$250kA). Four of the samples ($\bar{\nu}_\mu$, ν_μ, $\bar{\nu}_e$ and ν_e) were generated using the nominal JNUBEAM flux without oscillations. The fifth sample is $\bar{\nu}_e$ interactions generated with the $\bar{\nu}_\mu$ flux, and will be denoted the "oscillated $\bar{\nu}_e$" sample. The sixth sample is ν_e interactions generated with the ν_μ flux, and will be denoted the "oscillated ν_e" sample.

Running mode	Neutrino type	Number of predicted events (flux averaged) per 22.5 kt fiducial per 10^{21} POT	Number of generated events in true fiducial volume in all available MC
Neutrino	$\bar{\nu}_\mu$	50.964	634600
	ν_μ	1396.245	634881
	$\bar{\nu}_e$	2.884	126728
	ν_e	26.663	126899
	Oscillated $\bar{\nu}_e$	52.939	126984
	Oscillated ν_e	1483.994	126916
Antineutrino	$\bar{\nu}_\mu$	353.789	633988
	ν_μ	208.369	634659
	$\bar{\nu}_e$	6.811	127013
	ν_e	8.941	127262
	Oscillated $\bar{\nu}_e$	376.420	126812
	Oscillated ν_e	215.099	127278

Table 1: Normalisation of input Super-K MC samples [34]. Oscillations are applied only to "Oscillated" samples which correspond to fully oscillated samples, e.g. assuming all ν_μ oscillated into ν_e.

2.2.2. Normalisation of Monte Carlo samples

The normalisation (integrated exposure in terms of POT) of each event sample is calculated from the number of events with a MC truth interaction vertex within the 22.5 kt fiducial volume.

$$N = \int dSdIdE \cdot \frac{d^3\Phi_{SK}}{dSdIdE_\nu} \cdot \sigma_{H_2O} \cdot \frac{N_A}{A} \cdot \rho \cdot L \tag{2}$$

where $d^3\Phi_{SK}/dSdIdE$ is the number of flux particles for the given neutrino species per neutrino energy bin dE_ν, per unit area dS and per POT, σ_{H_2O} is the total interaction cross section on water for the given neutrino species, I is the beam exposure in terms of POT, N_A is Avogadro's number, A is the mass number for water, ρ is the water density and L is the neutrino path length in the water volume.

The Super-K MC samples are normalised to the Run 1-6 POT using the numbers in **??**. Assuming that all the MC files are used, each sample is first normalised to 10^{21} POT by weighting it with the ratio of the number of events per 22.5 kt per 10^{21} POT (second column) to the number of generated events in the true fiducial volume in all the available MC (third column). Then each sample is normalised to 6.914×10^{20} POT for neutrino and 4.011×10^{20} for antineutrino mode, by weighting the result respectively by 0.6914 and 0.4011 (the Run 1-6 POT in units of 10^{21} POT).

2.2.3. Single μ-like ring event selection cuts

The Super-K MC templates contain events from the ν_μ, $\bar{\nu}_\mu$, ν_e, $\bar{\nu}_e$, oscillated ν_e and oscillated $\bar{\nu}_e$ samples passing the standard Super-K selection cuts for single μ-like ring events. These cuts are:

- the number of PMT hits in the highest-charge OD cluster is less than or equal to 15, and the event passes the flasher rejection cuts (`evclass == 1`),
- the reconstructed vertex is at least 2 metres away from the ID wall (`wall > 200`),
- the visible energy in the event is above 30 MeV (`evis > 30`),
- there is exactly 1 reconstructed ring in the event (`nring == 1`),
- the reconstructed ring is μ-like (`ip[0] = 3`),
- the reconstructed muon momentum is above 200 MeV (`amomm[0] > 200`), and
- there is at most 1 decay e^- in the event (`nmue <= 1`).

The same selection is applied to both neutrino and antineutrino mode samples.

2.2.4. Single e-like ring event selection cuts

The Super-K MC templates contain events from the ν_μ, $\bar{\nu}_\mu$, ν_e, $\bar{\nu}_e$, oscillated ν_e and oscillated $\bar{\nu}_e$ samples passing the standard Super-K selection cuts for single e-like ring events. These cuts are:

- the number of PMT hits in the highest-charge OD cluster is less than or equal to 15, and the event passes the flasher rejection cuts (`evclass == 1`),
- the reconstructed vertex is at least 2 metres away from the ID wall (`wall > 200`),
- the visible energy in the event is above 30 MeV (`evis > 30`),
- there is exactly 1 reconstructed ring in the event (`nring == 1`),
- the reconstructed ring is e-like (`ip[0] = 2`),
- number of decay electrons is zero (`nmue = 0`),
- the reconstructed neutrino energy is below 1250 MeV
- fiTQun π^0 rejection cut: $\ln(L_{\pi^0}/L_e) < 175 - 0.875 \times m_{\gamma\gamma}$

The same selection is applied to both neutrino and antineutrino mode samples [33].

2.2.5. List of Monte Carlo templates

For each Super-K MC sample, a number of different MC templates is constructed corresponding to different true reaction modes. The template number depends on the type of oscillation analysis and the specific systematic parameters considered in the analysis. 12 MC templates were used for our 2010 Run 1+2 2-flavour disappearance analysis [2], while 15 MC templates were used for our 2010 Run 1+2 3-flavour disappearance analysis [3], 28 MC templates were used for our 2012 Run 1+2+3 3-flavour disappearance analysis [4] and 32 MC templates were used for our 2014 Run 1+2+3+4 3-flavour disappearance analysis [6].

In our previous $\bar{\nu}_\mu$-disappearance and $\bar{\nu}_e$ appearance analyses the NC interaction modes were subdivided into NC1$\pi^\pm$, NC1π^0, NC coherent and NC other. In this analysis the new mode NC1γ, which was previously included in NC other, is used. We have also added 5 CC 2p-2h templates for $\bar{\nu}_\mu$, ν_μ, $\bar{\nu}_e$, ν_e and oscillated ν_e and 5 templates for the oscillated $\bar{\nu}_e$ MC sample. This gives a total of 50 templates:

- $\bar{\nu}_\mu$ CCQE,
- $\bar{\nu}_\mu$ CC1π,
- $\bar{\nu}_\mu$ CC coherent,
- $\bar{\nu}_\mu$ CC 2p-2h,
- $\bar{\nu}_\mu$ CC other,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC1π^0,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC1$\pi^\pm$,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC coherent,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC other,
- $\bar{\nu}_\mu$(and $\rightarrow \bar{\nu}_e$, $\bar{\nu}_\tau$) NC 1γ,
- ν_μ CCQE,
- ν_μ CC1π,
- ν_μ CC coherent,
- ν_μ CC 2p-2h,

- ν_μ CC other,
- ν_μ (and $\rightarrow \nu_e$, ν_τ) NC1π^0,
- ν_μ (and $\rightarrow \nu_e$, ν_τ) NC1$\pi^\pm$,
- ν_μ (and $\rightarrow \nu_e$, ν_τ) NC coherent,
- ν_μ (and $\rightarrow \nu_e$, ν_τ) NC other,
- ν_μ (and $\rightarrow \nu_e$, ν_τ) NC 1γ,
- $\bar{\nu}_e$ CCQE,
- $\bar{\nu}_e$ CC1π,
- $\bar{\nu}_e$ CC coherent,
- $\bar{\nu}_e$ CC 2p-2h,
- $\bar{\nu}_e$ CC other,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC1π^0,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC1$\pi^\pm$,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC coherent,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC other,
- $\bar{\nu}_e$ (and $\rightarrow \bar{\nu}_\mu$, $\bar{\nu}_\tau$) NC 1γ,
- ν_e CCQE,
- ν_e CC1π,
- ν_e CC coherent,
- ν_e CC 2p-2h,
- ν_e CC other,
- ν_e (and $\rightarrow \nu_\mu$, ν_τ) NC1π^0,
- ν_e (and $\rightarrow \nu_\mu$, ν_τ) NC1$\pi^\pm$,
- ν_e (and $\rightarrow \nu_\mu$, ν_τ) NC coherent,
- ν_e (and $\rightarrow \nu_\mu$, ν_τ) NC other,
- ν_e (and $\rightarrow \nu_\mu$, ν_τ) NC 1γ,
- oscillated $\nu_\mu \rightarrow \nu_e$ CCQE,
- oscillated $\nu_\mu \rightarrow \nu_e$ CC1π,
- oscillated $\nu_\mu \rightarrow \nu_e$ CC coherent,
- oscillated $\nu_\mu \rightarrow \nu_e$ CC 2p-2h,
- oscillated $\nu_\mu \rightarrow \nu_e$ CC other,
- oscillated $\bar{\nu}_\mu \rightarrow \bar{\nu}_e$ CCQE,
- oscillated $\bar{\nu}_\mu \rightarrow \bar{\nu}_e$ CC1π,

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	66.077	0.035	4.291	0.002	0.225	0.001	70.631
CC1pi	16.973	0.017	1.566	0.001	0.067	0.000	18.625
CCcoherent	0.326	0.000	0.080	0.000	0.001	0.000	0.408
2p-2h	19.038	0.010	0.845	0.000	0.050	0.000	19.944
CCother	7.735	0.008	0.494	0.001	0.002	0.000	8.240
NC1pi0	0.537	0.016	0.017	0.001	N/A	N/A	0.572
NC1piPM	3.944	0.089	0.142	0.008	N/A	N/A	4.183
NCcoherent	0.016	0.000	0.001	0.000	N/A	N/A	0.018
NCother	2.993	0.121	0.165	0.012	N/A	N/A	3.290
NC1gamma	0.004	0.000	0.000	0.000	N/A	N/A	0.004
Total	117.643	0.297	7.602	0.025	0.345	0.002	125.914

Table 2: Predicted number of events in the neutrino mode μ-like sample obtained after applying the BANFF weight for a total exposure of 6.914×10^{20} POT. The Asimov data set A in table 12 is used.

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	6.993	0.004	12.792	0.005	0.006	0.014	19.813
CC1pi	2.707	0.003	3.033	0.002	0.002	0.003	5.750
CCcoherent	0.054	0.000	0.169	0.000	0.000	0.000	0.223
2p-2h	2.503	0.001	2.053	0.001	0.002	0.002	4.562
CCother	1.280	0.001	0.885	0.000	0.000	0.000	2.168
NC1pi0	0.060	0.002	0.050	0.001	N/A	N/A	0.114
NC1piPM	0.378	0.015	0.380	0.010	N/A	N/A	0.784
NCcoherent	0.001	0.000	0.006	0.000	N/A	N/A	0.008
NCother	0.530	0.025	0.296	0.012	N/A	N/A	0.862
NC1gamma	0.000	0.000	0.000	0.000	N/A	N/A	0.000
Total	14.508	0.051	19.664	0.032	0.010	0.019	34.284

Table 3: Predicted number of events in the antineutrino mode μ-like sample obtained after applying the BANFF weight for a total exposure of 4.011×10^{20} POT. The Asimov data set A in table 12 is used.

- oscillated $\bar{\nu}_\mu \to \bar{\nu}_e$ CC coherent,
- oscillated $\bar{\nu}_\mu \to \bar{\nu}_e$ CC 2p-2h,
- oscillated $\bar{\nu}_\mu \to \bar{\nu}_e$ CC other.

In table 2, 3, 4 and 5 the predicted number of events is shown for all the selected samples and each flavour and reaction mode.

2.2.6. Binning of Monte Carlo templates

As already explained, this analyses uses an updated binning scheme for reconstructed variables in order to improve the separation of the different reaction / flavour components, in particular $\bar{\nu}_\mu \to \bar{\nu}_e$ and $\nu_\mu \to \nu_e$, fundamental in measuring δ_{CP}. The ν_e intrinsic beam and NC components can also be better separated from signal. Though the new variable θ increases the number of template bins and so the CPU time, the number of reconstructed neutrino energy bins has not been reduced, in order to continue to have a very good sensitivity to the oscillation pattern. For μ-like samples the templates are built as a function of the reconstructed neutrino energy as in the past analyses.

μ-like templates have 84 true energy bins and 73 reconstructed energy bins. e-like templates have 84 true energy bins, 25 reconstructed energy bins and 15 reconstructed lepton angle bins.

For all templates the 84 true energy bins of constant width are the following:

- 6 50-MeV bins from 0-0.3 GeV,

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	0.056	2.129	0.001	0.088	15.875	0.099	18.249
CC1pi	0.016	0.348	0.000	0.027	1.688	0.017	2.096
CCcoherent	0.000	0.002	0.000	0.002	0.015	0.002	0.021
2p-2h	0.008	0.710	0.000	0.022	3.901	0.019	4.659
CCother	0.001	0.045	0.000	0.003	0.029	0.001	0.080
NC1pi0	0.524	0.012	0.017	0.001	N/A	N/A	0.553
NC1piPM	0.099	0.002	0.003	0.000	N/A	N/A	0.105
NCcoherent	0.174	0.004	0.018	0.001	N/A	N/A	0.197
NCother	0.126	0.005	0.008	0.001	N/A	N/A	0.139
NC1gamma	0.464	0.007	0.021	0.001	N/A	N/A	0.494
Total	1.469	3.264	0.069	0.145	21.508	0.138	26.594

Table 4: Predicted number of events in the neutrino mode e-like sample obtained after applying the BANFF weight for a total exposure of 6.914×10^{20} POT. The Asimov data set A in table 12 is used.

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	0.002	0.189	0.003	0.339	0.375	1.156	2.065
CC1pi	0.002	0.040	0.001	0.060	0.057	0.137	0.298
CCcoherent	0.000	0.000	0.000	0.006	0.000	0.019	0.026
2p-2h	0.001	0.069	0.000	0.063	0.114	0.185	0.433
CCother	0.001	0.008	0.000	0.004	0.004	0.002	0.019
NC1pi0	0.045	0.002	0.049	0.001	N/A	N/A	0.098
NC1piPM	0.011	0.000	0.008	0.000	N/A	N/A	0.020
NCcoherent	0.021	0.001	0.081	0.001	N/A	N/A	0.103
NCother	0.022	0.001	0.012	0.000	N/A	N/A	0.035
NC1gamma	0.037	0.002	0.081	0.002	N/A	N/A	0.121
Total	0.141	0.313	0.235	0.478	0.551	1.500	3.218

Table 5: Predicted number of events in the antineutrino mode μ-like sample obtained after applying the BANFF weight for a total exposure of 4.011×10^{20} POT. The Asimov data set A in table 12 is used.

- 28 25-MeV bins from 0.3-1 GeV,
- 40 50-MeV bins from 1-3 GeV,
- 5 100-MeV bin from 3-3.5 GeV,
- 1 bin from 3.5-4 GeV,
- 1 bin from 4-5 GeV,
- 1 bin from 5-7 GeV,
- 1 bin from 7-10 GeV and
- 1 bin from 10-30 GeV

For μ-like samples the 73 reconstructed energy bins are the following:

- 60 50-MeV bins from 0-3 GeV,
- 4 250-MeV bins from 3-4 GeV,
- 4 500-MeV bins from 4-6 GeV,
- 4 1000-MeV bins from 6-10 GeV and

- 1 bin from 10-30 GeV.

For e-like samples the 25 reconstructed energy bins with constant width are the following:

- 25 50-MeV bins from 0-1.25 GeV.

which is identical to the past analyses.

The θ binning used for e-like samples is

- 14 10° bins from 0° to 140°,
- 1 bin for the range $140^\circ - 180^\circ$.

that is the same as used by the $p-\theta$ analysis group [15].

For μ-like samples a single θ bin, that covers the whole range 0° - 180°, is used. The predicted e-like 2-dimensional spectra are shown in appendix B for each flavour component and compared with the 1-dimensional reconstructed energy spectra. It is clear that the components $\nu_\mu \to \nu_e$ and $\bar{\nu}_\mu \to \bar{\nu}_e$ are better separated by the reconstructed energy vs θ templates as shown in fig. 36 and 37.

A study was performed to decide the optimum binning scheme for reconstructing neutrino energy balancing the needs for accuracy and speed. Details are given in Appendix I of Ref. [4]. The reconstructed neutrino energy binning described here is unchanged with respect to the Run 1+2+3+4 joint fit analysis. The reconstructed lepton angle binning is the same use by the $p-\theta$ group for all their analyses [15].

2.3. Applying flux tuning based on NA61/SHINE data

In this analysis, the nominal MC templates, which are constructed using a MC event sample generated with the 13a flux, are reweighted to the 13av1.1 flux which is tuned on NA61/SHINE hadron production measurements [37][49]. The reweighted MC templates are denoted the *NA61/SHINE-tuned* MC templates in this document. The applied weights are obtained from [37] and are shown as a function of the true neutrino energy in fig. 1 and 4 for neutrino mode. and in fig. 5 and 8 for antineutrino mode. The effect of this flux tuning on the predicted spectra events is discussed in section 2.6

2.4. Applying flux and cross section (BANFF) tuning based on ND280 data

The BANFF group has made a fit of the $\bar{\nu}_\mu$ and ν_μ ND280 samples to find the best-fit values of the flux normalisation and cross section parameters [28]. This analysis uses these best-fit values to reweight the NA61/SHINE-tuned MC templates, and the templates obtained after this second reweighting will be denoted the *BANFF-tuned* MC templates in this document. A list of BANFF parameters, their best-fit values and their 1σ prefit and postfit errors is given in table 6, 7 and 8.

The effects of the BANFF tune on the predicted spectra is discussed in section 2.6.

2.5. Applying neutrino oscillations

The MC templates are constructed from the unoscillated MC samples and must be weighted with the appropriate oscillation probability. Oscillations are applied as a function of true energy to the MC templates for all CC interactions.

The MC templates calculated from the $\overset{(-)}{\nu}_\mu$ MC sample are weighted with $\mathrm{P}(\overset{(-)}{\nu}_\mu \to \overset{(-)}{\nu}_\mu)$, with this survival probability being calculated using the oscillation parameters, namely $\sin^2\theta_{23}$, $\sin^2\theta_{12}$, $\sin^2\theta_{13}$, $|\Delta m^2_{32}|$ (normal hierarchy) or $|\Delta m^2_{31}|$ (inverted hierarchy), Δm^2_{21} and δ_{CP}. The $\overset{(-)}{\nu}_e$ templates are weighted with $\mathrm{P}(\overset{(-)}{\nu}_e \to \overset{(-)}{\nu}_e)$, while the oscillated $\overset{(-)}{\nu}_\mu \to \overset{(-)}{\nu}_e$ templates are weighted with $\mathrm{P}(\overset{(-)}{\nu}_\mu \to \overset{(-)}{\nu}_e)$ (the sample was generated assuming 100% of $\overset{(-)}{\nu}_\mu$ transform to $\overset{(-)}{\nu}_e$). All these oscillation probabilities are computed in a 3-flavour framework including matter effects in constant-density matter (assuming an Earth crust density of 2.6 g/cm^3). Custom oscillation probability calculation code was developed within the *VALOR* analysis framework. Details and estimates of the numerical accuracy of this code are first presented in [3], and is also shown in Appendix A of [4]. Comparisons between the *VALOR* oscillation probability calculation code and Prob3++ [75], which is typically used by the alternative T2K oscillation analyses, are shown in Appendix B of [4].

In the standard 3-flavour oscillation framework, oscillations of $\overset{(-)}{\nu}_e$ and $\overset{(-)}{\nu}_\mu$ can yield $\overset{(-)}{\nu}_\tau$. In this analysis, we neglect contributions from ν_τ-CC and $\bar{\nu}_\tau$-CC as their energy threshold is around 3.5 GeV and their effect

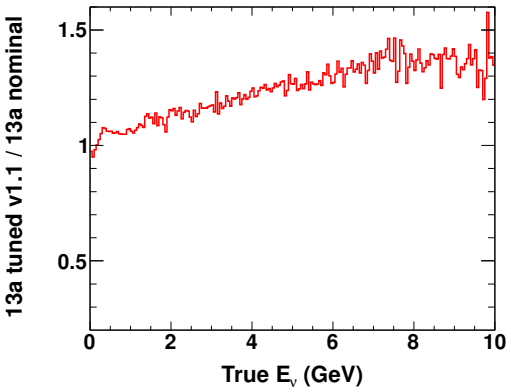


Figure 1: ν_μ 13a_v1.1 Run 1-6a flux tuning weights applied to nominal ν_μ and oscillated-ν_e MC templates [37] (neutrino mode).

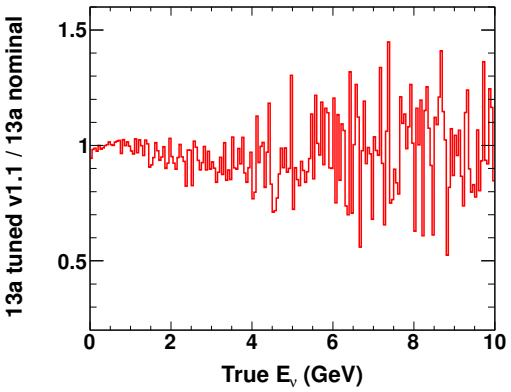


Figure 2: $\bar{\nu}_\mu$ 13a_v1.1 Run 1-6a flux tuning weights applied to nominal $\bar{\nu}_\mu$ and oscillated-$\bar{\nu}_e$ MC templates [37] (neutrino mode).

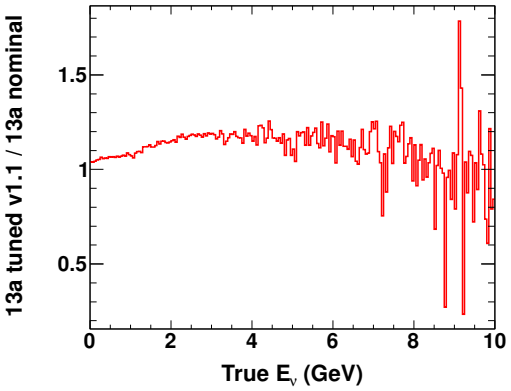


Figure 3: ν_e 13a_v1.1 Run 1-6a flux tuning weights applied to nominal ν_e MC templates [37] (neutrino mode).

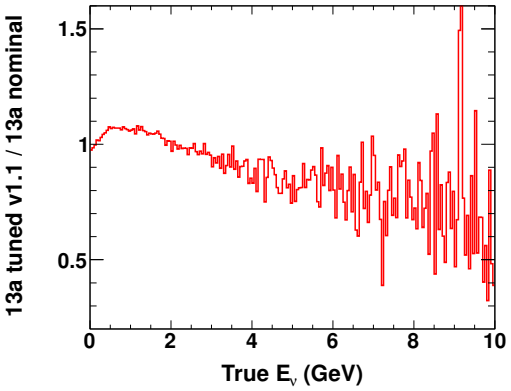


Figure 4: $\bar{\nu}_e$ 13a_v1.1 Run 1-6a flux tuning weights applied to nominal $\bar{\nu}_e$ MC templates [37] (neutrino mode).

is negligible[1]. Accordingly, this analysis uses no ν_τ-CC and $\bar{\nu}_\tau$-CC MC templates (and, in fact, no ν_τ and $\bar{\nu}_\tau$ Super-K MC samples are available).

It should be emphasized here that the ν_μ NC MC templates for a mode m contain the NC MC templates for the mixture of ν_e+ν_μ+ν_τ resulting from 3-flavour ν_μ oscillations for that mode m. The same applies to the $\bar{\nu}_\mu$, ν_e and $\bar{\nu}_e$ NC MC templates. These NC MC templates are unchanged under standard 3-flavour oscillations.

Also it should be noted that there are no explicit NC MC templates made from the oscillated $\bar{\nu}_e$ and ν_e samples. If they were used, the oscillated $\bar{\nu}_e$ (i.e. $\bar{\nu}_e$ coming from oscillations of $\bar{\nu}_\mu$) would be double counted since they are already included in the $\bar{\nu}_\mu$ NC MC templates. The same applies to oscillated ν_e.

2.6. Predictions of nominal and tuned-MC spectra

In this section, we present the expected spectra in Super-K for various oscillation scenarios. Unless explicitly stated otherwise, the oscillation parameters are set to the Asimov data set A shown in table 12 and the normal mass hierarchy is assumed. All plots are generated for an integrated exposure corresponding to Run 1-6.

Predicted single ring Super-K spectra are shown in fig. 9. By comparing neutrino and antineutrino mode

[1] A back-of-the-envelope calculation can be found in the replies to comments and revision history document of [4] that gives $\ll 1\%$ ν_τ contamination in the 1-ring μ-like Super-K sample. A Super-K MC study is required to determine whether the Super-K efficiency assumptions made there are valid.

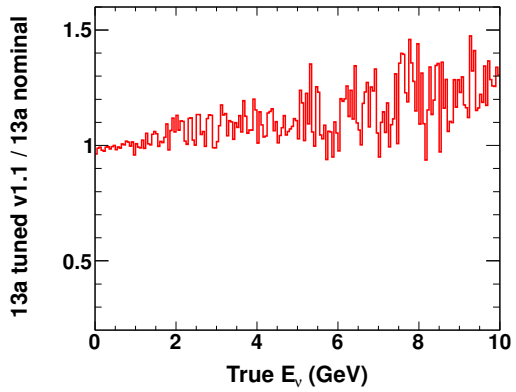


Figure 5: ν_μ 13a_v1.1 Run 5c-6b flux tuning weights applied to nominal ν_μ and oscillated-ν_e MC templates [37] (antineutrino mode).

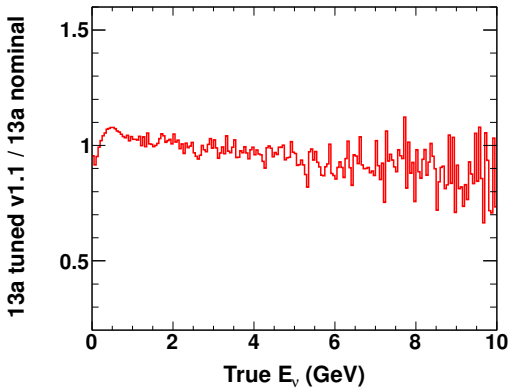


Figure 6: $\bar{\nu}_\mu$ 13a_v1.1 Run 5c-6b flux tuning weights applied to nominal $\bar{\nu}_\mu$ and oscillated-$\bar{\nu}_e$ MC templates [37] (antineutrino mode).

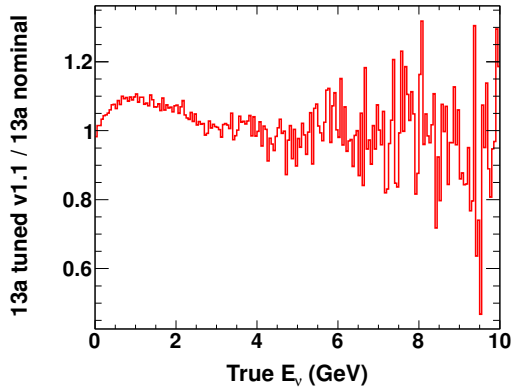


Figure 7: ν_e 13a_v1.1 Run 5c-6b flux tuning weights applied to nominal ν_e MC templates [37] (antineutrino mode).

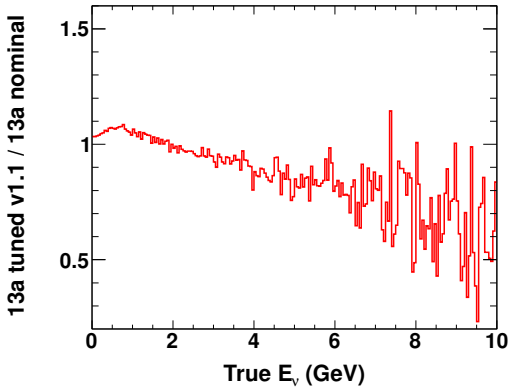
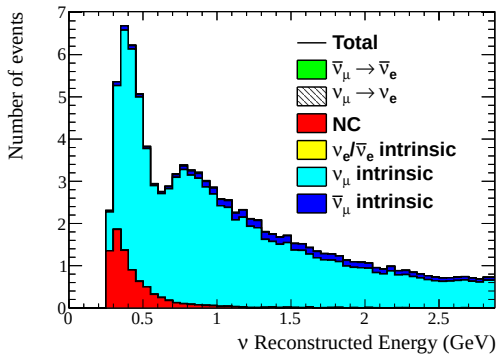
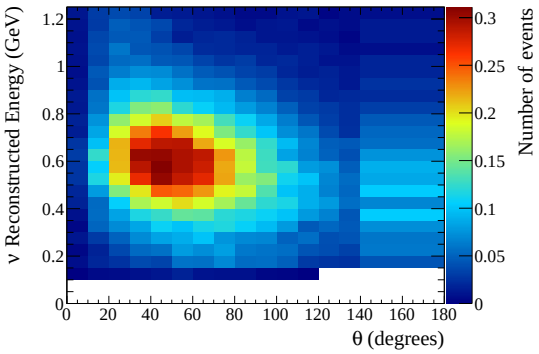


Figure 8: $\bar{\nu}_e$ 13a_v1.1 Run 5c-6b flux tuning weights applied to nominal $\bar{\nu}_e$ MC templates [37] (antineutrino mode).

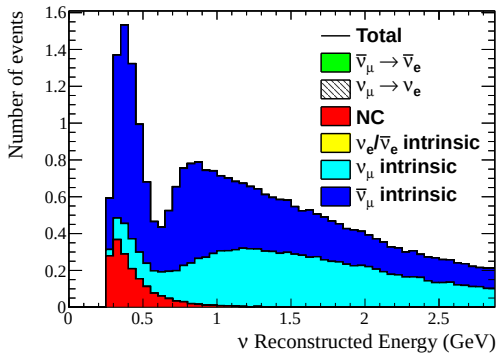
e-like plots it is clear that E_{reco} vs θ distributions are differently populated by $\nu_\mu \to \nu_e$ and $\bar{\nu}_\mu \to \bar{\nu}_e$ events. More details on the advantage of the 2-dimensional templates are given in appendix B.



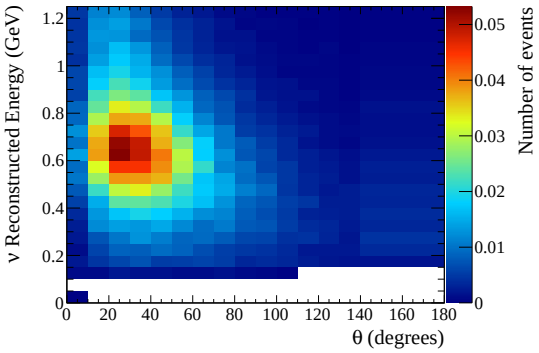
(a) Muon-like, neutrino mode



(b) Electron-like, neutrino mode



(c) Muon-like, antineutrino mode



(d) Electron-like antineutrino mode

Figure 9: Predicted spectra, muon-like distributions are function of the reconstructed neutrino energy, while the electron-like distributions are function of both the reconstructed neutrino energy and the reconstructed angle between the outgoing lepton and the neutrino direction. The distributions correspond to the statistics collected in the full Run 1-6 data set. The muon-like distributions are broken down by the following components: $\bar{\nu}_\mu \to \bar{\nu}_e$ (green), $\nu_\mu \to \nu_e$ (black/white), NC (red), $\nu_e + \bar{\nu}_e$ intrinsic beam (yellow), ν_μ intrinsic beam (light blue) and $\bar{\nu}_\mu$ intrinsic beam (blue). The spectra are generated with the systematic parameters measured with the ND280 data shown in table 6, 7, 8 and the nominal Super-K systematic parameters shown in table 9 and the oscillation parameters corresponding to the Asimov data set A shown in table 12.

3. Systematic effects in the prediction of the single μ-like ring event energy spectrum

3.1. Input systematic parameters

This analysis considers 102 flux, cross section and Super-K detector systematic parameters plus 7 oscillation parameters, which can be either treat as systematic parameters or as parameters of interest, depending on the analysis that is performed.

1. **Flux and cross-section parameters** (65 parameters).

In this analysis, all the flux and cross-section systematic parameters are included in the ND280 data (BANFF) fit. However, since ND280 is not sensitive to some cross section systematics they remain unconstrained (detailed later). Parameters fit are 11 ν_μ, 5 $\bar{\nu}_\mu$, 7 ν_e and 2 $\bar{\nu}_e$ neutrino mode (FHC) flux parameters, 5 ν_μ, 11 $\bar{\nu}_\mu$, 2 ν_e and 7 $\bar{\nu}_e$ antineutrino mode (RHC) flux parameters and 15 cross-section parameters. Best-fit values of these parameters and a covariance matrix giving their correlated uncertainties are obtained by the BANFF group from a fit of the $\bar{\nu}_\mu$ and ν_μ ND280 samples as described in Ref [28]. A list of the BANFF parameters, their best-fit values and their prefit and postfit errors, taken from TN-230 [29], is given in table 6, 7, 8.

The best-fit values are used to reweight the templates as described in section 2.4, and the postfit errors in the covariance matrix are used as prior uncertainties in this analysis. The Super-K systematic parameters are shown in table 9. All the cross section parameters set to 1 correspond to the NEUT nominal value, except "CC other shape" which NEUT nominal value is 0. For all the normalisation parameters, 1 is the nominal pre-banff fit value.

The effects of these correlated uncertainties are evaluated using the Cholesky method, that allows to throw the systematic parameters according to their multivariate distribution taking into account the existing correlations.

Some of the cross-section parameters have complicated, often non-linear, effects on true and reconstructed neutrino energy bin normalisations; the effects of uncertainties in these parameters are evaluated by reweighting the MC templates. Weights are calculated for tweaks from -3σ to +3σ in steps of 0.5σ, and interpolations are made between those weights using cubic splines (for MaQE only, weights are calculated from -7σ to +3σ). Cubic splines are made for the following cross-section uncertainties:

- CCQE axial-mass scaling factor (MaQE)
- Fermi momentum for ^{16}O
- Binding energy for ^{16}O
- Nucleon to Δ transition axial form factor (C_5^A)
- Resonance-production axial-mass scaling factor (MaRes)
- Scale of isospin 1/2 nonresonant background
- CC other shape

The splines are made in bins of theta, true and reconstructed energy, and the binning is given in Ref. [36]. The θ binning is the same as for the analysis templates, shown in section 2.2.6.

The effects of the flux and cross-section normalisation uncertainties are also evaluated by reweighting the MC templates, but no splines are made for them since the weight is identical to the tweak for these uncertainties (each flux parameter is a normalisation for a limited range of true energy). The cross section parameters which simply normalise the overall spectrum component are:

- 2p-2h, applied to all the 2p-2h events (both neutrino and antineutrino)
- 2p-2h-$\bar{\nu}$, applied to all the antineutrino 2p-2h events
- CC coherent for ^{16}O normalisation
- NC coherent normalisation

- NC other normalisation
- NC 1γ normalisation
- CC ν_e normalisation
- CC $\bar{\nu}_e$ normalisation

Uncertainties due to "second class currents" are implemented as an additional uncertainty on CC $\bar{\nu}_e$ and CC ν_e parameters with a 2% anti-correlated constraint between them. The BANFF parameters are applied to the MC templates as specified in appendix D in table 38, 39, 40 and 41. The normalisation parameter of the NC1γ component was not used in the previous analyses, since it was included in NC other.

In the BANFF fit a bias on the best-fit flux parameters is present due to the degeneracy between the neutrino flux and ND280 detector parameters. For this reason a covariance matrix that accounts for the bias of the flux as well as all the other parameters is summed in quadrature to the total post-fit covariance. Furthermore the best-fit values have been corrected in order to remove the bias. The description of the procedure is given in [28]. As shown in appendix D the impact of the bias on the systematic parameters fitted by the BANFF analysis is small compared to the other systematic uncertainties.

3.2. Comparison of predicted number of events in neutrino mode e-like sample between current and 2015 analysis

The predicted number of events in antineutrino mode e-like sample is now smaller compared to the $\bar{\nu}_e$ appearance analysis presented at the 2015 summer conferences [10], 3.218 instead of 3.729. The differences between the current and previous oscillation analyses is mainly due to:

(a) new value of $\sin^2\theta_{13}$ corresponding to the PDG 2015 release, as shown in table 12, which is 0.0217 while in the previous analysis it was 0.0250. If the old PDG value is used the expected number of antineutrino mode e-like events becomes 3.542.

(b) Subsequently, if the cross section parameters are set to the old post-BANFF fit values, the expected number of events becomes 3.63. Indeed all the post-BANFF 2015 fit cross section parameters are higher compared to the 2016 BANFF fit result. The effect of the flux parameters is smaller, indeed if they are set to BANFF 2015 result, the expected number of events becomes 3.66.

(c) Finally, after all the changes listed above, the rescaling of CC coherent and NC1γ components (see section 2.2.1) is not applied, the agreement between the 2015 and 2016 analysis is obtained. The first one increases the number of events up to 3.772 while the latter one reduces down to 3.711, which is approximately the number of events expected in the 2015 analysis.

Index	Parameter	Description	Best fit	1σ pre/postfit fractional error
0	$f^{banff}_{0;t,r}$	FHC ν_μ flux normalisation, E = 0.0 - 0.4 GeV	1.128	0.099 / 0.064
1	$f^{banff}_{1;t,r}$	FHC ν_μ flux normalisation, E = 0.4 - 0.5 GeV	1.156	0.103 / 0.061
2	$f^{banff}_{2;t,r}$	FHC ν_μ flux normalisation, E = 0.5 - 0.6 GeV	1.148	0.096 / 0.051
3	$f^{banff}_{3;t,r}$	FHC ν_μ flux normalisation, E = 0.6 - 0.7 GeV	1.128	0.087 / 0.043
4	$f^{banff}_{4;t,r}$	FHC ν_μ flux normalisation, E = 0.7 - 1.0 GeV	1.104	0.113 / 0.046
5	$f^{banff}_{5;t,r}$	FHC ν_μ flux normalisation, E = 1.0 - 1.5 GeV	1.100	0.092 / 0.045
6	$f^{banff}_{6;t,r}$	FHC ν_μ flux normalisation, E = 1.5 - 2.5 GeV	1.127	0.070 / 0.044
7	$f^{banff}_{7;t,r}$	FHC ν_μ flux normalisation, E = 2.5 - 3.5 GeV	1.124	0.074 / 0.048
8	$f^{banff}_{8;t,r}$	FHC ν_μ flux normalisation, E = 3.5 - 5.0 GeV	1.121	0.087 / 0.049
9	$f^{banff}_{9;t,r}$	FHC ν_μ flux normalisation, E = 5.0 - 7.0 GeV	1.075	0.098 / 0.053
10	$f^{banff}_{10;t,r}$	FHC ν_μ flux normalisation, E = 7.0 - 30.0 GeV	1.064	0.114 / 0.065
11	$f^{banff}_{11;t,r}$	FHC $\bar{\nu_\mu}$ flux normalisation, E = 0.0 - 0.7 GeV	1.100	0.103 / 0.081
12	$f^{banff}_{12;t,r}$	FHC $\bar{\nu_\mu}$ flux normalisation, E = 0.7 - 1.0 GeV	1.112	0.079 / 0.048
13	$f^{banff}_{13;t,r}$	FHC $\bar{\nu_\mu}$ flux normalisation, E = 1.0 - 1.5 GeV	1.111	0.084 / 0.060
14	$f^{banff}_{14;t,r}$	FHC $\bar{\nu_\mu}$ flux normalisation, E = 1.5 - 2.5 GeV	1.116	0.086 / 0.070
15	$f^{banff}_{15;t,r}$	FHC $\bar{\nu_\mu}$ flux normalisation, E = 2.5 - 30.0 GeV	1.162	0.086 / 0.069
16	$f^{banff}_{16;t,r}$	FHC ν_e flux normalisation, E = 0.0 - 0.5 GeV	1.134	0.090 / 0.052
17	$f^{banff}_{17;t,r}$	FHC ν_e flux normalisation, E = 0.5 - 0.7 GeV	1.135	0.090 / 0.049
18	$f^{banff}_{18;t,r}$	FHC ν_e flux normalisation, E = 0.7 - 0.8 GeV	1.135	0.086 / 0.047
19	$f^{banff}_{19;t,r}$	FHC ν_e flux normalisation, E = 0.8 - 1.5 GeV	1.119	0.081 / 0.043
20	$f^{banff}_{20;t,r}$	FHC ν_e flux normalisation, E = 1.5 - 2.5 GeV	1.115	0.079 / 0.046
21	$f^{banff}_{21;t,r}$	FHC ν_e flux normalisation, E = 2.5 - 4.0 GeV	1.111	0.084 / 0.050
22	$f^{banff}_{22;t,r}$	FHC ν_e flux normalisation, E = 4.0 - 30.0 GeV	1.118	0.094 / 0.067
23	$f^{banff}_{23;t,r}$	FHC $\bar{\nu_e}$ flux normalisation, E = 0.0 - 2.5 GeV	1.121	0.074 / 0.057
24	$f^{banff}_{24;t,r}$	FHC $\bar{\nu_e}$ flux normalisation, E = 2.5 - 30.0 GeV	1.153	0.128 / 0.117

Table 6: Summary of neutrino mode flux systematics included in the *VALOR* joint fit analysis. A description of how the BANFF fit constrained these systematics can be found in Ref. [28].

Index	Parameter	Description	Best fit	1σ fractional error
25	$f^{banff}_{0;t,r}RHC$	RHC ν_μ flux normalisation, E = 0.0 - 0.7 GeV	1.098	0.094 / 0.072
26	$f^{banff}_{1;t,r}RHC$	RHC ν_μ flux normalisation, E = 0.7 - 1.0 GeV	1.121	0.079 / 0.052
27	$f^{banff}_{2;t,r}RHC$	RHC ν_μ flux normalisation, E = 1.0 - 1.5 GeV	1.130	0.077 / 0.048
28	$f^{banff}_{3;t,r}RHC$	RHC ν_μ flux normalisation, E = 1.5 - 2.5 GeV	1.155	0.081 / 0.054
29	$f^{banff}_{4;t,r}RHC$	RHC ν_μ flux normalisation, E = 2.5 - 30.0 GeV	1.111	0.080 / 0.055
30	$f^{banff}_{5;t,r}RHC$	RHC $\bar{\nu_\mu}$ flux normalisation, E = 0.0 - 0.4 GeV	1.118	0.104 / 0.071
31	$f^{banff}_{6;t,r}RHC$	RHC $\bar{\nu_\mu}$ flux normalisation, E = 0.4 - 0.5 GeV	1.127	0.102 / 0.060
32	$f^{banff}_{7;t,r}RHC$	RHC $\bar{\nu_\mu}$ flux normalisation, E = 0.5 - 0.6 GeV	1.117	0.096 / 0.052
33	$f^{banff}_{8;t,r}RHC$	RHC $\bar{\nu_\mu}$ flux normalisation, E = 0.6 - 0.7 GeV	1.121	0.085 / 0.044
34	$f^{banff}_{9;t,r}RHC$	RHC $\bar{\nu_\mu}$ flux normalisation, E = 0.7 - 1.0 GeV	1.155	0.125 / 0.066
35	$f^{banff}_{10;t,r}RHC$	RHC $\bar{\nu_\mu}$ flux normalisation, E = 1.0 - 1.5 GeV	1.132	0.105 / 0.057
36	$f^{banff}_{11;t,r}RHC$	RHC $\bar{\nu_\mu}$ flux normalisation, E = 1.5 - 2.5 GeV	1.139	0.080 / 0.053
37	$f^{banff}_{12;t,r}RHC$	RHC $\bar{\nu_\mu}$ flux normalisation, E = 2.5 - 3.5 GeV	1.141	0.074 / 0.054
38	$f^{banff}_{13;t,r}RHC$	RHC $\bar{\nu_\mu}$ flux normalisation, E = 3.5 - 5.0 GeV	1.151	0.094 / 0.071
39	$f^{banff}_{14;t,r}RHC$	RHC $\bar{\nu_\mu}$ flux normalisation, E = 5.0 - 7.0 GeV	1.133	0.093 / 0.070
40	$f^{banff}_{15;t,r}RHC$	RHC $\bar{\nu_\mu}$ flux normalisation, E = 7.0 - 30.0 GeV	1.082	0.130 / 0.110
41	$f^{banff}_{16;t,r}RHC$	RHC ν_e flux normalisation, E = 0.0 - 2.5 GeV	1.117	0.069 / 0.051
42	$f^{banff}_{17;t,r}RHC$	RHC ν_e flux normalisation, E = 2.5 - 30.0 GeV	1.112	0.085 / 0.071
43	$f^{banff}_{18;t,r}RHC$	RHC $\bar{\nu_e}$ flux normalisation, E = 0.0 - 0.5 GeV	1.126	0.095 / 0.058
44	$f^{banff}_{19;t,r}RHC$	RHC $\bar{\nu_e}$ flux normalisation, E = 0.5 - 0.7 GeV	1.127	0.091 / 0.051
45	$f^{banff}_{20;t,r}RHC$	RHC $\bar{\nu_e}$ flux normalisation, E = 0.7 - 0.8 GeV	1.133	0.091 / 0.052
46	$f^{banff}_{21;t,r}RHC$	RHC $\bar{\nu_e}$ flux normalisation, E = 0.8 - 1.5 GeV	1.132	0.084 / 0.046
47	$f^{banff}_{22;t,r}RHC$	RHC $\bar{\nu_e}$ flux normalisation, E = 1.5 - 2.5 GeV	1.125	0.080 / 0.056
48	$f^{banff}_{23;t,r}RHC$	RHC $\bar{\nu_e}$ flux normalisation, E = 2.5 - 4.0 GeV	1.119	0.089 / 0.071
49	$f^{banff}_{24;t,r}RHC$	RHC $\bar{\nu_e}$ flux normalisation, E = 4.0 - 30.0 GeV	1.166	0.156 / 0.141

Table 7: Summary of antineutrino mode flux systematics included in the *VALOR* joint fit analysis. A description of how the BANFF fit constrained these systematics can be found in Ref. [28].

Index	Parameter	Description	Best fit	1σ pre/postfit fractional error
50	$f^{banff}_{Norm_{2p-2h}}$	Meson exchange current normalisation for ^{16}O	1.546	- / 0.343
51	$f^{banff}_{C_5^A}$	C_5^A nucleon to Δ transition axial form factor	0.789	0.119 / 0.062
52	$f^{banff}_{Bg_{RES}}$	Scale of isospin 1/2 nonresonant background	1.042	0.154 / 0.134
53	$f^{banff}_{M_A^{QE}}$	CCQE axial-mass scaling factor	0.931	- / 0.028
54	$f^{banff}_{M_A^{RES}}$	Resonance-production axial-mass scaling factor	0.889	0.158 / 0.041
55	$f^{banff}_{p_f}$	Fermi momentum for ^{16}O	1.041	- / 0.105
56	$f^{banff}_{Shape_{CCoth}}$	CC other shape	-0.022	0.400 / 0.208
57	$f^{banff}_{E_B}$	Binding energy for ^{16}O	0.882	0.333 / 0.282
58	$f^{banff}_{Norm_{CCcoh}}$	CC coherent for ^{16}O normalisation	0.858	0.300 / 0.228
59	$f^{banff}_{Norm_{NCcoh}}$	NC coherent normalisation	0.931	0.300 / 0.298
60	$f^{banff}_{Norm_{NCother}}$	NC other normalisation	1.000	0.300 / 0.300
61	$f^{banff}_{Norm_{\nu_e\to\nu_\mu}}$	CC ν_e normalisation	1.000	0.028 / 0.028
62	$f^{banff}_{Norm_{NC1\gamma}}$	NC 1γ normalisation	1.000	1.000 / 1.000
63	$f^{banff}_{Norm_{\bar{\nu}_e\to\bar{\nu}_\mu}}$	CC $\bar{\nu}_e$ normalisation	1.000	0.028 / 0.028
64	$f^{banff}_{Norm_{2p-2h-Bar}}$	Antineutrino Meson exchange current normalisation for ^{16}O	0.578	- / 0.177

Table 8: Summary of cross section systematics included in the *VALOR* joint fit analysis. Parameters with no prefit error were unconstrained. A description of how the BANFF fit constrained these systematics can be found in Ref. [28].

Index	Parameter	Description	1σ frac. error
0	$f^{SK+FSI}_{0;t,r}$	SKDet + FSI/SI 0; E_{reco} 0.00 - 0.40 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CCQE (1Rmu)	0.018
1	$f^{SK+FSI}_{1;t,r}$	SKDet + FSI/SI 1; Ereco 0.40 - 1.10 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CCQE (1Rmu)	0.017
2	$f^{SK+FSI}_{2;t,r}$	SKDet + FSI/SI 2; Ereco 1.10 - 30.00 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CCQE (1Rmu)	0.018
3	$f^{SK+FSI}_{3;t,r}$	SKDet + FSI/SI 3; Ereco 0.00 - 30.00 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CCnQE (1Rmu)	0.077
4	$f^{SK+FSI}_{4;t,r}$	SKDet + FSI/SI 4; Ereco 0.00 - 30.00 GeV; $\nu_e/\nu_{\bar{e}}$/sigν_e CC (1Rmu)	1.004
5	$f^{SK+FSI}_{5;t,r}$	SKDet + FSI/SI 5; Ereco 0.00 - 30.00 GeV; all NC (1Rmu)	0.598
6	$f^{SK+FSI}_{6;t,r}$	SKDet + FSI/SI 6; Ereco 0.00 - 0.35 GeV; osc ν_e CC (1Re)	0.100
7	$f^{SK+FSI}_{7;t,r}$	SKDet + FSI/SI 7; Ereco 0.35 - 0.80 GeV; osc ν_e CC (1Re)	0.030
8	$f^{SK+FSI}_{8;t,r}$	SKDet + FSI/SI 8; Ereco 0.80 - 1.25 GeV; osc ν_e CC (1Re)	0.033
9	$f^{SK+FSI}_{9;t,r}$	SKDet + FSI/SI 9; Ereco 0.00 - 0.35 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CC (1Re)	1.225
10	$f^{SK+FSI}_{10;t,r}$	SKDet + FSI/SI 10; Ereco 0.35 - 0.80 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CC (1Re)	1.166
11	$f^{SK+FSI}_{11;t,r}$	SKDet + FSI/SI 11; Ereco 0.80 - 1.25 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CC (1Re)	1.028
12	$f^{SK+FSI}_{12;t,r}$	SKDet + FSI/SI 12; Ereco 0.00 - 0.35 GeV; $\nu_e/\nu_{\bar{e}}$ CC (1Re)	0.063
13	$f^{SK+FSI}_{13;t,r}$	SKDet + FSI/SI 13; Ereco 0.35 - 0.80 GeV; $\nu_e/\nu_{\bar{e}}$ CC (1Re)	0.040
14	$f^{SK+FSI}_{14;t,r}$	SKDet + FSI/SI 14; Ereco 0.80 - 1.25 GeV; $\nu_e/\nu_{\bar{e}}$ CC (1Re)	0.061
15	$f^{SK+FSI}_{15;t,r}$	SKDet + FSI/SI 15; Ereco 0.00 - 0.35 GeV; all NC (1Re)	0.488
16	$f^{SK+FSI}_{16;t,r}$	SKDet + FSI/SI 16; Ereco 0.35 - 0.80 GeV; all NC (1Re)	0.194
17	$f^{SK+FSI}_{17;t,r}$	SKDet + FSI/SI 17; Ereco 0.80 - 1.25 GeV; all NC (1Re)	0.590
18	$f^{SK+FSI}_{0;t,r}RHC$	SKDet + FSI/SI 0; Ereco 0.00 - 0.40 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CCQE (1Rmu); RHC	0.018
19	$f^{SK+FSI}_{1;t,r}RHC$	SKDet + FSI/SI 1; Ereco 0.40 - 1.10 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CCQE (1Rmu); RHC	0.016
20	$f^{SK+FSI}_{2;t,r}RHC$	SKDet + FSI/SI 2; Ereco 1.10 - 30.00 GeV; $\nu_\mu/\nu_{\bar{\mu}}$CCQE (1Rmu); RHC	0.016
21	$f^{SK+FSI}_{3;t,r}RHC$	SKDet + FSI/SI 3; Ereco 0.00 - 30.00 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CCnQE (1Rmu); RHC	0.093
22	$f^{SK+FSI}_{4;t,r}RHC$	SKDet + FSI/SI 4; Ereco 0.00 - 30.00 GeV; $\nu_e/\nu_{\bar{e}}$/sigν_e CC (1Rmu); RHC	1.004
23	$f^{SK+FSI}_{5;t,r}RHC$	SKDet + FSI/SI 5; Ereco 0.00 - 30.00 GeV; all NC (1Rmu); RHC	0.597
24	$f^{SK+FSI}_{6;t,r}RHC$	SKDet + FSI/SI 6; Ereco 0.00 - 0.35 GeV; osc ν_e CC (1Re); RHC	0.060
25	$f^{SK+FSI}_{7;t,r}RHC$	SKDet + FSI/SI 7; Ereco 0.35 - 0.80 GeV; osc ν_e CC (1Re); RHC	0.027
26	$f^{SK+FSI}_{8;t,r}RHC$	SKDet + FSI/SI 8; Ereco 0.80 - 1.25 GeV; osc ν_e CC (1Re); RHC	0.056
27	$f^{SK+FSI}_{9;t,r}RHC$	SKDet + FSI/SI 9; Ereco 0.00 - 0.35 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CC (1Re); RHC	1.244
28	$f^{SK+FSI}_{10;t,r}RHC$	SKDet + FSI/SI 10; Ereco 0.35 - 0.80 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CC (1Re); RHC	1.123
29	$f^{SK+FSI}_{11;t,r}RHC$	SKDet + FSI/SI 11; Ereco 0.80 - 1.25 GeV; $\nu_\mu/\nu_{\bar{\mu}}$ CC (1Re); RHC	0.904
30	$f^{SK+FSI}_{12;t,r}RHC$	SKDet + FSI/SI 12; Ereco 0.00 - 0.35 GeV; $\nu_e/\nu_{\bar{e}}$ CC (1Re); RHC	0.044
31	$f^{SK+FSI}_{13;t,r}RHC$	SKDet + FSI/SI 13; Ereco 0.35 - 0.80 GeV; $\nu_e/\nu_{\bar{e}}$ CC (1Re); RHC	0.032
32	$f^{SK+FSI}_{14;t,r}RHC$	SKDet + FSI/SI 14; Ereco 0.80 - 1.25 GeV; $\nu_e/\nu_{\bar{e}}$ CC (1Re); RHC	0.068
33	$f^{SK+FSI}_{15;t,r}RHC$	SKDet + FSI/SI 15; Ereco 0.00 - 0.35 GeV; all NC (1Re); RHC	0.482
34	$f^{SK+FSI}_{16;t,r}RHC$	SKDet + FSI/SI 16; Ereco 0.35 - 0.80 GeV; all NC (1Re); RHC	0.171
35	$f^{SK+FSI}_{17;t,r}RHC$	SKDet + FSI/SI 17; Ereco 0.80 - 1.25 GeV; all NC (1Re); RHC	0.538
0	$f^{SK}_{E;r}$	SKDet Energy Scale; Ereco 0.00 - 30.00 GeV; All	0.024

Table 9: Summary of SK detector + FSI +SI + PN systematics included in the *VALOR* joint fit analysis.

Parameter	Sample	Category	Energy range (GeV)
0 | 18	electron-like	oscillated $\nu_e/\bar{\nu}_e$ CC	$E_{reco} < 0.35$
1 | 19	electron-like	oscillated $\nu_e/\bar{\nu}_e$ CC	$0.35 < E_{reco} < 0.80$
2 | 20	electron-like	oscillated $\nu_e/\bar{\nu}_e$ CC	$0.80 < E_{reco} < 1.25$
3 | 21	electron-like	$\nu_\mu/\bar{\nu}_\mu$ CC	$E_{reco} < 0.35$
4 | 22	electron-like	$\nu_\mu/\bar{\nu}_\mu$ CC	$0.35 < E_{reco} < 0.80$
5 | 23	electron-like	$\nu_\mu/\bar{\nu}_\mu$ CC	$0.80 < E_{reco} < 1.25$
6 | 24	electron-like	$\nu_e/\bar{\nu}_e$ CC	$E_{reco} < 0.35$
7 | 25	electron-like	$\nu_e/\bar{\nu}_e$ CC	$0.35 < E_{reco} < 0.80$
8 | 26	electron-like	$\nu_e/\bar{\nu}_e$ CC	$0.80 < E_{reco} < 1.25$
9 | 27	electron-like	all NC	$E_{reco} < 0.35$
10 | 28	electron-like	all NC	$0.35 < E_{reco} < 0.80$
11 | 29	electron-like	all NC	$0.80 < E_{reco} < 1.25$
12 | 30	muon-like	$\nu_\mu/\bar{\nu}_\mu$ CCQE	$E_{reco} < 0.40$
13 | 31	muon-like	$\nu_\mu/\bar{\nu}_\mu$ CCQE	$0.40 < E_{reco} < 1.10$
14 | 32	muon-like	$\nu_\mu/\bar{\nu}_\mu$ CCQE	$1.10 < E_{reco} < 30.0$
15 | 33	muon-like	all $\nu_\mu/\bar{\nu}_\mu$ CCnQE	$E_{reco} < 30.0$
16 | 34	muon-like	$\nu_e/\bar{\nu}_e$ CC	$E_{reco} < 30.0$
17 | 35	muon-like	all NC	$E_{reco} < 30.0$

Table 10: Binning of Super-K selection efficiency systematic parameters in covariance matrices for both electron- and muon-like samples for each category and reconstructed neutrino energy range. The same parametrization is used for both neutrino and antineutrino mode samples. The parameter number is shown for both neutrino (0-17) and antineutrino (18-35) mode.

2. **Super-K efficiencies & effects of intranuclear and secondary re-interactions** (36 parameters).

The Super-K detector uncertainties include the efficiencies of the fiducial volume and reduction chain, and the OD, ring-counting, PID, momentum and decay-electron cuts. These uncertainties have been evaluated by comparing 1412 days of SK-IV atmospheric data with atmospheric neutrino Monte Carlo. Ring counting errors are evaluated using ν_μ CCQE, ν_μ CCnQE, NC, and ν_e CC-enriched control samples from atmospheric neutrinos. The NC PID error is evaluated using a NC-enriched control sample, obtained by not using the PID information during the selection. All the Monte Carlo control samples are fitted simultaneously to the data with a χ^2 function using the selection efficiencies as fit parameters [30] [31].

Intranuclear final-state interactions (FSI) have significant effects in the T2K energy range and result in event topologies, for scattering from nuclear targets, which are drastically different from those created in neutrino scattering from free nucleons. Uncertainties due to FSI were estimated by simultaneously varying the NEUT parameters that scale the interaction probabilities for quasi-elastic scattering, absorption, charge exchange and pion production using the method described in Sec. 2 in [20]. The NIWG group produced 16 parameter sets that span the 1σ surface in the parameter space; this was done by varying the NEUT parameters when running the pion scattering simulations, and comparing the results with external pion-carbon scattering data. These 16 parameter sets are shown in Tab. 2 in [20]. Uncertainties due to secondary pion interactions (SI) were also evaluated. The NIWG group adapted the NEUT nuclear cascade model to simulate SI in the Super-K detector simulation (SKDETSIM) and to allow variations in the interaction probabilities. Details of this are given in [23]. Since the same model is used for FSI and SI, it is possible to evaluate the uncertainties in both FSI and SI simultaneously. This was done by reweighting a single e-like or μ-like ring selected SK MC with the 16 FSI parameter sets using T2KReweight. The reweighting method for FSI is described in Sec. 8 in [19]. Variations in the SI interaction probabilities were included for each FSI parameter set. Subsequently a covariance matrix was calculated using equation 1 in [20]; the quantity p in that equation is the average event weight for a FSI parameter set, and the sum is over the parameter sets.

We consider the Super-K detector efficiency and FSI+SI systematics together, and add their covariance matrices linearly. Each of these covariance matrices have 36 correlated parameters, corresponding

Parameter(s)	Prior	Range
$\sin^2\theta_{23}$	uniform	$[0.3; 0.7]$
$\sin^2 2\theta_{13}$ ($\sin^2\theta_{13}$) reactors	gauss	0.085 ± 0.005
$\sin^2 2\theta_{13}$ ($\sin^2\theta_{13}$) T2K only	uniform	$[0; 0.4]$
$\sin^2 2\theta_{12}$	gauss	0.846 ± 0.021
$\lvert\Delta m^2_{32}\rvert$ (NH) / $\lvert\Delta m^2_{13}\rvert$ (IH)	uniform	$[2; 3] \times 10^{-3}$ eV^2/c^4
Δm^2_{21}	gauss	$(7.53 \pm 0.18) \times 10^{-5}$ eV^2/c^4
δ_{CP}	uniform	$[-\pi; +\pi]$
Mass Hierarchy	fixed	NH or IH

Table 11: Treatment of the oscillation parameters in the $\nu/\bar{\nu}$ joint analysis. All the gaussian priors are from [71]. The parameter $\sin^2 2\theta_{13}$ can be constrained with the measurement of reactor experiments using the gaussian prior, otherwise a uniform prior is used and the measurement relies only on the T2K data. The nominal values of $\sin^2 2\theta_{13}$, $\sin^2 2\theta_{12}$ and Δm^2_{21} are from [71], while all the other oscillation parameter values corresponds to the most probable values obtained by the Bayesian analysis on the T2K run 1-4 neutrino mode data [14]. The mass hierarchy is not marginalised but fixed to either NH or IH.

to different modes and reconstructed energy bins, as shown in table 10. The effects of these uncertainties are also evaluated using the Cholesky method.

3. **Super-K energy scale** (1 parameter).

The systematic parameter $f^{SK}_{E;r}$ is included to estimate the effects of uncertainty in the Super-K reconstructed energy scale. This uncertainty is estimated to be 2.4% [32]; its effects are calculated by scaling the bin edges of the MC templates and, assuming uniform distribution of events within the bins, calculating the number of events gained from / lost to neighbouring bins. The effect of the energy scale uncertainty is propagated to all MC templates as shown in table 38, 39, 40 and 41 and is common to all the Super-K samples.

4. **Oscillation parameters and mass hierarchy** (7 parameters).

The 3-flavour oscillation probability is used. All the nuisance oscillation parameters, i.e. those parameters which are not shown in a given contour, are marginalised (see section 4) except the mass hierarchy which is fixed to either normal or inverted hierarchy. The treatment of the oscillation parameters in the analysis is shown in table 11 while the nominal values are shown in table 12. Two different set of oscillation parameters were chosen. The Asimov data set A is the best-fit of the Run 1-4 data Mach3 analysis [45]. The Asimov data set B is a variation of Asimov data set A, useful to test the sensitivity to the opposite $\sin^2\theta_{23}$ octant, which reduces the number of expected events in the e-like samples, as well as the sensitivity to δ_{CP}.

For $\sin^2\theta_{13}$ two different priors can be used, depending on whether the reactor constraint is applied, denoted as "reactors", or not, denoted as "T2K only". In the first case a gaussian prior given by the combined measurement of reactor experiments is used, while in the latter case a flat prior which covers a large range of values is adopted. With respect to the older global fit [70], the value of $\sin^2\theta_{13}$ has been reduced by about 10% and the error reduced as well. Since an excess of events was seen in the T2K Run 1-4 neutrino mode e-like sample and not much data are collected in neutrino mode during the Run 5+6 (only 0.34×10^{20} POT), this change is expected to have an impact on the search for CP violation, making the confidence intervals tighter.

Parameter(s)	Asimov A	Asimov B
$\sin^2\theta_{23}$	0.528	0.45
$\sin^2 2\theta_{13}$ ($\sin^2\theta_{13}$) reactors	0.085 (0.0217)	0.085 (0.0217)
$\sin^2 2\theta_{12}$	0.846	0.846
$\lvert\Delta m^2_{32}\rvert$ (NH) / $\lvert\Delta m^2_{31}\rvert$ (IH)	2.509×10^{-3} eV^2/c^4	2.509×10^{-3} eV^2/c^4
Δm^2_{21}	7.53×10^{-5} eV^2/c^4	7.53×10^{-5} eV^2/c^4
δ_{CP}	-1.601	0
Mass Hierarchy	Normal	Normal

Table 12: Values of oscillation parameters used to compute the event rates, systematic effects and sensitivity studies. Each set of oscillation parameters correspond to a different Asimov data set, which is the MC expected distribution in a certain oscillation hypothesis. In the Asimov data set A the nominal values of $\sin^2 2\theta_{13}$, $\sin^2 2\theta_{12}$ and Δm^2_{21} are from [71], while all the other oscillation parameter values corresponds to the most probable values obtained by the Bayesian analysis on the T2K run 1-4 neutrino mode data [14]. The mass hierarchy is not marginalised but fixed to either NH or IH. The only difference between the Asimov data sets A and B is given by δ_{CP}, set to 0, and $\sin^2\theta_{23}$ which corresponds to an arbitrary point in the opposite octant.

3.3. Evaluation of the effects of systematic parameters on the predicted single e-like ring reconstructed energy spectrum

In this section, we present the effects of the systematics described in section 3.1 on the predicted number of single ring events and the predicted spectra in Super-K. All results are generated for an integrated exposure corresponding to Run 1-6. and the BANFF-tuned MC templates are used.

In table 13, 14, 15 and 16 the effect of each single systematic parameter obtained by varying its value by $\pm 1\sigma$ is shown.

Parameter	Nominal	BANFF tuned	BANFF tuned +1σ		BANFF tuned -1σ	
$f^{banff}_{0;t,r}$	120.349	125.914	126.429	0.40884 %	125.399	-0.40884 %
$f^{banff}_{1;t,r}$	120.349	125.914	126.409	0.39338 %	125.418	-0.39338 %
$f^{banff}_{2;t,r}$	120.349	125.914	126.033	0.094558 %	125.795	-0.094558 %
$f^{banff}_{3;t,r}$	120.349	125.914	126.012	0.078072 %	125.815	-0.078072 %
$f^{banff}_{4;t,r}$	120.349	125.914	126.834	0.73082 %	124.994	-0.73082 %
$f^{banff}_{5;t,r}$	120.349	125.914	126.771	0.68095 %	125.056	-0.68095 %
$f^{banff}_{6;t,r}$	120.349	125.914	126.733	0.65073 %	125.094	-0.65073 %
$f^{banff}_{7;t,r}$	120.349	125.914	126.452	0.42768 %	125.375	-0.42768 %
$f^{banff}_{8;t,r}$	120.349	125.914	126.471	0.44277 %	125.356	-0.44277 %
$f^{banff}_{9;t,r}$	120.349	125.914	126.126	0.16892 %	125.701	-0.16892 %
$f^{banff}_{10;t,r}$	120.349	125.914	125.96	0.036964 %	125.867	-0.036964 %
$f^{banff}_{11;t,r}$	120.349	125.914	125.941	0.021242 %	125.887	-0.021242 %
$f^{banff}_{12;t,r}$	120.349	125.914	125.934	0.015671 %	125.894	-0.015671 %
$f^{banff}_{13;t,r}$	120.349	125.914	126.002	0.069974 %	125.826	-0.069974 %
$f^{banff}_{14;t,r}$	120.349	125.914	126.08	0.13209 %	125.747	-0.13209 %
$f^{banff}_{15;t,r}$	120.349	125.914	126.063	0.1185 %	125.765	-0.1185 %
$f^{banff}_{16;t,r}$	120.349	125.914	125.914	0.00034323 %	125.913	-0.00034323 %
$f^{banff}_{17;t,r}$	120.349	125.914	125.915	0.00058236 %	125.913	-0.00058236 %
$f^{banff}_{18;t,r}$	120.349	125.914	125.914	0.00041762 %	125.913	-0.00041762 %
$f^{banff}_{19;t,r}$	120.349	125.914	125.916	0.0021262 %	125.911	-0.0021262 %
$f^{banff}_{20;t,r}$	120.349	125.914	125.917	0.0028071 %	125.91	-0.0028071 %
$f^{banff}_{21;t,r}$	120.349	125.914	125.917	0.0024707 %	125.911	-0.0024707 %
$f^{banff}_{22;t,r}$	120.349	125.914	125.916	0.0015978 %	125.912	-0.0015978 %
$f^{banff}_{23;t,r}$	120.349	125.914	125.914	0.00041569 %	125.913	-0.00041569 %
$f^{banff}_{24;t,r}$	120.349	125.914	125.915	0.0011942 %	125.912	-0.0011942 %
$f^{banff}_{Norm_{2p-2h}}$	120.349	125.914	130.333	3.5095 %	121.495	-3.5095 %
$f^{banff}_{C_5^A}$	120.349	125.914	127.621	1.3562 %	124.311	-1.2726 %
$f^{banff}_{Bg_{RES}}$	120.349	125.914	126.985	0.85109 %	124.971	-0.74846 %
$f^{banff}_{M_A^{QE}}$	120.349	125.914	127.832	1.5237 %	123.975	-1.5395 %
$f^{banff}_{M_A^{RES}}$	120.349	125.914	127.008	0.86933 %	124.837	-0.85554 %
$f^{banff}_{p_f}$	120.349	125.914	123.763	-1.7078 %	127.899	1.5763 %
$f^{banff}_{Shape_{CCoth}}$	120.349	125.914	126.435	0.41379 %	125.393	-0.41379 %
$f^{banff}_{E_B}$	120.349	125.914	125.899	-0.01212 %	125.93	0.012715 %
$f^{banff}_{Norm_{CCcoh}}$	120.349	125.914	126.022	0.086181 %	125.805	-0.086181 %
$f^{banff}_{Norm_{NCcoh}}$	120.349	125.914	125.919	0.0044519 %	125.908	-0.0044519 %
$f^{banff}_{Norm_{NCother}}$	120.349	125.914	126.901	0.78393 %	124.927	-0.78393 %
$f^{banff}_{Norm_{\nu_e\to\nu_\mu}}$	120.349	125.914	125.926	0.0093354 %	125.902	-0.0093354 %
$f^{banff}_{Norm_{NC1\gamma}}$	120.349	125.914	125.918	0.0029615 %	125.91	-0.0029615 %
$f^{banff}_{Norm_{\bar{\nu}_e\to\bar{\nu}_\mu}}$	120.349	125.914	125.914	0.00011837 %	125.914	-0.00011837 %
$f^{banff}_{Norm_{2p-2h-Bar}}$	120.349	125.914	126.172	0.20522 %	125.655	-0.20522 %
$f^{SK}_{E;r}$	120.349	125.914	125.914	-4.5145e-14 %	125.914	-7.9003e-14 %
$f^{SK+FSI}_{0;t,r}$	120.349	125.914	126.074	0.1271 %	125.754	-0.1271 %
$f^{SK+FSI}_{1;t,r}$	120.349	125.914	126.556	0.5104 %	125.271	-0.5104 %
$f^{SK+FSI}_{2;t,r}$	120.349	125.914	126.675	0.60427 %	125.153	-0.60427 %
$f^{SK+FSI}_{3;t,r}$	120.349	125.914	128.014	1.6677 %	123.814	-1.6677 %
$f^{SK+FSI}_{4;t,r}$	120.349	125.914	126.336	0.33551 %	125.491	-0.33551 %
$f^{SK+FSI}_{5;t,r}$	120.349	125.914	130.738	3.8311 %	121.09	-3.8311 %

Table 13: $\pm1\sigma$ effect of the systematic parameters on the total number of μ-like Super-K events with 6.914×10^{20} POT for neutrino mode. Only the systematic parameters which effect this sample are shown.

Parameter	Nominal	BANFF tuned	BANFF tuned $+1\sigma$		BANFF tuned -1σ	
$f^{banff}_{0;t,r}$	24.2563	26.5935	26.6217	0.10583 %	26.5654	-0.10583 %
$f^{banff}_{1;t,r}$	24.2563	26.5935	26.7186	0.47018 %	26.4685	-0.47018 %
$f^{banff}_{2;t,r}$	24.2563	26.5935	26.8418	0.93345 %	26.3453	-0.93345 %
$f^{banff}_{3;t,r}$	24.2563	26.5935	26.8365	0.91353 %	26.3506	-0.91353 %
$f^{banff}_{4;t,r}$	24.2563	26.5935	26.8768	1.0652 %	26.3102	-1.0652 %
$f^{banff}_{5;t,r}$	24.2563	26.5935	26.6316	0.14302 %	26.5555	-0.14302 %
$f^{banff}_{6;t,r}$	24.2563	26.5935	26.6004	0.025826 %	26.5867	-0.025826 %
$f^{banff}_{7;t,r}$	24.2563	26.5935	26.5978	0.015909 %	26.5893	-0.015909 %
$f^{banff}_{8;t,r}$	24.2563	26.5935	26.5992	0.021436 %	26.5878	-0.021436 %
$f^{banff}_{9;t,r}$	24.2563	26.5935	26.5962	0.010031 %	26.5909	-0.010031 %
$f^{banff}_{10;t,r}$	24.2563	26.5935	26.595	0.0054078 %	26.5921	-0.0054078 %
$f^{banff}_{11;t,r}$	24.2563	26.5935	26.5975	0.014836 %	26.5896	-0.014836 %
$f^{banff}_{12;t,r}$	24.2563	26.5935	26.5963	0.010428 %	26.5907	-0.010428 %
$f^{banff}_{13;t,r}$	24.2563	26.5935	26.596	0.0093637 %	26.591	-0.0093637 %
$f^{banff}_{14;t,r}$	24.2563	26.5935	26.595	0.00542 %	26.5921	-0.00542 %
$f^{banff}_{15;t,r}$	24.2563	26.5935	26.5948	0.0046444 %	26.5923	-0.0046444 %
$f^{banff}_{16;t,r}$	24.2563	26.5935	26.6185	0.093913 %	26.5685	-0.093913 %
$f^{banff}_{17;t,r}$	24.2563	26.5935	26.628	0.12954 %	26.5591	-0.12954 %
$f^{banff}_{18;t,r}$	24.2563	26.5935	26.6109	0.065228 %	26.5762	-0.065228 %
$f^{banff}_{19;t,r}$	24.2563	26.5935	26.6464	0.19869 %	26.5407	-0.19869 %
$f^{banff}_{20;t,r}$	24.2563	26.5935	26.598	0.016936 %	26.589	-0.016936 %
$f^{banff}_{21;t,r}$	24.2563	26.5935	26.594	0.0017641 %	26.5931	-0.0017641 %
$f^{banff}_{22;t,r}$	24.2563	26.5935	26.5937	0.00071333 %	26.5933	-0.00071333 %
$f^{banff}_{23;t,r}$	24.2563	26.5935	26.6008	0.027288 %	26.5863	-0.027288 %
$f^{banff}_{24;t,r}$	24.2563	26.5935	26.5937	0.00079831 %	26.5933	-0.00079831 %
$f^{banff}_{Norm_{2p-2h}}$	24.2563	26.5935	27.6259	3.882 %	25.5612	-3.882 %
$f^{banff}_{C^A_5}$	24.2563	26.5935	26.8014	0.78157 %	26.3972	-0.73812 %
$f^{banff}_{Bg_{RES}}$	24.2563	26.5935	26.6753	0.30745 %	26.5216	-0.27037 %
$f^{banff}_{M^{QE}_A}$	24.2563	26.5935	27.0363	1.6648 %	26.1369	-1.7172 %
$f^{banff}_{M^{RES}_A}$	24.2563	26.5935	26.6925	0.37237 %	26.4929	-0.37839 %
$f^{banff}_{p_f}$	24.2563	26.5935	25.9799	-2.3076 %	27.1501	2.093 %
$f^{banff}_{Shape_{CCoth}}$	24.2563	26.5935	26.6063	0.048094 %	26.5807	-0.048083 %
$f^{banff}_{E_B}$	24.2563	26.5935	26.5533	-0.15134 %	26.626	0.12201 %
$f^{banff}_{Norm_{CCcoh}}$	24.2563	26.5935	26.5991	0.021135 %	26.5879	-0.021135 %
$f^{banff}_{Norm_{NCcoh}}$	24.2563	26.5935	26.6567	0.23764 %	26.5303	-0.23764 %
$f^{banff}_{Norm_{NCother}}$	24.2563	26.5935	26.6352	0.15691 %	26.5518	-0.15691 %
$f^{banff}_{Norm_{\nu_e\to\nu_\mu}}$	24.2563	26.5935	27.2933	2.6315 %	25.8937	-2.6315 %
$f^{banff}_{Norm_{NC1\gamma}}$	24.2563	26.5935	27.0873	1.8568 %	26.0997	-1.8568 %
$f^{banff}_{Norm_{\bar{\nu}_e\to\bar{\nu}_\mu}}$	24.2563	26.5935	26.6014	0.029817 %	26.5856	-0.029817 %
$f^{banff}_{Norm_{2p-2h-Bar}}$	24.2563	26.5935	26.6061	0.047159 %	26.581	-0.047159 %
$f^{SK}_{E;r}$	24.2563	26.5935	26.5935	1.3359e-14 %	26.5935	-1.3359e-14 %
$f^{SK+FSI}_{6;t,r}$	24.2563	26.5935	26.8445	0.94359 %	26.3426	-0.94359 %
$f^{SK+FSI}_{7;t,r}$	24.2563	26.5935	27.0622	1.7624 %	26.1248	-1.7624 %
$f^{SK+FSI}_{8;t,r}$	24.2563	26.5935	26.7121	0.44605 %	26.4749	-0.44605 %
$f^{SK+FSI}_{9;t,r}$	24.2563	26.5935	26.6525	0.22178 %	26.5345	-0.22178 %
$f^{SK+FSI}_{10;t,r}$	24.2563	26.5935	26.6244	0.116 %	26.5627	-0.116 %
$f^{SK+FSI}_{11;t,r}$	24.2563	26.5935	26.6021	0.0321 %	26.585	-0.0321 %
$f^{SK+FSI}_{12;t,r}$	24.2563	26.5935	26.6184	0.093385 %	26.5687	-0.093385 %
$f^{SK+FSI}_{13;t,r}$	24.2563	26.5935	26.6657	0.27133 %	26.5214	-0.27133 %
$f^{SK+FSI}_{14;t,r}$	24.2563	26.5935	26.6662	0.27323 %	26.5209	-0.27323 %
$f^{SK+FSI}_{15;t,r}$	24.2563	26.5935	26.8604	1.0036 %	26.3266	-1.0036 %
$f^{SK+FSI}_{16;t,r}$	24.2563	26.5935	26.7358	0.53512 %	26.4512	-0.53512 %
$f^{SK+FSI}_{17;t,r}$	24.2563	26.5935	26.7166	0.46264 %	26.4705	-0.46264 %

Table 14: $\pm1\sigma$ effect of the systematic parameters on the total number of e-like Super-K events with 6.914×10^{20} POT for neutrino mode. Only the systematic parameters which effect this sample are shown.

Parameter	Nominal	BANFF tuned	BANFF tuned $+1\sigma$		BANFF tuned -1σ	
$f^{banff}_{0;t,r}RHC$	33.354	34.2841	34.3212	0.1084 %	34.2469	-0.1084 %
$f^{banff}_{1;t,r}RHC$	33.354	34.2841	34.3316	0.13851 %	34.2366	-0.13851 %
$f^{banff}_{2;t,r}RHC$	33.354	34.2841	34.4224	0.4034 %	34.1458	-0.4034 %
$f^{banff}_{3;t,r}RHC$	33.354	34.2841	34.5137	0.66983 %	34.0544	-0.66983 %
$f^{banff}_{4;t,r}RHC$	33.354	34.2841	34.5214	0.69216 %	34.0468	-0.69216 %
$f^{banff}_{5;t,r}RHC$	33.354	34.2841	34.4278	0.41934 %	34.1403	-0.41934 %
$f^{banff}_{6;t,r}RHC$	33.354	34.2841	34.3655	0.23759 %	34.2026	-0.23759 %
$f^{banff}_{7;t,r}RHC$	33.354	34.2841	34.301	0.049526 %	34.2671	-0.049526 %
$f^{banff}_{8;t,r}RHC$	33.354	34.2841	34.2964	0.036085 %	34.2717	-0.036085 %
$f^{banff}_{9;t,r}RHC$	33.354	34.2841	34.4627	0.52119 %	34.1054	-0.52119 %
$f^{banff}_{10;t,r}RHC$	33.354	34.2841	34.4614	0.51727 %	34.1067	-0.51727 %
$f^{banff}_{11;t,r}RHC$	33.354	34.2841	34.4688	0.53874 %	34.0994	-0.53874 %
$f^{banff}_{12;t,r}RHC$	33.354	34.2841	34.3882	0.30381 %	34.1799	-0.30381 %
$f^{banff}_{13;t,r}RHC$	33.354	34.2841	34.3968	0.32868 %	34.1714	-0.32868 %
$f^{banff}_{14;t,r}RHC$	33.354	34.2841	34.3136	0.086075 %	34.2546	-0.086075 %
$f^{banff}_{15;t,r}RHC$	33.354	34.2841	34.2898	0.016631 %	34.2784	-0.016631 %
$f^{banff}_{16;t,r}RHC$	33.354	34.2841	34.2853	0.0035678 %	34.2828	-0.0035678 %
$f^{banff}_{17;t,r}RHC$	33.354	34.2841	34.2856	0.0045846 %	34.2825	-0.0045846 %
$f^{banff}_{18;t,r}RHC$	33.354	34.2841	34.2841	0.00019776 %	34.284	-0.00019776 %
$f^{banff}_{19;t,r}RHC$	33.354	34.2841	34.2841	0.00023211 %	34.284	-0.00023211 %
$f^{banff}_{20;t,r}RHC$	33.354	34.2841	34.2841	0.00018231 %	34.284	-0.00018231 %
$f^{banff}_{21;t,r}RHC$	33.354	34.2841	34.2843	0.0007317 %	34.2838	-0.0007317 %
$f^{banff}_{22;t,r}RHC$	33.354	34.2841	34.2845	0.0011599 %	34.2837	-0.0011599 %
$f^{banff}_{23;t,r}RHC$	33.354	34.2841	34.2846	0.0015682 %	34.2835	-0.0015682 %
$f^{banff}_{24;t,r}RHC$	33.354	34.2841	34.2846	0.0016844 %	34.2835	-0.0016844 %
$f^{banff}_{Norm_{2p-2h}}$	33.354	34.2841	35.2949	2.9484 %	33.2732	-2.9484 %
$f^{banff}_{C_5^A}$	33.354	34.2841	34.7665	1.4073 %	33.8496	-1.2673 %
$f^{banff}_{Bg_{RES}}$	33.354	34.2841	34.6114	0.95474 %	33.9962	-0.8396 %
$f^{banff}_{M_A^{QE}}$	33.354	34.2841	34.7782	1.4412 %	33.7964	-1.4226 %
$f^{banff}_{M_A^{RES}}$	33.354	34.2841	34.557	0.79619 %	34.0229	-0.76184 %
$f^{banff}_{p_f}$	33.354	34.2841	33.7369	-1.596 %	34.8188	1.5598 %
$f^{banff}_{Shape_{CCoth}}$	33.354	34.2841	34.4278	0.41939 %	34.1403	-0.41939 %
$f^{banff}_{E_B}$	33.354	34.2841	34.2825	-0.0045403 %	34.2856	0.0043363 %
$f^{banff}_{Norm_{CCcoh}}$	33.354	34.2841	34.3434	0.17311 %	34.2247	-0.17311 %
$f^{banff}_{Norm_{NCcoh}}$	33.354	34.2841	34.2866	0.0073745 %	34.2815	-0.0073745 %
$f^{banff}_{Norm_{NCother}}$	33.354	34.2841	34.5427	0.75445 %	34.0254	-0.75445 %
$f^{banff}_{Norm_{\nu_e\to\nu_\mu}}$	33.354	34.2841	34.2846	0.0015892 %	34.2835	-0.0015892 %
$f^{banff}_{Norm_{NC1\gamma}}$	33.354	34.2841	34.2844	0.00099164 %	34.2837	-0.00099164 %
$f^{banff}_{Norm_{\bar{\nu}_e\to\bar{\nu}_\mu}}$	33.354	34.2841	34.2848	0.0022231 %	34.2833	-0.0022231 %
$f^{banff}_{Norm_{2p-2h-Bar}}$	33.354	34.2841	34.9123	1.8323 %	33.6559	-1.8323 %
$f^{SK}_{E;r}$	33.354	34.2841	34.2841	-2.0725e-14 %	34.2841	-2.0725e-14 %
$f^{SK+FSI}_{0;t,r}RHC$	33.354	34.2841	34.3274	0.12641 %	34.2407	-0.12641 %
$f^{SK+FSI}_{1;t,r}RHC$	33.354	34.2841	34.4079	0.36115 %	34.1602	-0.36115 %
$f^{SK+FSI}_{2;t,r}RHC$	33.354	34.2841	34.5168	0.67883 %	34.0513	-0.67883 %
$f^{SK+FSI}_{3;t,r}RHC$	33.354	34.2841	35.0403	2.2058 %	33.5278	-2.2058 %
$f^{SK+FSI}_{4;t,r}RHC$	33.354	34.2841	34.3304	0.13529 %	34.2377	-0.13529 %
$f^{SK+FSI}_{5;t,r}RHC$	33.354	34.2841	35.3403	3.0809 %	33.2278	-3.0809 %

Table 15: $\pm1\sigma$ effect of the systematic parameters on the total number of μ-like Super-K events with 4.011×10^{20} POT for antineutrino mode. Only the systematic parameters which effect this sample are shown.

Parameter	Nominal	BANFF tuned	BANFF tuned $+1\sigma$		BANFF tuned -1σ	
$f^{banff}_{0;t,r}RHC$	3.11707	3.21831	3.23425	0.49537 %	3.20237	-0.49537 %
$f^{banff}_{1;t,r}RHC$	3.11707	3.21831	3.22891	0.32934 %	3.20771	-0.32934 %
$f^{banff}_{2;t,r}RHC$	3.11707	3.21831	3.22364	0.16581 %	3.21297	-0.16581 %
$f^{banff}_{3;t,r}RHC$	3.11707	3.21831	3.22045	0.066577 %	3.21617	-0.066577 %
$f^{banff}_{4;t,r}RHC$	3.11707	3.21831	3.22085	0.078879 %	3.21577	-0.078879 %
$f^{banff}_{5;t,r}RHC$	3.11707	3.21831	3.22043	0.065975 %	3.21619	-0.065975 %
$f^{banff}_{6;t,r}RHC$	3.11707	3.21831	3.22509	0.21083 %	3.21152	-0.21083 %
$f^{banff}_{7;t,r}RHC$	3.11707	3.21831	3.23439	0.49965 %	3.20223	-0.49965 %
$f^{banff}_{8;t,r}RHC$	3.11707	3.21831	3.2365	0.56517 %	3.20012	-0.56517 %
$f^{banff}_{9;t,r}RHC$	3.11707	3.21831	3.25195	1.0453 %	3.18467	-1.0453 %
$f^{banff}_{10;t,r}RHC$	3.11707	3.21831	3.2239	0.17371 %	3.21272	-0.17371 %
$f^{banff}_{11;t,r}RHC$	3.11707	3.21831	3.21966	0.041857 %	3.21696	-0.041857 %
$f^{banff}_{12;t,r}RHC$	3.11707	3.21831	3.21896	0.020249 %	3.21766	-0.020249 %
$f^{banff}_{13;t,r}RHC$	3.11707	3.21831	3.21907	0.023667 %	3.21755	-0.023667 %
$f^{banff}_{14;t,r}RHC$	3.11707	3.21831	3.21857	0.0081707 %	3.21805	-0.0081707 %
$f^{banff}_{15;t,r}RHC$	3.11707	3.21831	3.21855	0.0074537 %	3.21807	-0.0074537 %
$f^{banff}_{16;t,r}RHC$	3.11707	3.21831	3.23255	0.44247 %	3.20407	-0.44247 %
$f^{banff}_{17;t,r}RHC$	3.11707	3.21831	3.21849	0.0055449 %	3.21813	-0.0055449 %
$f^{banff}_{18;t,r}RHC$	3.11707	3.21831	3.22275	0.13805 %	3.21387	-0.13805 %
$f^{banff}_{19;t,r}RHC$	3.11707	3.21831	3.22312	0.14949 %	3.2135	-0.14949 %
$f^{banff}_{20;t,r}RHC$	3.11707	3.21831	3.22095	0.081949 %	3.21567	-0.081949 %
$f^{banff}_{21;t,r}RHC$	3.11707	3.21831	3.22701	0.2705 %	3.2096	-0.2705 %
$f^{banff}_{22;t,r}RHC$	3.11707	3.21831	3.21896	0.020273 %	3.21766	-0.020273 %
$f^{banff}_{23;t,r}RHC$	3.11707	3.21831	3.21839	0.002446 %	3.21823	-0.002446 %
$f^{banff}_{24;t,r}RHC$	3.11707	3.21831	3.21838	0.002219 %	3.21824	-0.002219 %
$f^{banff}_{Norm_{2p-2h}}$	3.11707	3.21831	3.31425	2.9812 %	3.12237	-2.9812 %
$f^{banff}_{C^A_5}$	3.11707	3.21831	3.25166	1.0364 %	3.19031	-0.86988 %
$f^{banff}_{B_{gRES}}$	3.11707	3.21831	3.23459	0.5058 %	3.20399	-0.4448 %
$f^{banff}_{M^{QE}_A}$	3.11707	3.21831	3.26276	1.3812 %	3.17411	-1.3732 %
$f^{banff}_{M^{RES}_A}$	3.11707	3.21831	3.22955	0.34945 %	3.20849	-0.30505 %
$f^{banff}_{p_f}$	3.11707	3.21831	3.14236	-2.3599 %	3.29191	2.2869 %
$f^{banff}_{Shape_{CCoth}}$	3.11707	3.21831	3.22119	0.089515 %	3.21543	-0.089518 %
$f^{banff}_{E_B}$	3.11707	3.21831	3.21749	-0.025439 %	3.21891	0.018569 %
$f^{banff}_{Norm_{CCcoh}}$	3.11707	3.21831	3.22518	0.21355 %	3.21144	-0.21355 %
$f^{banff}_{Norm_{NCcoh}}$	3.11707	3.21831	3.2514	1.0282 %	3.18522	-1.0282 %
$f^{banff}_{Norm_{NCother}}$	3.11707	3.21831	3.22886	0.32779 %	3.20776	-0.32779 %
$f^{banff}_{Norm_{\nu_e\to\nu_\mu}}$	3.11707	3.21831	3.24258	0.75412 %	3.19404	-0.75412 %
$f^{banff}_{Norm_{NC1\gamma}}$	3.11707	3.21831	3.33932	3.76 %	3.0973	-3.76 %
$f^{banff}_{Norm_{\bar{\nu}_e\to\bar{\nu}_\mu}}$	3.11707	3.21831	3.27411	1.734 %	3.1625	-1.734 %
$f^{banff}_{Norm_{2p-2h-Bar}}$	3.11707	3.21831	3.2943	2.3612 %	3.14232	-2.3612 %
$f^{SK}_{E;r}$	3.11707	3.21831	3.21831	-1.3799e-14 %	3.21831	-4.1397e-14 %
$f^{SK+FSI}_{6;t,r}RHC$	3.11707	3.21831	3.22805	0.30255 %	3.20857	-0.30255 %
$f^{SK+FSI}_{7;t,r}RHC$	3.11707	3.21831	3.25713	1.2064 %	3.17948	-1.2064 %
$f^{SK+FSI}_{8;t,r}RHC$	3.11707	3.21831	3.24365	0.78746 %	3.19297	-0.78746 %
$f^{SK+FSI}_{9;t,r}RHC$	3.11707	3.21831	3.22378	0.16988 %	3.21284	-0.16988 %
$f^{SK+FSI}_{10;t,r}RHC$	3.11707	3.21831	3.2219	0.11164 %	3.21472	-0.11164 %
$f^{SK+FSI}_{11;t,r}RHC$	3.11707	3.21831	3.22052	0.068625 %	3.2161	-0.068625 %
$f^{SK+FSI}_{12;t,r}RHC$	3.11707	3.21831	3.22172	0.10612 %	3.21489	-0.10612 %
$f^{SK+FSI}_{13;t,r}RHC$	3.11707	3.21831	3.23071	0.38524 %	3.20591	-0.38524 %
$f^{SK+FSI}_{14;t,r}RHC$	3.11707	3.21831	3.23982	0.66827 %	3.1968	-0.66827 %
$f^{SK+FSI}_{15;t,r}RHC$	3.11707	3.21831	3.28217	1.9844 %	3.15444	-1.9844 %
$f^{SK+FSI}_{16;t,r}RHC$	3.11707	3.21831	3.25076	1.0083 %	3.18586	-1.0083 %
$f^{SK+FSI}_{17;t,r}RHC$	3.11707	3.21831	3.2479	0.91957 %	3.18871	-0.91957 %

Table 16: $\pm1\sigma$ effect of the systematic parameters on the total number of e-like Super-K events with 4.011×10^{20} POT for antineutrino mode. Only the systematic parameters which effect this sample are shown.

The effects of all the combined systematic uncertainties on the reconstructed energy spectrum of single ring events is shown in fig. 10 for the typical oscillation scenario with parameters of Asimov datasets A. In appendix D the same plots are shown for different oscillation scenarios. These plots show the total error envelope for the reconstructed energy spectra; this is calculated as the $\pm 1\sigma$ spread with respect to the median of bin contents from 10k toy MC experiments generated with randomized systematic parameters. All systematic parameters are considered and their correlations are taken into account. The spectra were tuned with the systematic parameters measured with the ND280 data shown in table 6, 7, 8, the nominal Super-K systematic parameters shown in table 9 and the oscillation parameter priors shown in table 11. In fig. 11, 12, 13, 14, 15 the error envelopes and fractional errors of reconstructed energy spectra for all the selected samples, given respectively by oscillations, Super-K detector + FSI/SI, flux only, cross section only, flux+cross section systematics are shown.

The effects of systematic uncertainties on the predicted number of events are summarised in table 17, 18, 19 and 20 with the typical parameter values given in table 12. For the measurement of δ_{CP} it is important to understand the effect of the systematic uncertainties on the ratio of the number of events between neutrino and antineutrino mode samples, which is the consequence of correlations between systematic parameters. If correlations do not reproduce the physics correctly, a fake asymmetry between neutrino and antineutrino oscillations can be seen. On the other hand if correlations exist but are not taken into account, the measurement of δ_{CP} could be biased. Depending on the value of δ_{CP}, correlations or anti-correlations between parameters are more conservative. If true $\delta_{CP} = \pm\pi/2$ the most conservative case is given by anti-correlation in the e-like samples, since the oscillation asymmetry could be hidden by the systematic uncertainties. If true $\delta_{CP} \sim 0$ uncorrelated parameters reduce the sensitivity to δ_{CP}. The effect of the systematic parameters on the ratio between neutrino and antineutrino mode events are shown in table 21 and 46 in appendix D. The most critical systematic parameters are σ_{ν_e} and $\sigma_{\bar{\nu}_e}$ which are applied respectively to ν_e and $\bar{\nu}_e$ events. Their fractional error is 2.83% and are partially anti-correlated with a non-diagonal elements in the covariance matrix of -0.0004 [40].

For the systematic effect study, in order to reduce the CPU time, the toys MC were produced using 1-dimensional templates E_{reco} e-like templates, instead of the 2-dimensional E_{reco} vs θ, used in the analysis. However it has been checked that using 1- or 2- dimensional templates does not make any difference in the drawing of the error envelopes and the effect of the systematic uncertainties on the total number of events.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ (Asimov A)
SKDet+FSI+SI	4.15%
Flux	3.59%
2p-2h (corr)	3.54%
2p-2h-bar (corr)	0.20%
NC other (uncorr)	0.77%
NC 1gamma (uncorr)	0.00%
XSec nue/numu (uncorr)	0.01%
XSec Tot (corr)	4.02%
XSec Tot	4.10%
Flux+XSec (ND280 constrained)	2.82%
Flux+XSec (All)	2.92%
Flux+XSec+SKDet+FSI+SI	5.09%
Oscillations	0.03%
All	5.13%
All (pre-fit)	11.9%

Table 17: Effect of 1σ variations of the groups of systematics parameters on total number of 1-ring muon-like events for neutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in table 6, 7 8 and 9 and the oscillation parameters of the Asimov data set A shown in table 12 are used. Since the parameters δ_{CP}, θ_{23} and Δm^2_{32} (Δm^2_{13}) and mass hierarchy will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. Pre-BANFF fit systematic uncertainties are used were stated.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ (Asimov A)
SKDet+FSI+SI	3.50%
Flux	3.67%
2p-2h (corr)	3.85%
2p-2h bar (corr)	0.05%
NC other (uncorr)	0.16%
NC 1gamma (uncorr)	1.47%
XSec nue/numu (uncorr)	2.60%
XSec Tot (corr)	4.25%
XSec Tot	5.20%
Flux+XSec (ND280 constrained)	2.84%
Flux+XSec (All)	4.13%
Flux+XSec+SKDet+FSI+SI	5.53%
Oscillations	4.16%
All	6.80%
All (pre-fit)	12.6%

Table 18: Effect of 1σ variations of the groups of systematics parameters on total number of 1-ring electron-like events for neutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in table 6, 7 8 and 9 and the oscillation parameters of the Asimov data set A shown in table 12 are used. Since the parameters δ_{CP}, θ_{23} and Δm^2_{32} (Δm^2_{13}) and mass hierarchy will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. Pre-BANFF fit systematic uncertainties are used were stated.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ (Asimov A)
SKDet+FSI+SI	3.94%
Flux	3.66%
2p-2h (corr)	2.96%
2p-2h bar (corr)	1.82%
NC other (uncorr)	0.75%
NC 1gamma (uncorr)	0.00%
XSec nue/numu (uncorr)	0.00%
XSec Tot (corr)	4.08%
XSec Tot	4.14%
Flux+XSec (ND280 constrained)	3.27%
Flux+XSec (All)	3.36%
Flux+XSec+SKDet+FSI+SI	5.18%
Oscillations	0.03%
All	5.12%
All (pre-fit)	12.7%

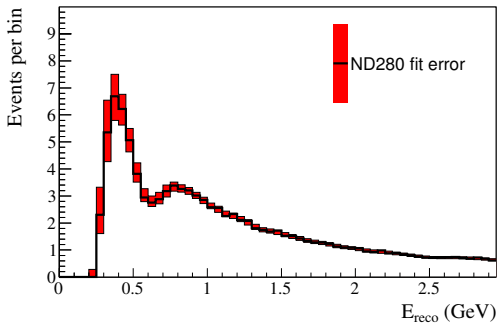
Table 19: Effect of 1σ variations of the groups of systematics parameters on total number of 1-ring muon-like events for antineutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in table 6, 7 8 and 9 and the oscillation parameters of the Asimov data set A shown in table 12 are used. Since the parameters δ_{CP}, θ_{23} and Δm^2_{32} (Δm^2_{13}) and mass hierarchy will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. Pre-BANFF fit systematic uncertainties are used were stated.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ (Asimov A)
SKDet+FSI+SI	3.97%
Flux	3.78%
2p-2h (corr)	3.00%
2p-2h bar (corr)	2.36%
NC other (uncorr)	0.33%
NC 1gamma (uncorr)	2.93%
XSec nue/numu (uncorr)	1.51%
XSec Tot (corr)	4.30%
XSec Tot	5.43%
Flux+XSec (ND280 constrained)	3.19%
Flux+XSec	4.61%
Flux+XSec+SKDet+FSI+SI	6.21%
Oscillations	4.00%
All	7.41%
All (pre-fit)	14.3%

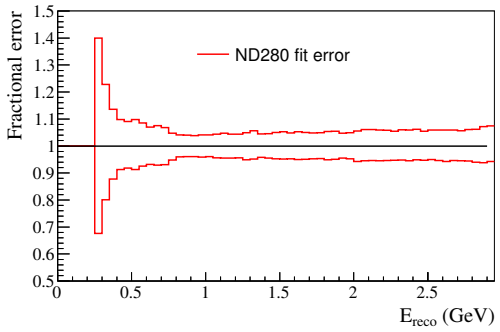
Table 20: Effect of 1σ variations of the groups of systematics parameters on total number of 1-ring electron-like events for antineutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in table 6, 7 8 and 9 and the oscillation parameters of the Asimov data set A shown in table 12 are used. Since the parameters δ_{CP}, θ_{23} and Δm^2_{32} (Δm^2_{13}) and mass hierarchy will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. Pre-BANFF fit systematic uncertainties are used were stated.

Source of uncertainty	μ-like $\delta\left(\frac{\#\nu\text{-mode}}{\#\bar{\nu}\text{-mode}}\right) / \left\langle\frac{\#\nu\text{-mode}}{\#\bar{\nu}\text{-mode}}\right\rangle$	e-like $\delta\left(\frac{\#\nu\text{-mode}}{\#\bar{\nu}\text{-mode}}\right) / \left\langle\frac{\#\nu\text{-mode}}{\#\bar{\nu}\text{-mode}}\right\rangle$
SKDet	0.7%	1.6%
FSI+SI	2.6%	3.5%
SKDet+FSI+SI	2.6%	3.8%
Flux	1.7%	1.7%
XSec NC other (uncorr)	0.0%	0.2%
XSec NC 1γ (uncorr)	0.0%	1.5%
XSec ν_e / ν_μ (uncorr)	0.0%	3.1%
XSec Tot (All)	1.7%	4.2%
Flux+XSec	1.8%	4.1%
Flux+XSec+SDet+FSI+SI	3.2%	5.7%
Oscillations	0.0%	0.8%
All	3.2%	5.8%

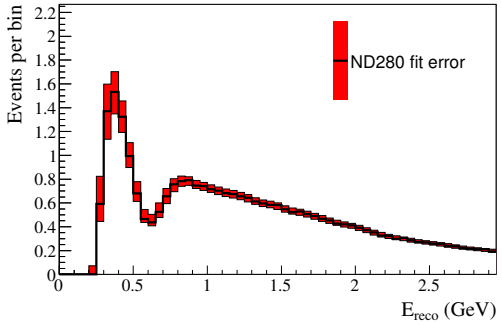
Table 21: Effect of 1σ variations of the several source of systematic uncertainties on the ratio between the total number of events in neutrino and antineutrino modes for 1-ring electron- and muon-like samples obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in table 6, 7 8 and 9 and the oscillation parameters of the Asimov data set A shown in table 12 are used. Since the parameters δ_{CP}, θ_{23} and Δm^2_{32} (Δm^2_{13}) and mass hierarchy will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.



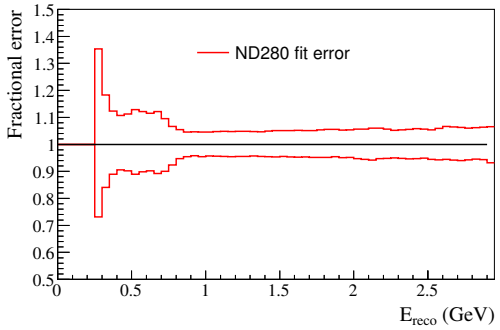
(a) Muon-like, neutrino mode



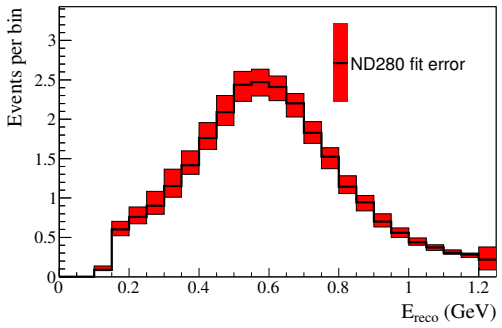
(b) Muon-like, neutrino mode



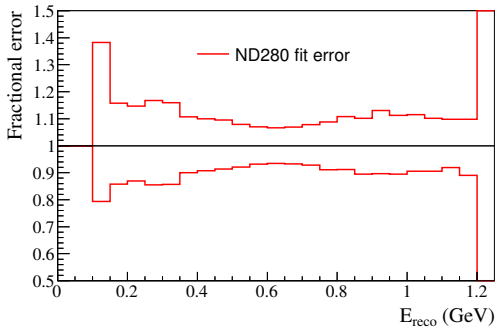
(c) Muon-like, antineutrino mode



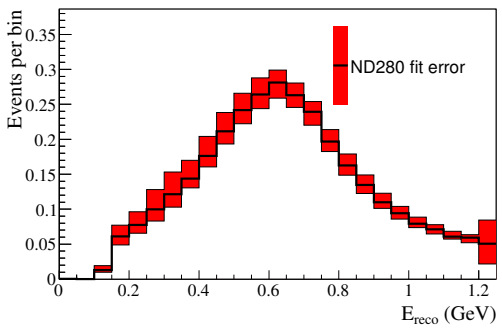
(d) Muon-like, antineutrino mode



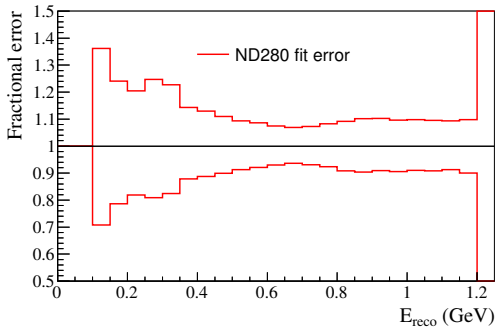
(e) Electron-like, neutrino mode



(f) Electron-like, neutrino mode

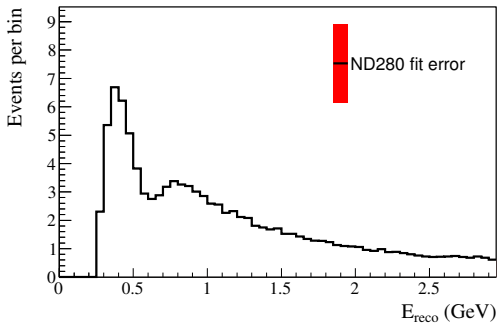


(g) Electron-like, antineutrino mode

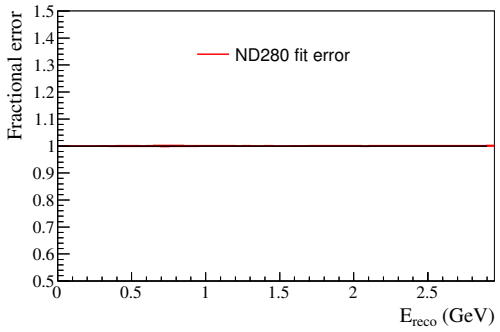


(h) Electron-like, antineutrino mode

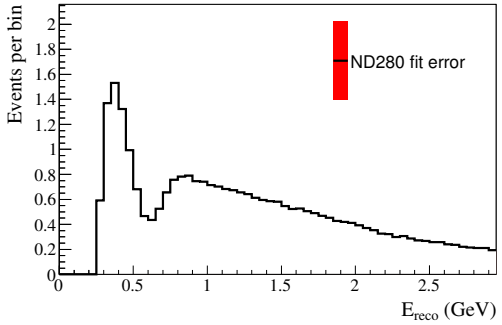
Figure 10: Error envelopes (left) and fractional errors (right) for the reconstructed neutrino energy spectrum for all the systematic parameters. 10k toy MC experiments are generated for the Asimov data set A shown in table 12 with randomized systematic parameters, taking into account their correlations. Since the parameters δ_{CP}, θ_{23} and Δm^2_{32} (Δm^2_{13}) and mass hierarchy will be measured, only the uncertainty on $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. BANFF tuned MC templates are used. The distributions correspond to the collected statistics in the full Run 1-6 data set.



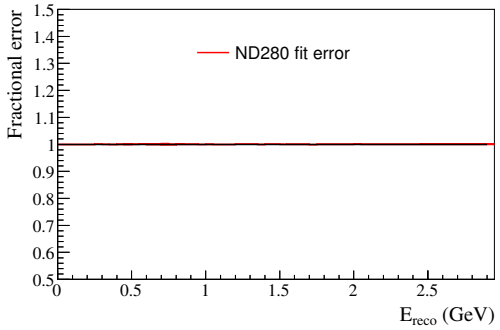
(a) Muon-like, neutrino mode



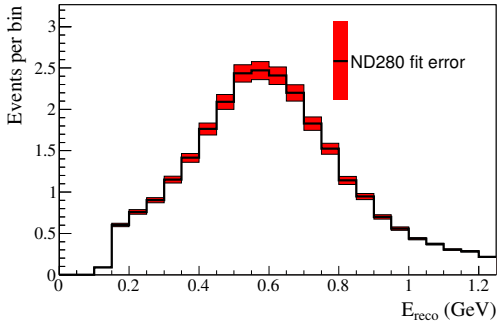
(b) Muon-like, neutrino mode



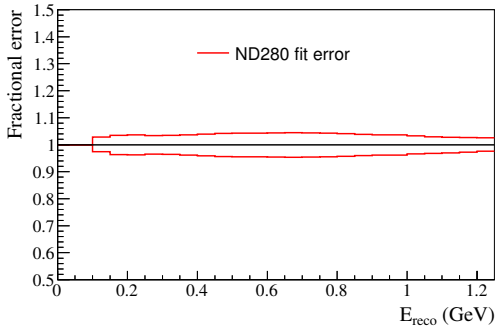
(c) Muon-like, antineutrino mode



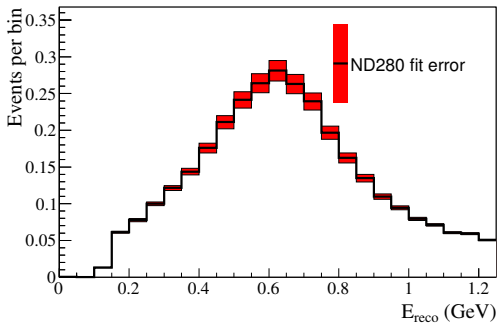
(d) Muon-like, antineutrino mode



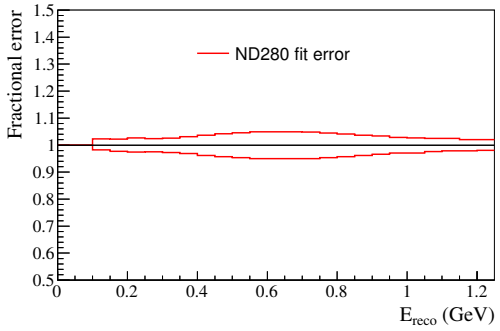
(e) Electron-like, neutrino mode



(f) Electron-like, neutrino mode

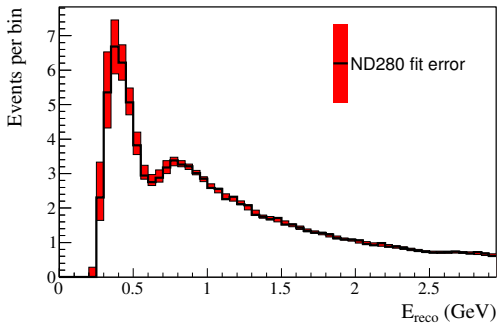


(g) Electron-like, antineutrino mode

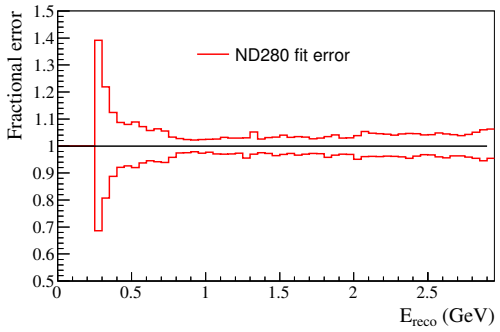


(h) Electron-like, antineutrino mode

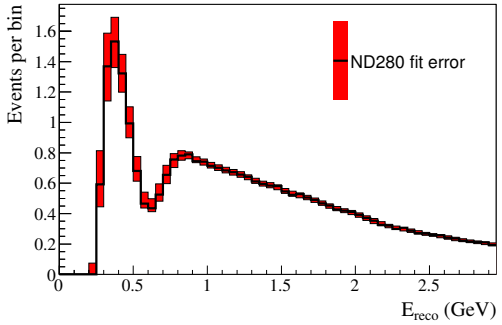
Figure 11: Error envelopes (left) and fractional errors (right) for the reconstructed neutrino energy spectrum as a result of the oscillation systematic parameters. 10k toy MC experiments are generated for the Asimov data set A shown in table 12 with randomized systematic parameters, taking into account their correlations. Since the parameters δ_{CP}, θ_{23} and Δm^2_{32} (Δm^2_{13}) and mass hierarchy will be measured, only the uncertainty on $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. BANFF tuned MC templates are used. The distributions correspond to the collected statistics in the full Run 1-6 data set.



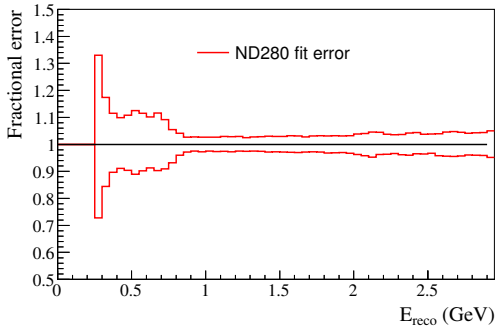
(a) Muon-like, neutrino mode



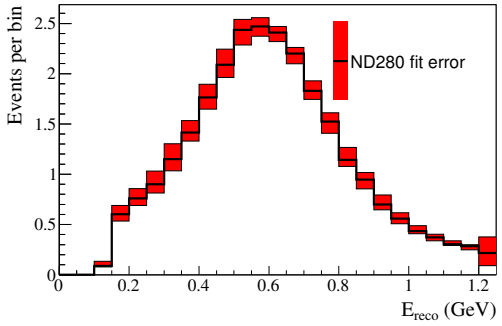
(b) Muon-like, neutrino mode



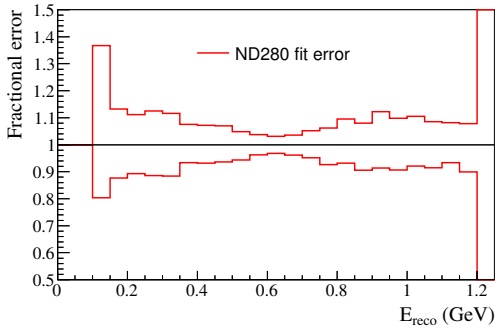
(c) Muon-like, antineutrino mode



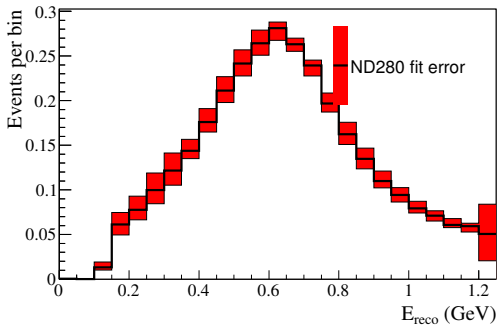
(d) Muon-like, antineutrino mode



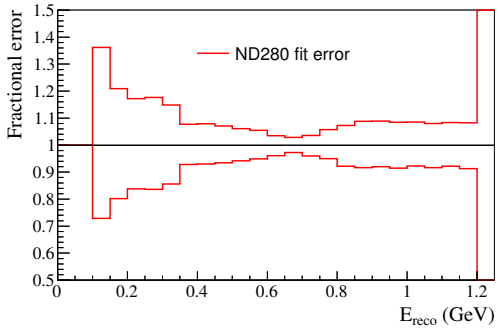
(e) Electron-like, neutrino mode



(f) Electron-like, neutrino mode

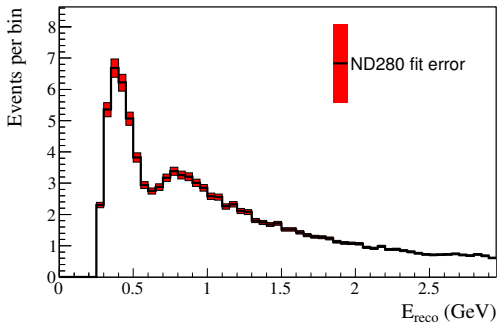


(g) Electron-like, antineutrino mode

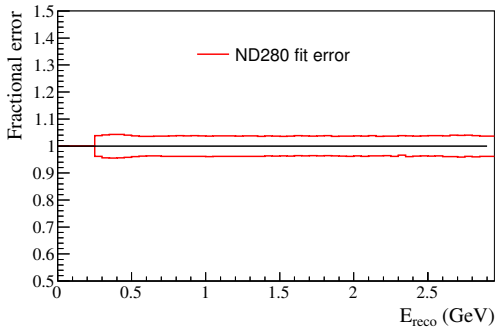


(h) Electron-like, antineutrino mode

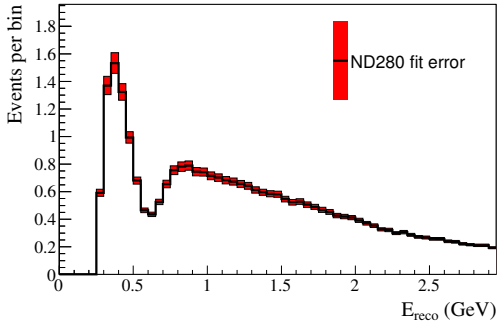
Figure 12: Error envelopes (left) and fractional errors (right) for the reconstructed neutrino energy spectrum as a result of the Super-K + FSI + SI + PN systematic parameters. 10k toy MC experiments are generated for the Asimov data set A shown in table 12 with randomized systematic parameters, taking into account their correlations. Since the parameters δ_{CP}, θ_{23} and Δm^2_{32} (Δm^2_{13}) and mass hierarchy will be measured, only the uncertainty on $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. BANFF tuned MC templates are used. The distributions correspond to the collected statistics in the full Run 1-6 data set.



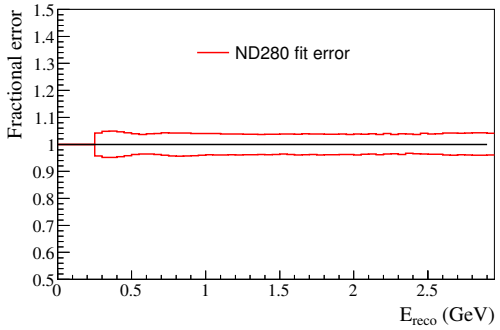
(a) Muon-like, neutrino mode



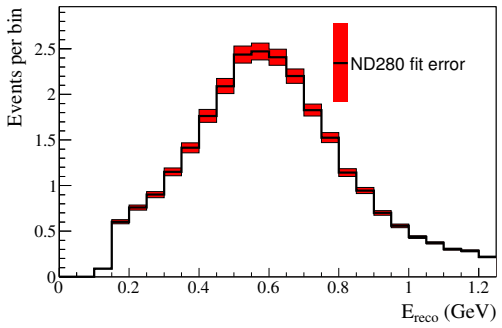
(b) Muon-like, neutrino mode



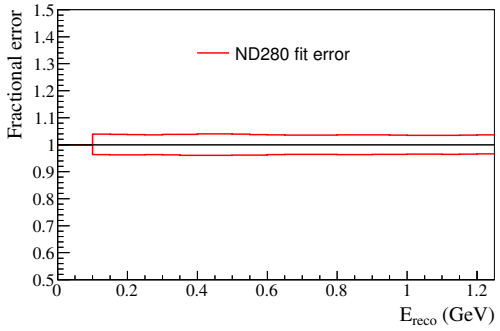
(c) Muon-like, antineutrino mode



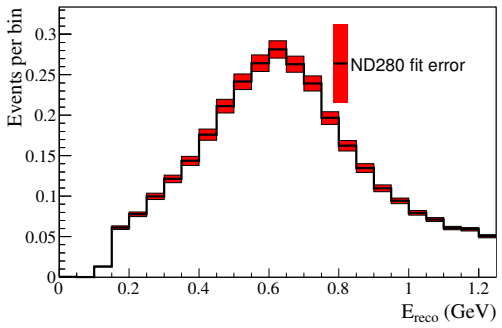
(d) Muon-like, antineutrino mode



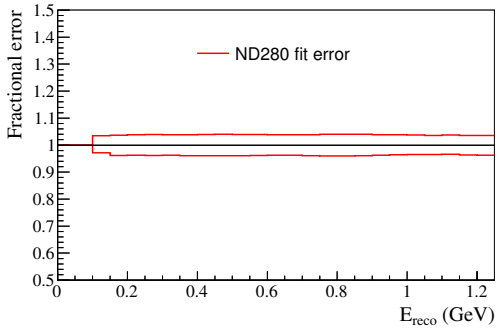
(e) Electron-like, neutrino mode



(f) Electron-like, neutrino mode

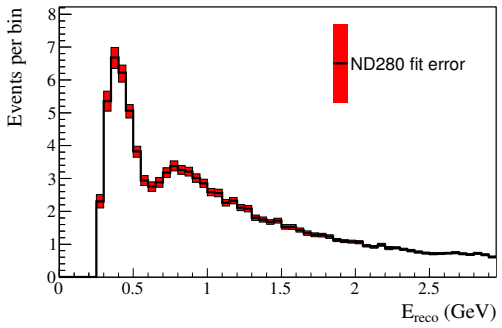


(g) Electron-like, antineutrino mode

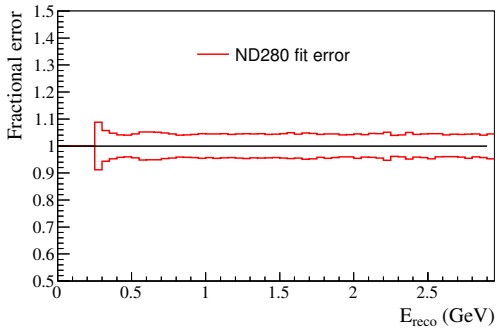


(h) Electron-like, antineutrino mode

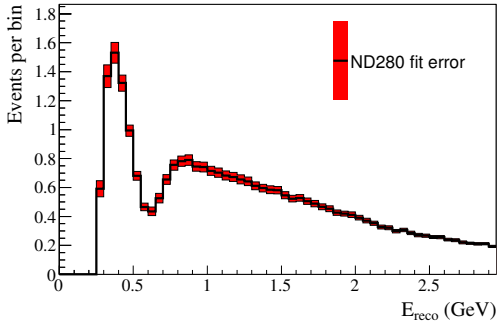
Figure 13: Error envelopes (left) and fractional errors (right) for the reconstructed neutrino energy spectrum as a result of the flux systematic parameters. 10k toy MC experiments are generated for the Asimov data set A shown in table 12 with randomized systematic parameters, taking into account their correlations. Since the parameters δ_{CP}, θ_{23} and Δm^2_{32} (Δm^2_{13}) and mass hierarchy will be measured, only the uncertainty on $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. BANFF tuned MC templates are used. The distributions correspond to the collected statistics in the full Run 1-6 data set.



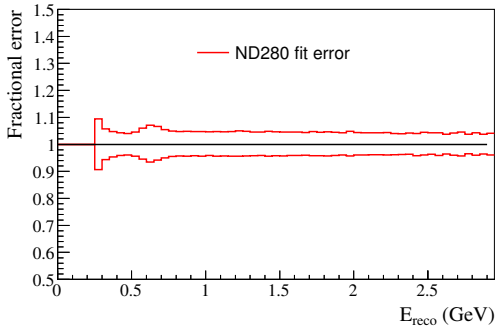
(a) Muon-like, neutrino mode



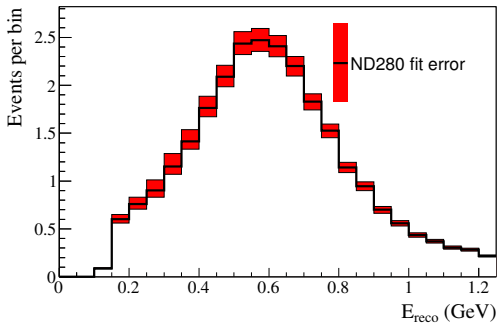
(b) Muon-like, neutrino mode



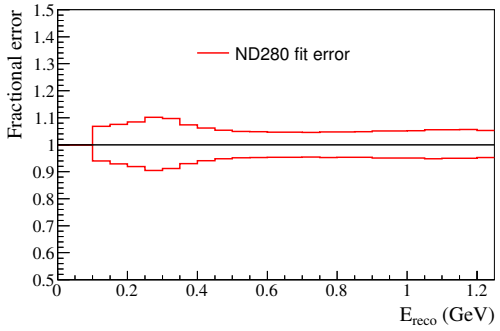
(c) Muon-like, antineutrino mode



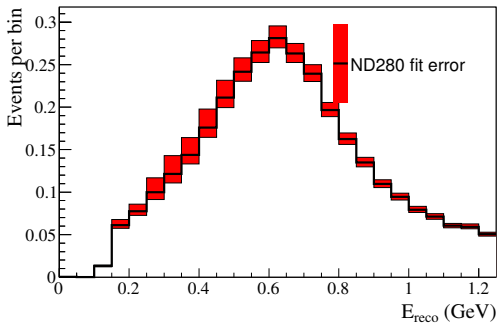
(d) Muon-like, antineutrino mode



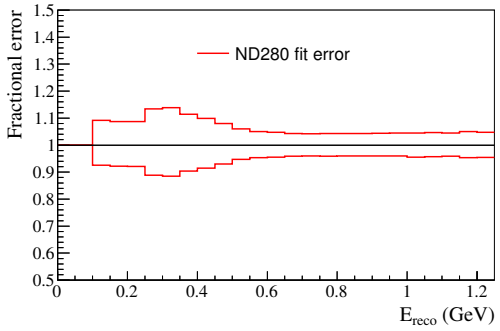
(e) Electron-like, neutrino mode



(f) Electron-like, neutrino mode

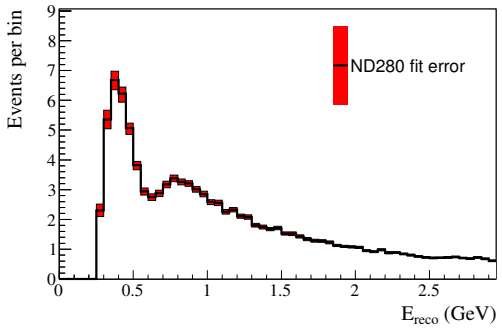


(g) Electron-like, antineutrino mode

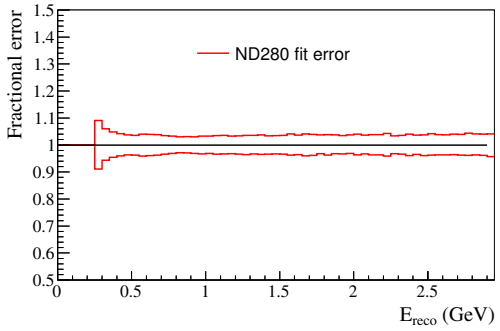


(h) Electron-like, antineutrino mode

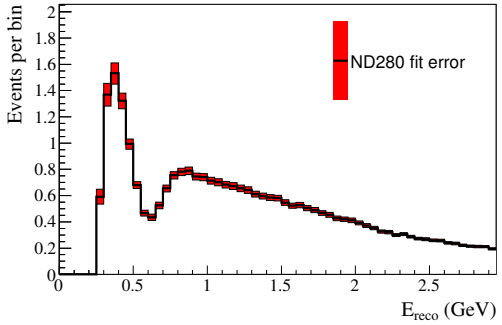
Figure 14: Error envelopes (left) and fractional errors (right) for the reconstructed neutrino energy spectrum as a result of the cross section systematic parameters. 10k toy MC experiments are generated for the Asimov data set A shown in table 12 with randomized systematic parameters, taking into account their correlations. Since the parameters δ_{CP}, θ_{23} and Δm^2_{32} (Δm^2_{13}) and mass hierarchy will be measured, only the uncertainty on $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. BANFF tuned MC templates are used. The distributions correspond to the collected statistics in the full Run 1-6 data set.



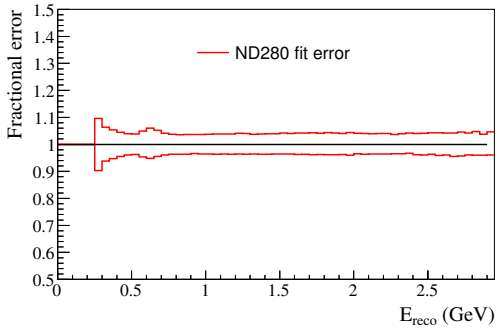
(a) Muon-like, neutrino mode



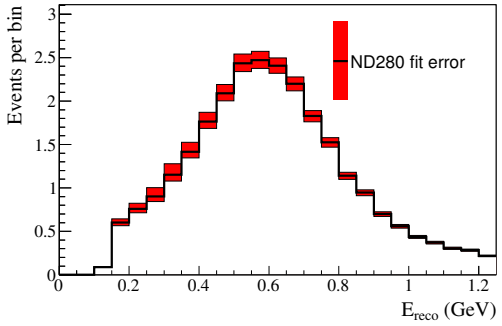
(b) Muon-like, neutrino mode



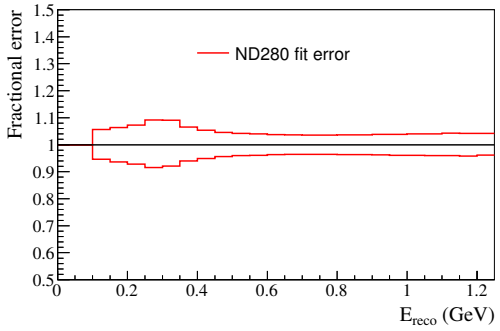
(c) Muon-like, antineutrino mode



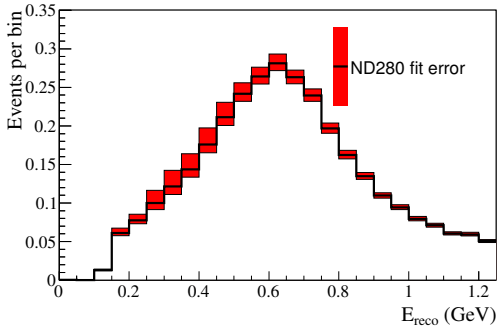
(d) Muon-like, antineutrino mode



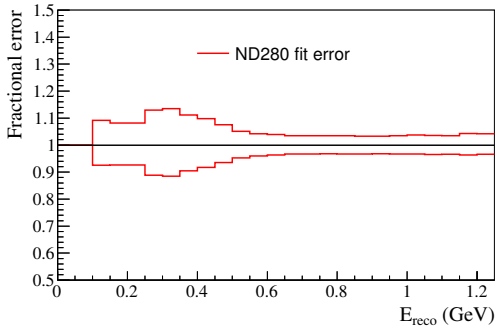
(e) Electron-like, neutrino mode



(f) Electron-like, neutrino mode



(g) Electron-like, antineutrino mode



(h) Electron-like, antineutrino mode

Figure 15: Error envelopes (left) and fractional errors (right) for the reconstructed neutrino energy spectrum as a result of the flux plus cross section systematic parameters. 10k toy MC experiments are generated for the Asimov data set A shown in table 12 with randomized systematic parameters, taking into account their correlations. Since the parameters δ_{CP}, θ_{23} and Δm^2_{32} (Δm^2_{13}) and mass hierarchy will be measured, only the uncertainty on $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. BANFF tuned MC templates are used. The distributions correspond to the collected statistics in the full Run 1-6 data set.

4. Analysis procedure

4.1. Fit method

The goal of this analysis is to measure the oscillation parameters $\sin^2\theta_{13}$, $\sin^2\theta_{23}$, Δm^2_{32} and search for CP violation by setting limits on δ_{CP} for a given hypothesis of mass hierarchy, which is fixed to either normal or inverted. All the systematics, as well as the oscillation parameters which are not measured in that particular analysis and treated as nuisance, are marginalised. The treatment of the oscillation parameters is shown in table 11. The parameters of interest will be denoted with the vector $\vec{\theta}$, since more than 1 parameter at a time can be measured. $\vec{\theta}$ can include the combination of all the oscillation parameters listed above.

The *VALOR* analysis uses a binned likelihood-ratio method where the estimation of $\vec{\theta}$ is obtained by comparing the observed and predicted Super-K E_{reco} vs θ_l spectra for 1-ring events.

Let N be the number of reconstructed energy and lepton angle bins and $\mathbf{a}$ be a $1 \times N_s$ - dimensional array of systematic (nuisance) parameters that affect the Super-K reconstructed energy spectrum prediction (e.g. parameters that control the beam, neutrino interaction simulations, detector response and oscillations).

The log-likelihood function used in the analysis is

$$-2\ln\lambda(\vec{\theta},\mathbf{a}) = 2\cdot\sum_{i=0}^{N-1}\left(n_i^{obs}\cdot\ln(n_i^{obs}/n_i^{exp}) + (n_i^{exp}-n_i^{obs})\right) \tag{3}$$

where n_i^{obs} is the observed number of events in the ith bin and $n_i^{exp} = n_i^{exp}(\vec{\theta};\mathbf{a})$ is the corresponding expected number of events.

The log-likelihood function obtained by adding a gaussian penatly term can be written as

$$-2\ln\lambda(\vec{\theta},\mathbf{a}) = 2\cdot\sum_{i=0}^{N-1}\left(n_i^{obs}\cdot\ln(n_i^{obs}/n_i^{exp}) + (n_i^{exp}-n_i^{obs})\right) + (\mathbf{a}-\mathbf{a_0})^T\cdot\mathbf{C}^{-1}\cdot(\mathbf{a}-\mathbf{a_0}) \tag{4}$$

where $\mathbf{a_0}$ is an $1 \times N_s$ - dimensional array with the default values of the systematic parameters, $\mathbf{a}^T$ is the transpose of $\mathbf{a}$, $\mathbf{C}$ is the systematic parameter covariance matrix of dimension $N_s \times N_s$, n_i^{obs} is the observed number of events in the ith bin, and $n_i^{exp}(\vec{\theta};\mathbf{a})$ is the corresponding expected number of events.

The total likelihood is given by the product of the likelihood terms of each selected sample,

$$\lambda(\vec{\theta};\mathbf{a}) = \prod_{s=1}^{N_{samples}}\lambda_s(\vec{\theta};\mathbf{a}) \tag{5}$$

where $N_{samples}$ is the number of selected samples used in the analysis, or by the sum of $-2\ln\lambda(\vec{\theta},\mathbf{a})$ from all the Super-K samples

$$-2\ln\lambda(\vec{\theta},\mathbf{a}) = \sum_{s=1}^{N_{samples}} -2\ln\lambda_s(\vec{\theta},\mathbf{a}) \tag{6}$$

The advantage of the used binned likelihood ratio method is that, in the large-sample limit, the quantity $-2\ln\lambda(\vec{\theta};\mathbf{a})$ in eq. (3) has a χ^2 distribution and it can therefore be used as a goodness-of-fit test.

Values of systematic parameters that give a negative predicted number of events in any reconstructed energy bin in any interaction mode are not allowed. If a systematic throw gives a negative number of events, that toy experiment is rejected.

4.2. Marginalization

In the previous analyses, except [10], a fully frequentist approach, in which all nuisance parameters were profiled, was used. Now a hybrid Bayesian-frequentist method is used. The nuisance parameters are marginalised over while the parameters of interest are minimized with respect to the likelihood distribution. The marginalization of the nuisance parameters give a so-called marginal likelihood that depends on only on a vector of parameters of interest, $\vec{\theta}$.

The marginal likelihood is derived using the following equation:

$$\lambda_{marg}(\vec{\theta}) = \int_A \lambda(\vec{\theta};\mathbf{a})\pi(\mathbf{a})d\mathbf{a} = \frac{1}{n}\sum_{i=1}^{n}\lambda(\vec{\theta};\mathbf{a}_i) \tag{7}$$

where $\mathbf{a}$ is the set of nuisance parameters in the parameter space A, $\pi(\mathbf{a})$ is the prior of the nuisance parameters. n is the number of toys MC produced for the marginalization and $\mathbf{a_i}$ is the single set of nuisance parameters extracted in the "i-th" toy MC. Many sets of $\mathbf{a}$ should be extracted in order to effectively sample the space of the nuisance parameters. The number of extractions depends on the number of nuisance parameters as well as the precision that is required by the measurement. In all the analyses shown in this note $n = 10k$ toys MC were used for the marginalisation of the systematic parameters.

We use the marginal log-likelihood, which will be denoted as χ^2_{marg} from now on, since it follows a χ^2 distribution when the Poisson statistic converges to the Gaussian:

$$\chi^2_{marg}(\vec{\theta}) = -2\ln\lambda_{marg}(\vec{\theta}) = -2\ln\left(\frac{1}{n}\sum_{i=1}^{n}\lambda(\vec{\theta};\mathbf{a}_i)\right) \tag{8}$$

It is important to note that this numerical integration is a weighted average of the likelihood evaluated at the n throws, not a weighted average of the log-likelihood. During the marginalization process, random values of the nuisance parameters are drawn according to the prior constraints which come from fits to near detector data and control samples at Super-K. Therefore, each throw is already weighted based on the prior information, and as a result the nuisance parameter values are not included in the calculation of the likelihood. For this reason eq. (3), which doesn't include the penalty term, is used.

Sometimes the free parameters, which are not marginalised, are constrained with a penalty term. In this case a modified version of eq. (8) is used:

$$\chi^2_{constr}(\vec{\theta}) = \chi^2_{marg}(\vec{\theta}) + (\vec{\theta} - \vec{\theta_0})^T V^{-1}(\vec{\theta} - \vec{\theta_0}) \tag{9}$$

where $\vec{\theta_0}$ is the vector of nominal values of the parameters of interest, $\vec{\theta}$ is the vector of its fitted values and V is the covariance matrix which relates all the free parameters. If the constrained parameter θ_{constr} is not correlated with the other parameters of interest, V is reduced to the uncertainty $\Delta\theta^{constr}$, and eq. (9) becomes

$$\chi^2_{constr}(\vec{\theta}) = \chi^2_{marg}(\vec{\theta}) + \frac{(\theta^{constr} - \theta_0^{constr})^2}{(\Delta\theta^{constr})^2} \tag{10}$$

This is the case of $\sin^2\theta_{13}$ when it is measured, i.e. it is parameter of interest, but at the same time it is constrained with the measurement of the reactor experiments. When a penalty term is applied to a parameter of interest it will be specified.

The toys MC used for the marginalization of the nuisance parameters are performed exactly in the same way as described in section 3.1, taking into account the correlations, but without statistical fluctuations. First $n = 10k$ distributions of E_{reco} vs θ_l for e-like and E_{reco} only for μ-like samples are produced using the extracted systematic parameters for a given point of the parameter of interest grid $\vec{\theta}$ for 1×10^{20} POT so that they may be scaled appropriately when comparing to datasets of differing POT. Finally they are stored in output files which can be used for whatever needed fit. A fit on a single dataset is performed as following:

- all the n toy MC distributions are read from the files,
- each distribution is normalised to the number of POT of the data set,
- for each toy distribution eq. (3) is calculated and $\lambda(\vec{\theta},\mathbf{a}) = e^{-(-2\ln\lambda(\vec{\theta},\mathbf{a}))/2}$ is obtained,
- the χ^2 value is finally obtained, using eq. (8), for a given set of parameter values of interest, $\vec{\theta}$. If $\vec{\theta}$ is constrained, eq. (10) is used.

This procedure is followed for each point in the $\vec{\theta}$ grid, where the binning is so fine that it does not impact on the precision of the measurement. Finally the best fit is found by looking for the $\vec{\theta}$ grid point with the minimum χ^2 value, χ^2_{bf}. This point corresponds to the best-fit parameters $\vec{\theta}_{bf}$. The minimization can be

also performed by using a discrete descent gradient algorithm [58] implemented in the VALOR framework that allows the minimum χ^2 to be found without the need of calculating the full χ^2 distribution across all grid points. This method is employed when a large number of fits is required, such as the calculation of critical values for Feldman-Cousins contours, as it drastically reduces the amount of CPU time required. A validation of the minimization algorithm will be shown in section 5 for the analyses where it is used.

4.3. Best-fit spectra with marginalization

When profiling, the best fit spectra are defined by the MC prediction that gives the largest likelihood values.

However, since the nuisance parameters are marginalised, there does not exist a single set of parameters which corresponds to the best fit. Therefore a new method has been developed to represent the best fit spectra which is consistent with the oscillation results.

The following procedure is used. From each toy MC i, used to marginalise the nuisance parameters, a different spectrum, with $N^j(\vec{\theta}_{bf})$ number of events in each bin j, is produced as well as a likelihood $\lambda(\vec{\theta}_{bf}, \mathbf{a_i})$ value, calculated using eq. (7), where $\vec{\theta}_{bf}$ is the best-fit value of the free parameters vector and $\mathbf{a_i}$ is the vector of nuisance parameters corresponding to the toy i. The best-fit number of events in the bin j of the spectrum is calculated as

$$N^j_{marg} = \frac{\sum_{i=1}^{n} \lambda(\vec{\theta}_{bf}; \mathbf{a_i}) \cdot N^j(\vec{\theta}_{bf}; \mathbf{a_i})}{\sum_{i=1}^{n} \lambda(\vec{\theta}_{bf}; \mathbf{a_i})} \tag{11}$$

where n is the number of toys MC.

One can easily find a relation between the profiled and marginalised best-fit spectra. In the limit of infinite number of toys MC, i.e. the full nuisance parameters space is scanned, the likelihood can be written as

$$\lambda_{marg}(\vec{\theta}_{bf}) = \frac{1}{n} \sum_{i=1,\ i\neq bf}^{n} \Big[\lambda(\vec{\theta}_{bf}; \mathbf{a_i}) + \lambda_{\mathbf{prof}}\Big] \tag{12}$$

where $\lambda_{prof} = \lambda(\vec{\theta}_{bf}; \mathbf{a_{bf}})$ is the profiled best-fit likelihood. If we denote as $N^j_{prof} = N^j_i(\vec{\theta}_{bf}; \mathbf{a_{bf}})$ the number of events in the bin j obtained with profiling, eq. (11) can be written as

$$N^j_{marg} = \frac{\sum_{i=1,\ i\neq bf}^{n} \lambda(\vec{\theta}_{bf}; \mathbf{a_i}) \cdot \mathbf{N^j}(\tilde{\theta}_{\mathbf{bf}}; \mathbf{a_i}) + \lambda_{\mathbf{prof}} \cdot \mathbf{N^j_{prof}}}{\sum_{i=1}^{n} \lambda(\vec{\theta}_{bf}; \mathbf{a_i})} \tag{13}$$

4.4. Confidence intervals

After the fit is completed, the full χ^2 distribution is then shifted with respect to $\chi^2(\vec{\theta}_{bf})$:

$$\Delta\chi^2(\vec{\theta}) = \chi^2(\vec{\theta}) - \chi^2(\vec{\theta}_{bf}) \tag{14}$$

Confidence intervals are set on $\Delta\chi^2(\vec{\theta})$. If

$$\Delta\chi^2(\vec{\theta}) > \Delta\chi^2_{crit} \tag{15}$$

where $\Delta\chi^2_{crit}$ is the critical value for a certain confidence level X%(CL), that value of $\vec{\theta}$ is excluded at the X% CL. In the gaussian approximation constant values of $\Delta\chi^2_{crit}$ can be used. It will be called "constant $\Delta\chi^2$ method".

However in some cases the gaussian approximation is not reliable and the "Feldman-Cousins method" [59] must be used. In this method a different critical value for each point of the $\vec{\theta}$ grid $\Delta\chi^2(\vec{\theta})_{crit}$ is calculated. The procedure consists of the following steps:

1. many toys MC (e.g. 10k) are produced assuming the oscillation hypothesis of the grid point $\vec{\theta}$ and taking into account both the statistical and systematic uncertainties;
2. for each toy first $\chi^2(\vec{\theta}_{fix})$ is obtained by minimizing $\chi^2(\vec{\theta})$ where the oscillation parameters $\vec{\theta}$ are kept fixed to the values corresponding to the grid point and the nuisance parameters are marginalised. Then $\chi^2(\vec{\theta})$ is minimized, leaving the oscillation parameters $\vec{\theta}$ as free and $\chi^2(\vec{\theta}_{bf})$ is obtained.

3. the test statistic is calculated as

$$\Delta\chi^2 = \chi^2(\vec{\theta}_{fix}) - \chi^2(\vec{\theta}_{bf}) \tag{16}$$

4. The steps 2 and 3 are repeated for each toy MC and $f(\Delta\chi^2)$, the distribution of $\Delta\chi^2(\vec{\theta})$, is obtained.

5. Several critical values for $X\%$ CL (e.g. 68%, 90%...), $\Delta\chi^2(\vec{\theta})_{crit}$, can be defined as:

$$\Delta\chi^2_{crit} : \int_{-\infty}^{\Delta\chi^2_{crit}} f(\Delta\chi^2)d(\Delta\chi^2) = X\% \tag{17}$$

where $f(\Delta\chi^2)$ is the p.d.f. of $\Delta\chi^2$ and $d(\Delta\chi^2)$ is the differential of the integral.

6. This procedure is repeated for all the points of the $\vec{\theta}$ grid and several critical values are computed

Once a critical value, $\Delta\chi^2(\vec{\theta})_{crit}$, is computed for each oscillation hypothesis in the $\vec{\theta}$ grid, the confidence intervals are set by the condition of eq. (15).

Several analyses are performed and confidence intervals are computed for:

- 1D δ_{CP} with reactor constraint: the parameter of interest is δ_{CP} and all the other oscillation parameters are marginalised; for $\sin^2\theta_{13}$, that is marginalised, the gaussian prior of reactor measurements shown in table 11, is used.
- 1D δ_{CP} T2K only: the parameter of interest is δ_{CP} and all the other oscillation parameters are marginalised; for $\sin^2\theta_{13}$, that is marginalised, the uniform prior, shown in table 11, is used.
- 1D $\sin^2\theta_{13}$ T2K only: the parameter of interest is $\sin^2\theta_{13}$ and all the other oscillation parameters are marginalised;
- 1D Δm^2_{32} (Δm^2_{13}) with reactor constraint: the parameter of interest is Δm^2_{32} (Δm^2_{13}) and all the other oscillation parameters are marginalised; for $\sin^2\theta_{13}$, that is marginalised, the gaussian prior of reactor measurements shown in table 11, is used.
- 1D $\sin^2\theta_{23}$ with reactor constraint: the parameter of interest is $\sin^2\theta_{23}$ and all the other oscillation parameters are marginalised; for $\sin^2\theta_{13}$, that is marginalised, the gaussian prior of reactor measurements shown in table 11, is used.
- 2D $\sin^2\theta_{13}$ vs δ_{CP} with reactor constraint: the parameters of interest are $\sin^2\theta_{13}$ and δ_{CP} while all the other oscillation parameters are marginalised; for $\sin^2\theta_{13}$, that is a free parameter, the gaussian prior of reactor measurements shown in table 11, is used, through a penalty term as in eq. (10).
- 2D $\sin^2\theta_{13}$ vs δ_{CP} T2K only: the parameters of interest are $\sin^2\theta_{13}$ and δ_{CP} while all the other oscillation parameters are marginalised;
- 2D $\sin^2\theta_{23}$ vs δ_{CP} T2K only: the parameters of interest are $\sin^2\theta_{23}$ and δ_{CP} while all the other oscillation parameters are marginalised;
- 2D Δm^2_{32} (Δm^2_{13}) vs δ_{CP} T2K only: the parameters of interest are Δm^2_{32} and δ_{CP} while all the other oscillation parameters are marginalised;
- 2D $\sin^2\theta_{23}$ vs Δm^2_{32} (Δm^2_{13}) with reactor constraint: the parameters of interest are $\sin^2\theta_{23}$ and Δm^2_{32} (Δm^2_{13}) while all the other oscillation parameters are marginalised; for $\sin^2\theta_{13}$, that is marginalised, the gaussian prior of reactor measurements shown in table 11, is used.

Each confidence interval is calculated for both normal and inverted hierarchy.

The 1D χ^2 confidence intervals with "constant $\Delta\chi^2$ method" are used to set the $\pm 1\sigma$ ranges.

Each point of the $\vec{\theta}$ grid corresponds to the center of the histogram bins, i.e. the χ^2 values are calculated at each histogram bin center. The ranges and binnings used for the analyses listed above are shown in table 22. All the 1-dimensional analyses, where the precision on the best-fit point is important and must be quoted, are performed with a fine binning. Since 2-dimensional analyses are performed only to set contours with "constant $\Delta\chi^2$ method" and a fit with a very high precision is not needed to find the best-fit point, coarser binning are used, without affecting the contour. Furthermore a very fine 2-dimensional binning would require much more CPU time and storage space. In the future $\bar{\nu}/\nu$ joint analysis releases when the "Feldman-Cousins method" will be needed also for 2-dimensional fits, a finer binning will be used.

Parameter	Min	Max	Step	Bins
	1-dimensional fit			
δ_{CP}	$-\pi$	$+\pi$	$\pi/200$	201
$\sin^2\theta_{13}$ (reactors)	0.015	0.036	0.021 / 81	201
$\sin^2\theta_{13}$ (T2K only)	0	0.113	0.113 / 200	201
$\sin^2\theta_{23}$	0.3	0.7	0.4 / 200	201
$\lvert\Delta m^2_{32}\rvert$ ($\mathrm{eV}^2/\mathrm{c}^4$)	2×10^{-3}	3×10^{-3}	$10^{-3}/200$	201
	2-dimensional fit			
δ_{CP}	$-\pi$	$+\pi$	$\pi/50$	51
$\sin^2\theta_{13}$ (reactors)	0.015	0.036	0.021 / 81	81
$\sin^2\theta_{13}$ (T2K only)	0	0.113	0.113 / 81	81
$\sin^2\theta_{23}$	0.3	0.7	0.4 / 81	81
$\lvert\Delta m^2_{32}\rvert$ ($\mathrm{eV}^2/\mathrm{c}^4$)	2×10^{-3}	3×10^{-3}	$10^{-3}/21$	21

Table 22: Grid limits and spacing used for all the 1-dimensional and 2-dimensional marginalization fits and confidence regions. The minimum and maximum values correspond to the center of the χ^2 histogram first and last bin.

4.5. Contours with profiled mass hierarchy

As already pointed out, the constant $\Delta\chi^2$ method gives the correct coverage when the gaussian approximation is valid and is very useful because it does not require a large amount of CPU time. However, we know that in some cases the constant $\Delta\chi^2$ method fails to give the correct coverage and can provide only approximate confidence intervals. An example of this is the case when the constraint on a parameter is not precise enough to keep it away from a physical boundary. For cases such as these a toy MC study is required in order to estimate the correct critical values (see section 4.4).

Also, when considering discrete parameters, such as the mass hierarchy, that is distributed as binomial, the $\Delta\chi^2$ is not expected to follow a χ^2 distribution but more likely to follow a gaussian distribution as found in [64]. This is an important distinction as the exclusion significance one can interpret from a constant $\Delta\chi^2$ contour is heavily dependant on the distribution the $\Delta\chi^2$ is expected to take.

This means that when the constant $\Delta\chi^2$ method is used in order to set limits as a function of some oscillation parameters as well as the mass hierarchy, the constant critical values we usually use do not give the right coverage for the disfavored mass hierarchy hypothesis contour.

For this reason the data fit 2-dimensional confidence intervals, which are calculated for both the hierarchy hypotheses, will be computed in two different ways:

- normal and inverted hierarchy contours are obtained independently with respect to their own minimum ("independent")
- the global minimum between the normal and inverted hierarchy hypotheses is used to set both the contours ("global");

These different ways to show the confidence intervals correspond to the two extremes, where the "independent" and "global" contours provide respectively the most and the least conservative results.

Since for the 1-dimensional fits the $\Delta\chi^2$ distribution are shown, there is no need to provide results using both "independent" and "global" approaches. However the reader should remember that the constant critical

values shown in the figures hold only for the $\Delta\chi^2$ distribution corresponding to the best-fit mass hierarchy hypothesis. Indeed the constant $\Delta\chi^2$ method does not give the proper coverage for discrete parameters and consequently the contour of the disfavored hierarchy has bad coverage.

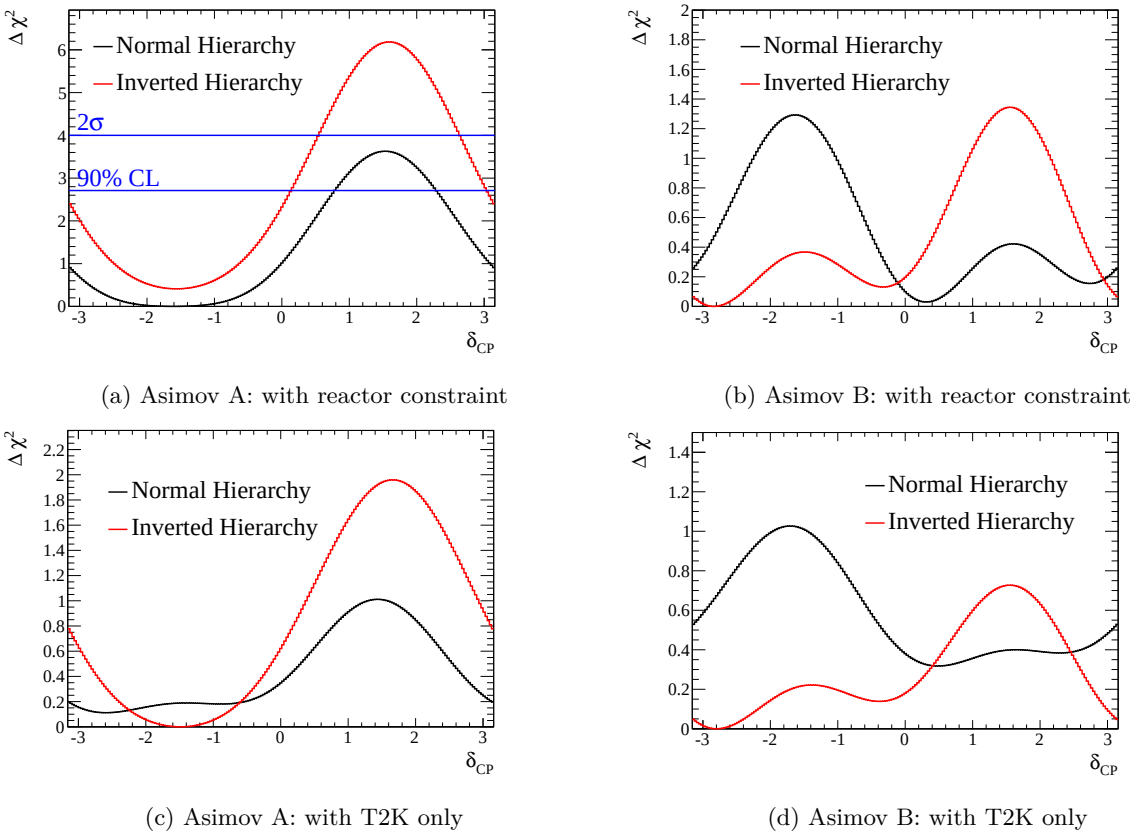
5. Validation of the $\bar{\nu}/\nu$ joint analysis

The validation of the fit is performed first by comparing the confidence intervals, obtained with Asimov data sets corresponding to the oscillation parameters in table 12 with the other group analyses [46]. Then the coverage study is performed for 1-dimensional δ_{CP} with reactor constraint fits, in order to verify if the "constant $\Delta\chi^2$" method is reliable or the "Feldman-Cousins" method is needed.

5.1. Sensitivity studies

The critical values of constant $\Delta\chi^2$ method in 1-dimensional confidence intervals are given by $Z\sigma = \sqrt{\Delta\chi^2}$ where Z is the number of standard deviations that defines the confidence level. For 90% CL the critical value is 2.71. The critical values for 2-dimensional fits are 2.3 and 4.61 respectively for 68% and 90% CL. A detailed table with the critical values for 1- and 2-dimensional confidence intervals can be found in [72].

In fig. 16 the sensitivity to δ_{CP} with reactor constraint and T2K only is shown for the Asimov data sets A and B.



(a) Asimov A: with reactor constraint

(b) Asimov B: with reactor constraint

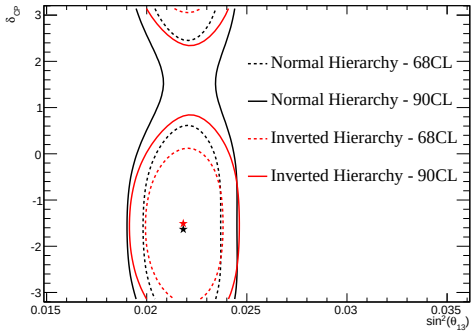
(c) Asimov A: with T2K only

(d) Asimov B: with T2K only

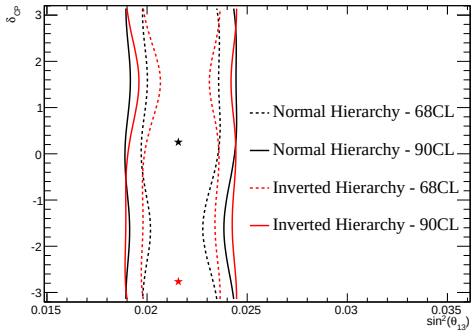
Figure 16: The expected $\Delta\chi^2$ distribution as a function of δ_{CP} for the Asimov data sets A and B of table 12 corresponding to the Run 1-6 statistics is shown. The mass hierarchy is minimized: both the normal and inverted hierarchy χ^2 distributions are shifted to the same global best-fit χ^2 value, which is taken as the minimum between normal and inverted hierarchy.

The sensitivity to $\sin^2\theta_{13}$ Vs δ_{CP} with reactor constraint and T2K only data, is shown in fig. 17, while $\sin^2\theta_{23}$ Vs Δm^2_{32} (Δm^2_{13}) with reactor constraint is shown in fig. 18. In both cases the normal and inverted hierarchy contours are set independently, i.e. each calculated with respect to its own minimum χ^2 value and not the global minimum as a function of the mass hierarchy.

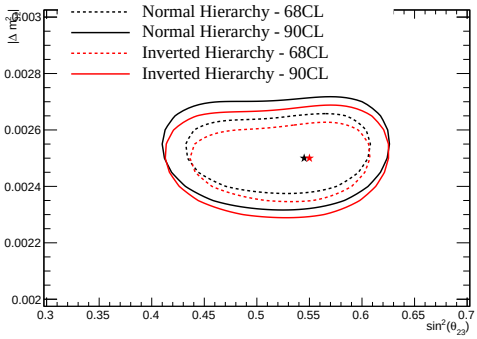
fig. 17 shows the sensitivity to $\sin^2\theta_{13}$ Vs δ_{CP} and $\sin^2\theta_{23}$ Vs Δm^2_{32} (Δm^2_{13}) with the reactor constraint. fig. 18 shows the sensitivity to $\sin^2\theta_{13}$ Vs δ_{CP} with T2K data only. In both fig. 17 and 18, normal and inverted hierarchy contours are set independently, i.e. each calculated with respect to its own minimum χ^2 value and not the global minimum as a function of the mass hierarchy.



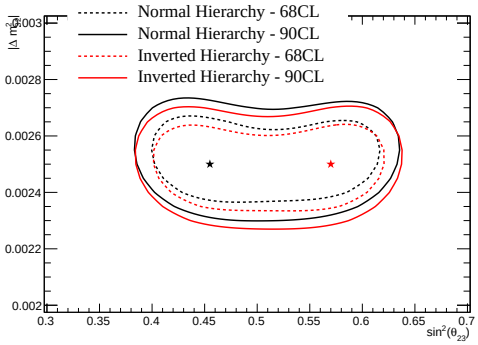
(a) Asimov A: $\sin^2\theta_{13}$ Vs δ_{CP}. Reactor constraint.



(b) Asimov B: $\sin^2\theta_{13}$ Vs δ_{CP}. Reactor constraint.

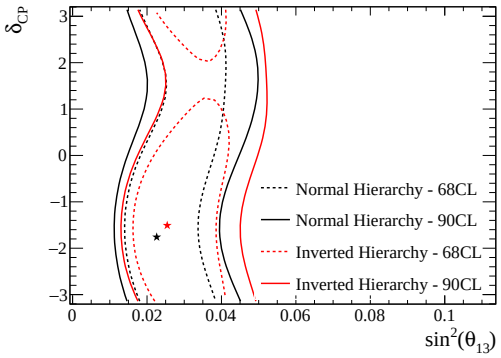


(c) Asimov A: $\sin^2\theta_{23}$ Vs Δm^2_{32} (m^2_{13}). Reactor constraint.

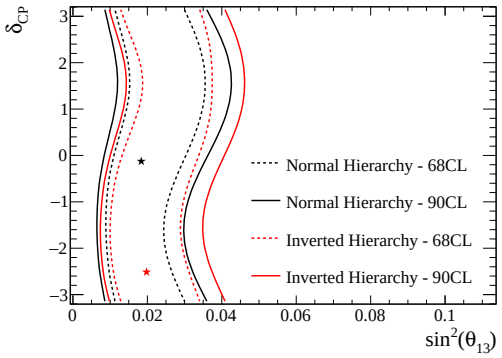


(d) Asimov B: $\sin^2\theta_{23}$ Vs Δm^2_{32} (m^2_{13}). Reactor constraint.

Figure 17: The contours at 68% and 90% CL with reactor constraint for the Asimov data sets A and B of table 12 corresponding to the Run 1-6 statistics is shown. Normal and inverted hierarchy contours are independent.



(a) Asimov A with T2K data only.



(b) Asimov B with T2K data only.

Figure 18: The contours at 68% and 90% CL for $\sin^2\theta_{13}$ Vs δ_{CP} for the Asimov data sets A and B of table 12 corresponding to the Run 1-6 statistics is shown. Normal and inverted hierarchy contours are independent. Only T2K data are used.

Gaussian approximation	
CL	**Critical value**
1σ	1
90%	2.71
2σ	4

Feldman-Cousins: $\delta_{CP}^{true} = -\pi/2$, MH^{true} = normal		
CL	**Critical value**	**Coverage**
1σ	0.582	0.779
90%	2.094	0.935
2σ	3.242	0.972

Feldman-Cousins: $\delta_{CP}^{true} = 0$, MH^{true} = normal		
CL	**Critical value**	**Coverage**
1σ	1.334	0.570
90%	2.460	0.923
2σ	3.198	0.981

Feldman-Cousins: $\delta_{CP}^{true} = +\pi/2$, MH^{true} = normal		
CL	**Critical value**	**Coverage**
1σ	1.178	0.605
90%	2.224	0.938
2σ	3.080	0.977

Table 23: The critical values of $\Delta\chi^2$ used to set the confidence intervals are shown for both the gaussian approximation (i.e. constant $\Delta\chi^2$) as well as the ones obtained with the Feldman-Cousins method. 10k toy experiments were used for each true point of δ_{CP}. Furthermore the coverage, defined by eq. (19), provided by the gaussian approximation for different true values δ_{CP} is shown. The true mass hierarchy is set to normal.

5.2. Coverage study of the simultaneous fit of δ_{CP} and mass hierarchy

The fit validation is performed also by making coverage studies for the fit of the mass hierarchy (MH) and δ_{CP} with the reactor constraint, the main result of this release of the $\bar{\nu}/\nu$ joint fit, in order to understand if the "Feldman-Cousins method" is needed. The coverage is defined as the number of times the true values of δ_{CP}^{true} and MH^{true}, used to generate the toys MC, is contained in the confidence interval defined by eq. (15):

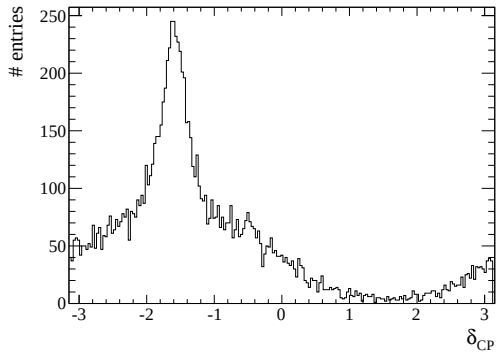
$$\Delta\chi^2(\delta_{CP}, \mathrm{MH}) = \chi^2(\delta_{CP}^{true}, \mathrm{MH}^{true}) - \chi^2(\delta_{CP}^{bf}, \mathrm{MC}^{bf}) \tag{18}$$

where δ_{CP}^{bf} and MH^{bf} are the best-fit values. Adopting the "constant $\Delta\chi^2$" method, corresponding to the gaussian approximation, the critical value $\Delta\chi^2_{crit} = \Delta\chi^2_{const}$, pre-computed analytically, at a certain $X\%$ confidence level is used for any value of δ_{CP} and MH. Many toys MC are performed (10k) and for each one eq. (18) is tested. The condition used to estimate the coverage is

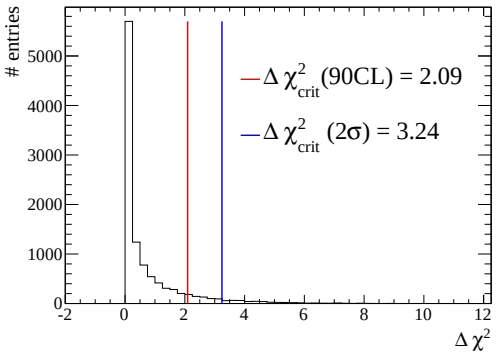
$$\Delta\chi^2(\delta_{CP}, \mathrm{MH}) < \Delta\chi^2_{crit} \tag{19}$$

The fit has a good coverage if the number of toys MC for which eq. (19) is satisfied (N^X_{toys}) is equal or close to $X\%$ within the statistical uncertainty of the number of toys MC. If $N^X_{toys} > X\%$, then the fit has over-coverage, that means the confidence intervals should be tighter than what obtained with the "constant $\Delta\chi^2$ method", while if $N^X_{toys} < X\%$, then the fit has under-coverage and excludes a wider range of values than in the correct case. In fig. 19 the $\Delta\chi^2$ distribution of δ_{CP} 1-dimensional fit with reactor constraint, obtained by using eq. (18), is shown for the true values $\delta_{CP} = -\pi/2, 0, +\pi/2$. The critical values computed using the Feldman-Cousins method are shown in table 23. It can be seen that the coverage given by the constant critical values is not correct, so, in order to provide correct confidence intervals, the Feldman-Cousins method is required.

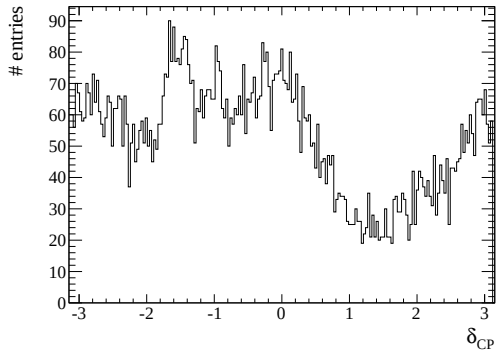
The constant $\Delta\chi^2$ method does not give the proper coverage when the parameter of interest is not gaussian distributed and when the relation with the likelihood is not linear. Another reason why the correct coverage could not be assured by the constant $\Delta\chi^2$ method is the presence of physical boundaries. δ_{CP} is continuous because it is periodic, however it shows up physical boundaries at $\pm\pi/2$, since they correspond



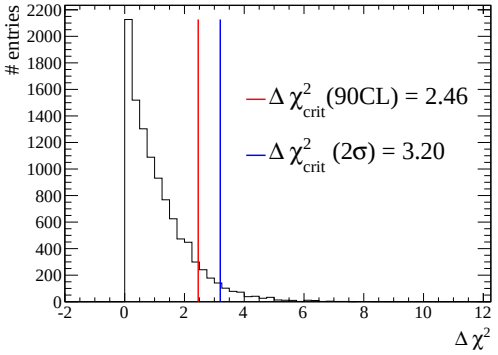
(a) Best-fit distribution of δ_{CP}. $\delta_{CP}^{true} = -\pi/2$.



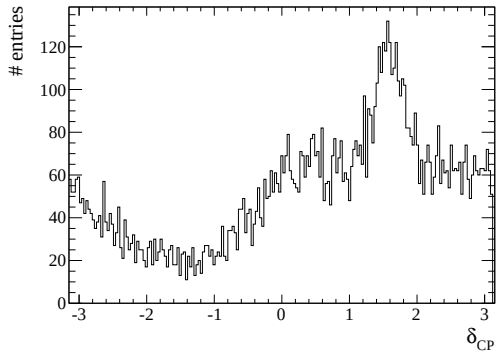
(b) $\Delta\chi^2$ distribution. $\delta_{CP}^{true} = -\pi/2$.



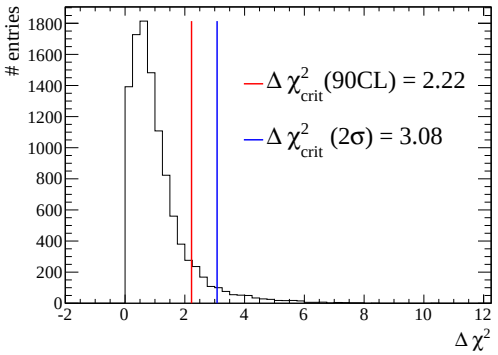
(c) Best-fit distribution of δ_{CP}. $\delta_{CP}^{true} = 0$.



(d) $\Delta\chi^2$ distribution. $\delta_{CP}^{true} = 0$.



(e) Best-fit distribution of δ_{CP}. $\delta_{CP}^{true} = +\pi/2$.



(f) $\Delta\chi^2$ distribution. $\delta_{CP}^{true} = +\pi/2$.

Figure 19: Distributions of best-fit δ_{CP} (left) and $\Delta\chi^2$ (right). $10k$ toy experiments are performed with true values of $\delta_{CP} = -\pi/2, 0, +\pi/2$, assuming normal hierarchy and applying the reactor constraint. Both the mass hierarchy and δ_{CP} are profiled in the fit. All the other oscillation parameters are marginalised. The statistics corresponds to the Run 1-6 data set.

to the values that maximize or minimize the number of events in the e-like samples. In general the gaussian approximation gives over-coverage for all the listed confidence intervals, except for 1σ CL of $\delta_{CP}^{true} = +\pi/2$ and 0 with $MH^{true} =$ normal.

This can be understood by analyzing the best-fit result of 10k toy experiments with $\delta_{CP}^{true} = +\pi/2$ and normal hierarchy. In this case the constant $\Delta\chi^2$ method gives under-coverage for 1σ CL (the constant critical value is larger than the one computed with Feldman-Cousins) but over-coverage for 90% and 2σ CL (the constant critical value is smaller than the one computed with Feldman-Cousins). In general a critical value estimated with Feldman-Cousins is larger (smaller) when the $\Delta\chi^2$ distribution is wider (tighter), that is favored when the best-fit parameter values are different from (same as) the true values (see eq. (18)). Wider

$\Delta\chi^2$ distributions are due to degeneracy with other parameters, in this case the mass hierarchy. Indeed in fig. 34 it is shown that the inverted hierarchy increases the expected number of events in antineutrino mode e-like sample and reduces it in neutrino mode e-like sample, similarly to the effect of $\delta_{CP} = +\pi/2$ (see table 24). On average the best-fit mass hierarchy is chosen to be inverted $\sim 70\%$ of the times and the best-fit distribution of δ_{CP} is peaked at $+\pi/2$. This is due to asymmetric fluctuations: upward statistical fluctuations in the antineutrino mode e-like sample and downward statistical fluctuations in the neutrino mode e-like sample. One would expect the mass hierarchy best-fit to be inverted $\sim 50\%$ of the times. However it should be considered that $\delta_{CP} = +\pi/2$, the true point, behaves as a physical boundary. So about half of the extreme asymmetric fluctuations would need more events in antineutrino mode and less in antineutrino mode than what $\delta_{CP} = +\pi/2$ can provide. For this reason the only possible way to fit these fluctuations is to choose inverted hierarchy, which has an effect similar to $\delta_{CP} = +\pi/2$. Another way to see it is by looking the $\Delta\chi^2$ distribution corresponding to fits where the best-fit mass hierarchy is normal, shown in fig. 20. This fits correspond to opposite asymmetric fluctuations: downward statistical fluctuation in the antineutrino mode e-like sample and upward statistical fluctuation in the neutrino mode e-like sample. If the best-fit mass hierarchy is normal, the $\Delta\chi^2$ distribution of δ_{CP} is quite flat, which means that to fit this asymmetric fluctuation the δ_{CP} parameter is enough, because the other physical boundary is at $-\pi/2$ which is far from the true point.

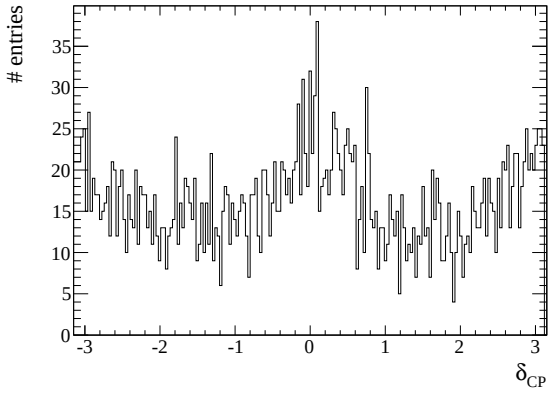
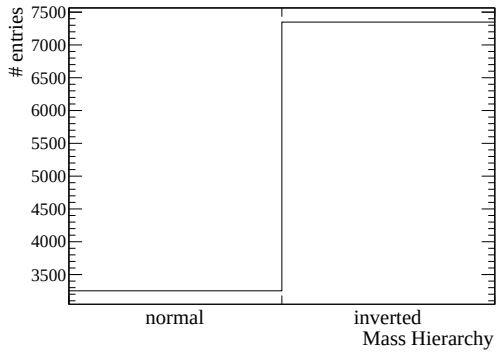


Figure 20: Best-fit δ_{CP} distribution for fits where the best-fit mass hierarchy is inverted The true values correspond to $\delta_{CP} = +\pi/2$ and normal hierarchy.

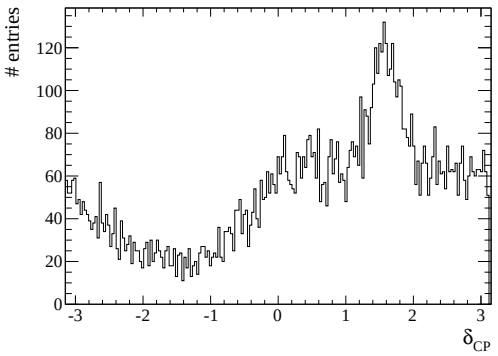
The under-coverage at 68% CL, i.e. estimated critical value is larger than 1, is due to the degeneracy between δ_{CP} and mass hierarchy as well as the fact that the true point is at the physical boundary. The true point corresponds to values of mass hierarchy (normal) and δ_{CP} $(+\pi/2)$ that have opposite effects and balance between each other. So if $\delta_{CP}^{bf} = +\pi/2$, in order to compensate the physical boundary in δ_{CP}, the best-fit mass hierarchy is inverted, which is different from the true one. Consequently eq. (18) will not have $\Delta\chi^2 \sim 0$ values. On the other hand the opposite happens when the true point is $\delta_{CP}^{true} = -\pi/2$ and normal hierarchy. Indeed in this case there is over-coverage, i.e. the critical value is smaller. This happens because δ_{CP}^{true} is at a physical boundary but the true mass hierarchy has the same effect on the spectra. So extreme asymmetric statistical fluctuations cannot be accomodated by neither δ_{CP} nor mass hierarchy, which best-fit values will correspond most of the time to the true point, as shown in fig. 21. For this reason eq. (18) will give more values of $\Delta\chi^2$ close to 0, producing a tighter distribution and consequently smaller critical values, as shown in fig. 19.

In order to test confidence intervals for higher significance the tail of the $\Delta\chi^2$ distribution is more important. In fig. 21 the best-fit mass hierarchy and δ_{CP} distributions are shown only for those toy experiments which best-fit $\Delta\chi^2$ is larger than 2. This toy experiments correspond to the following asymmetric fluctuations of the data (upward for neutrino mode e-like sample and downward for antineutrino e-like sample) that prefer values of $\delta_{CP} \sim -\pi/2$ and normal hierarchy. Since normal hierarchy corresponds to the true point and there is no other possible value of mass hierarchy that can better fit the statistical fluctuation, eq. (18) becomes relatively small enough to produce a tighter distribution in the range of large $\Delta\chi^2$.

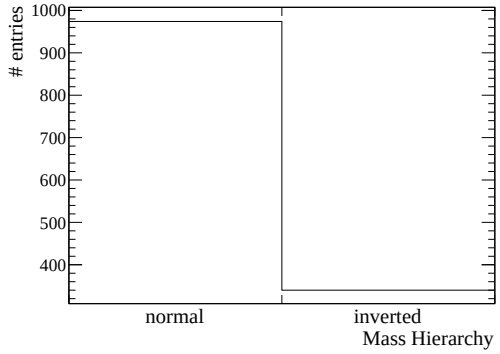
Another proof of this behavior can be given by looking to fig. 22, that shows the best-fit mass hierarchy obtained with 10k toy experiments with $\delta_{CP}^{true} = -\pi/2$ and normal hierarchy. In this case also for the 1σ CL the constant $\Delta\chi^2$ method gives over-coverage, because the $\Delta\chi^2$ distribution is tighter. The reason is that $\delta_{CP} = -\pi/2$ is degenerate with normal hierarchy (they have similar effects on the e-like spectra), which is the true mass hierarchy, and there is no other possible value of mass hierarchy able to fit extreme fluctuations. For this reason eq. (18) becomes relatively tight in the tail.



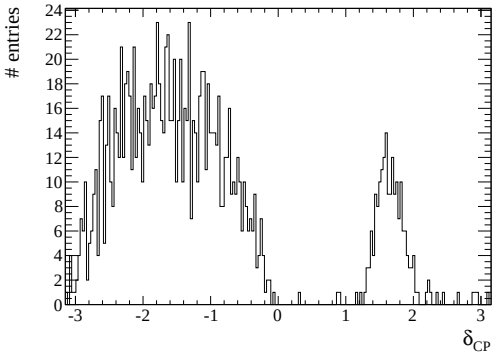
(a) Best-fit mass hierarchy distribution.



(b) Best-fit distribution of δ_{CP}.



(c) Best-fit mass hierarchy distribution. Only fits with best-fit $\Delta\chi^2 > 2$ are selected.

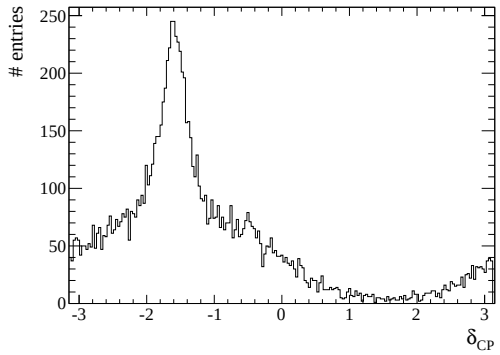


(d) Best-fit δ_{CP} distribution. Only fits with best-fit $\Delta\chi^2 > 2$ are selected.

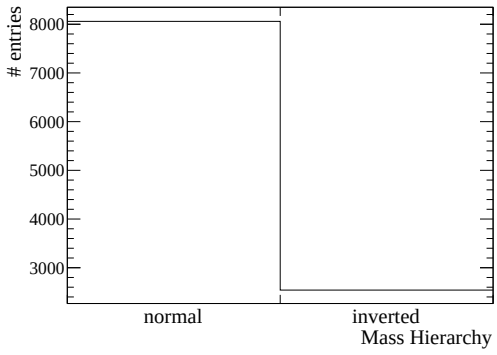
Figure 21: Best-fit distribution of mass hierarchy and δ_{CP} from the simultaneous fit of δ_{CP} and mass hierarchy. $10k$ toy experiments are produced with $\delta_{CP}^{true} = +\pi/2$ and normal hierarchy.

From the best-fit distributions of δ_{CP} we can see that the current statistic is still not enough for a precise measurement of δ_{CP}. However if $\delta_{CP}^{true} = \pm\pi/2$ the best-fit distribution shows a peak around the true value. For $\delta_{CP}^{true} = 0$ the fit prefers values smaller than 0, though the true mass hierarchy is normal, it is chosen as best-fit only about 54% of the times. This is, again, due to the degeneracy between the mass hierarchy and δ_{CP}. The normal hierarchy (true value) is not enough to fit extreme upward fluctuations in neutrino mode and downward fluctuations in antineutrino mode e-like sample, since the mass hierarchy is in a discrete boundary. So values of $\delta_{CP} \sim -\pi/2$ are favored in order to balance it. Furthermore in these fits the normal hierarchy is preferred with respect to inverted hierarchy about the 65% of the times. Indeed if we only look to the fits where the normal hierarchy is preferred, i.e. the true value, the best-fit distribution of δ_{CP} is symmetrically distributed around 0. as shown in fig. 23.

In general most of the strange behaviors present in the best-fit and $\Delta\chi^2$ distributions can be explained with a still low sensitivity to both δ_{CP} and mass hierarchy and the strong degeneracy between the two oscillation parameters.



(a) Best-fit distribution of δ_{CP}.



(b) Best-fit distribution of mass hierarchy.

Figure 22: Best-fit distribution of mass hierarchy from the simultaneous fit of δ_{CP} and mass hierarchy. $10k$ toy experiments are produced with $\delta_{CP}^{true} = -\pi/2$ and normal hierarchy.

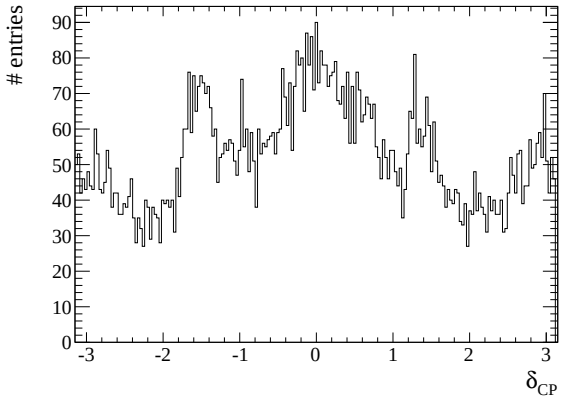


Figure 23: Best-fit distribution of δ_{CP} obtained by fixing the mass hierarchy to normal and not profiled. $10k$ toy experiments are produced with $\delta_{CP}^{true} = 0$ and normal hierarchy.

e-like neutrino mode			
	$\delta_{CP} = -\pi/2$	$\delta_{CP} = 0$	$\delta_{CP} = +\pi/2$
Normal Hierarchy	26.596	22.407	18.179
Inverted Hierarchy	23.554	19.704	15.891
e-like antineutrino mode			
	$\delta_{CP} = -\pi/2$	$\delta_{CP} = 0$	$\delta_{CP} = +\pi/2$
Normal Hierarchy	3.219	3.704	4.122
Inverted Hierarchy	3.475	3.987	4.499

Table 24: Expected number of events for neutrino and antineutrino mode e-like sample for different values of δ_{CP} and mass hierarchy.

5.3. Validation of the minimization algorithm

As explained in section 4.4, the best-fit parameters are found by setting up a grid of points of the parameters of interest and calculating the $\Delta\chi^2$ for each of them. The grid is fine enough in order to have a very small variation of the $\Delta\chi^2$ distribution as a function of the parameter of interest. The best-fit point can be found by computing the entire χ^2 distribution and looking for the lowest χ^2 value. However since the Feldman-Cousins implies a large number of toy experiments and fits for several points of δ_{CP}, this can be very CPU time consuming. For this reason a minimization algorithm has been developed. Similar to MINUIT [57], this algorithm uses the descent gradient: at each point of the parameter of interests it calculates the χ^2

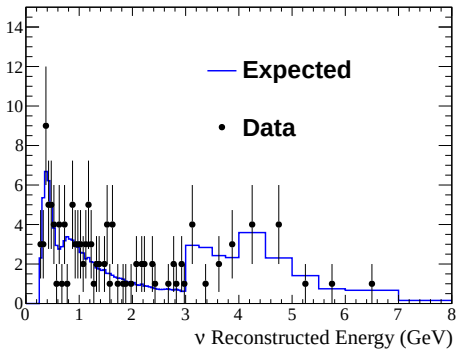
in the adjacent points along all the possible directions (depending on the number of dimensions) and move toward the lower one, until it stops in a minimum. It is worth to note the MINUIT cannot be used because the grid of points of the free parameters is discrete, while it needs to work in a continuous space.

However the descent gradient algorithm is strongly affected by the presence of local minima and depends on the point of the free parameters from where the fit starts. In order to avoid this problem the fit is performed 4 times starting from 4 different starting points: $-\pi$, $-\pi/2$, $+\pi/2$ and $+\pi$. The method has been tested by performing $10k$ toys for normal hierarchy and $4k$ toys for inverted hierarchy for the true points $\delta_{CP} = -\pi/2, 0, +\pi/2, +\pi$. For all the tests the number of times where the fit stopped in a local minimum is $\leq 1/10000$. Similar numbers have been found for inverted hierarchy. For all the critical values the minimization algorithm is used helping to reduce the CPU time of about 30% - 50%, depending on the toy experiment.

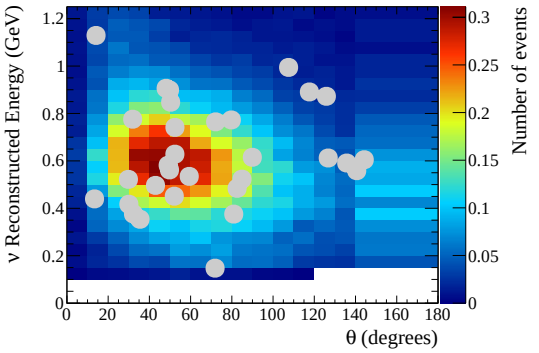
6. Results of the $\bar{\nu}/\nu$ joint analysis with the Run 1-6 dataset

The results of the Run 1-6 data set fit, for an exposure of 6.914×10^{20}POT in neutrino mode and of 4.011×10^{20}POT in antineutrino mode, are shown in this chapter. In fig. 24 the predicted spectra of μ-like and e-like samples are compared to the observed data. In table 25 the expected and observed number of events in Run 1-6 data set are shown. The predicted number of events are shown for the oscillation parameters in table 12 of Asimov data sets A and B. All the predicted and observed events are in good agreement. A small excess of events with respect to the expectation is observed in the neutrino mode e-like samples, though it is still within 1σ of the statistical uncertainty. This trend is mostly inherited from the Run 1-4 data set, since only 2 neutrino mode e-like events are observed in Run 5+6. Furthermore the latest global fit of the reactor experiments gives a lower value of $\sin^2\theta_{13}$ with a much smaller uncertainty (see section 3.1 and table 12). It is interesting to see that the number of events of antineutrino mode e-like events is in perfect agreement with the assumption of $\delta_{CP} = -1.601$, which is the best-fit value of the Run 1-4 data set. This means that, for what concerns the event rate, we expect to see agreement between neutrino and antineutrino mode data in the fit of δ_{CP}, though the statistics are not expected to be enough to reject $\delta_{CP} = 0$ at any significant confidence level.

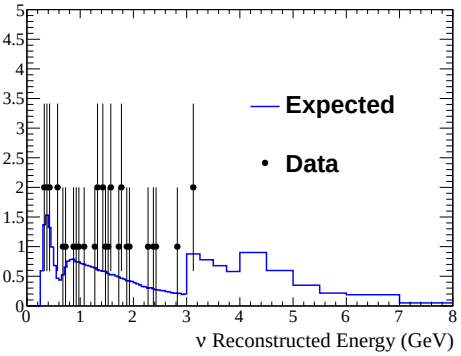
The comparison of the confidence intervals obtained by the fit of the Run 1-6 data set among all the oscillation analysis groups is shown in [46].



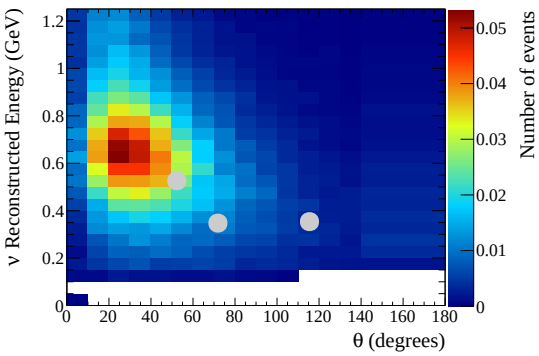
(a) Muon-like, neutrino mode



(b) Electron-like, neutrino mode



(c) Muon-like, antineutrino mode



(d) Electron-like, antineutrino mode

Figure 24: The predicted spectra compared to the Run 1-6 dataset (dots), which corresponds to an exposure of 6.914×10^{20}POT in neutrino mode and of 4.011×10^{20}POT in antineutrino mode, are shown. The expected distribution corresponds to the oscillation parameters of the Asimov data set A shown in table 12. The spectra were generated with the systematic parameters measured with the ND280 data shown in table 6, 7, 8 and the nominal Super-K systematic parameters shown in table 9.

Beam mode	Sample	Exp. Asimov A	Exp. Asimov B	Exp. Not Osc	Observed	Best fit
neutrino	μ-like	125.914	130.464	484.737	124	125.688
neutrino	e-like	26.594	19.954	5.698	31	26.969
antineutrino	μ-like	34.284	35.032	99.02	34	34.114
antineutrino	e-like	3.218	3.355	1.253	3	3.344

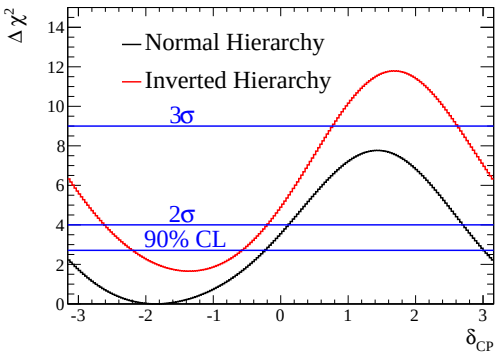
Table 25: The observed, expected and best-fit number of events in Run 1-6 data set for an exposure of 6.914×10^{20}POT in neutrino mode and of 4.011×10^{20}POT in antineutrino mode are shown for each selected sample. The prediction is produced using the BANFF tuning and the oscillation parameters shown in table 12. The total number of events of the best-fit spectra is obtained by using eq. (11).

6.1. 1-dimensional χ^2 distributions and best-fit

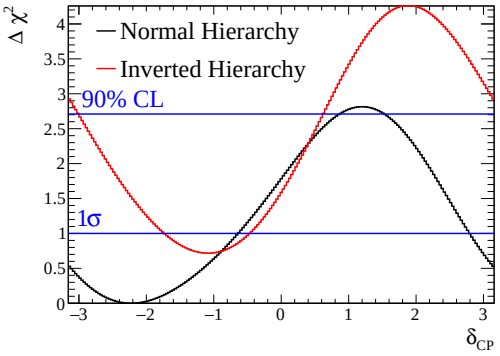
The same 1-dimensional χ^2 distributions as in section 5.1 are produced and shown in fig. 25. The best-fit value of δ_{CP} when the reactor constraint is applied is -1.854 not as close to $-\pi/2$ (-1.571) as the Run 1-4 fit result, which best-fit value was -1.601 (see table 12). However the difference of χ^2 between δ_{CP}^{bf} (best-fit value) and $\delta_{CP} = -\pi/2$ is smaller than 0.1. So the result is consistent with $\delta_{CP} = -\pi/2$ as well as the Run 1-4 data fit.

Looking at the χ^2 distribution of $\sin^2\theta_{13}$ with T2K only data in fig. 25 we can see that the reactor measurements are in agreement with the T2K data at about 1σ level if normal hierarchy is assumed.

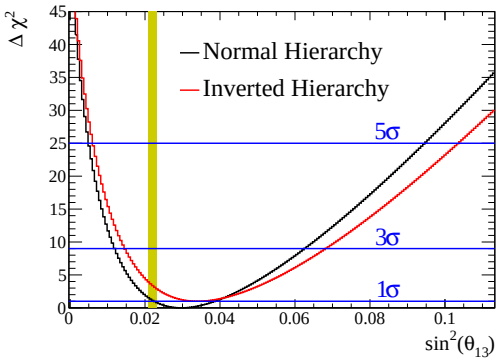
In fig. 26 the χ^2 distributions as a function of δ_{CP} with reactor constraint is obtained by running the analysis using either only neutrino or antineutrino mode samples for the Run 1-6 data set, in order to check the compatibility of neutrino and antineutrino mode data. The result is mainly driven by the excess of



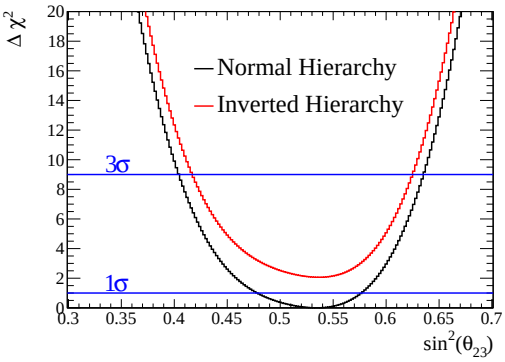
(a) δ_{CP} with reactor constraint



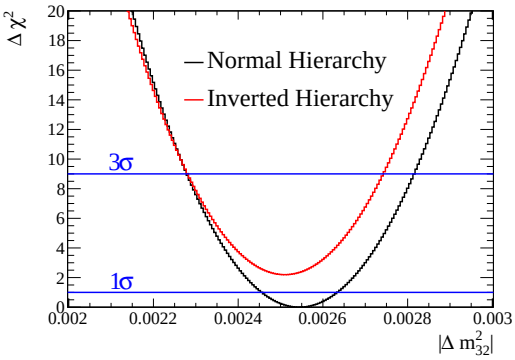
(b) δ_{CP} with T2K data only



(c) $\sin^2\theta_{13}$ with T2K data only



(d) $\sin^2\theta_{23}$ with reactor constraint



(e) Δm^2_{32} (Δm^2_{13}) with reactor constraint

Figure 25: Measured $\Delta\chi^2$ distributions with the full Run 1-6 data set. Both normal (black) and inverted (red) hierarchies are shown. The *constant* $\Delta\chi^2$ method is used. Both normal and inverted hierarchy $\Delta\chi^2$ distributions are shifted with respect to the same global minimum. All the oscillation parameters which are not parameters of interest are marginalised using the prior distributions shown in table 22. The measured $\Delta\chi^2$ distribution as a function of $\sin^2\theta_{13}$ is compared with the value given by the global analysis of the reactor measurements, $\sin^2\theta_{13} = 0.0219 \pm 0.0012$ [71], shown by the yellow bar.

the observed number of events in e-like neutrino mode sample compared to what is expected. Though the significance given by the antineutrino mode samples is still low, also given by the fact that $\sin^2\theta_{23}$ is not constrained by the antineutrino mode μ-like sample as much as by the neutrino mode μ-like sample, it is interesting to note that the antineutrino data also prefer a value of δ_{CP} close to $-\pi/2$.

In fig. 27 the χ^2 distributions as a function of δ_{CP} is obtained by performing a rate-only analysis; instead of using the shape information of the E_{reco} vs θ_l spectra, the total number of events of each sample is fitted, reducing the E_{reco} vs θ_l template to 1 bin. Then the best-fit value of δ_{CP} is -1.602 closer to $-\pi/2$ with respect to the case where the shape information is used.

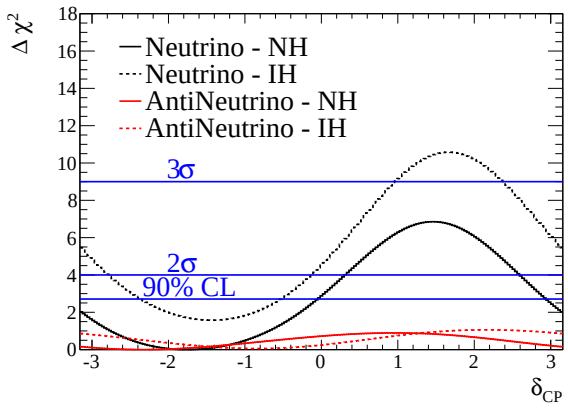


Figure 26: Measured $\Delta\chi^2$ distributions with the full Run 1-6 data set as a function of δ_{CP} with reactor constraint independently for only neutrino and antineutrino mode data. Both normal (black) and inverted (red) hierarchies are shown. The *constant* $\Delta\chi^2$ method is used. Both normal and inverted hierarchy $\Delta\chi^2$ distributions are shifted with respect to the same global minimum, independently for the neutrino and antineutrino mode data. All the oscillation parameters which are not parameters of interest are marginalised using the prior distributions shown in table 22.

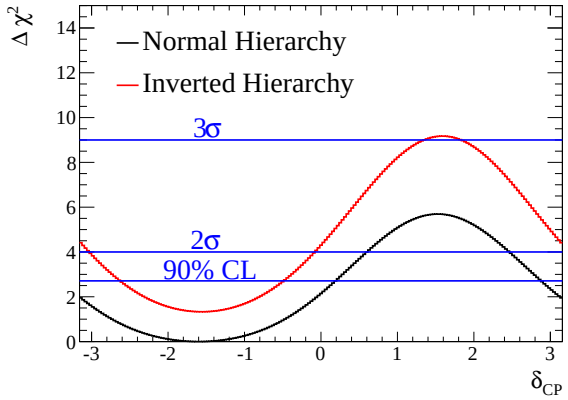


Figure 27: Measured $\Delta\chi^2$ distributions with a rate-only analysis using the full Run 1-6 data set as a function of δ_{CP} with reactor constraint. Both normal (black) and inverted (red) hierarchies are shown. The *constant* $\Delta\chi^2$ method is used. Both normal and inverted hierarchy $\Delta\chi^2$ distributions are shifted with respect to the same global minimum. All the oscillation parameters which are not parameters of interest are marginalised using the prior distributions shown in table 22.

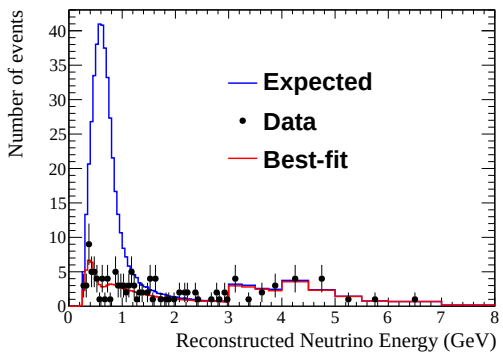
In table 26 the best-fit values as well as the 1σ ranges for each oscillation parameter are shown, for both normal and inverted hierarchy. The 1σ ranges are extracted for each oscillation parameter from the 1-dimensional $\Delta\chi^2$ distributions shown in fig. 25, where one oscillation parameter is free and all the other oscillation as well as the systematic parameters are marginalised. The mass hierarchy is minimized: for both normal and inverted hierarchy the 1σ ranges are calculated with respect to the same global minimum. Since this approach is not entirely correct and there does not exist any procedure to set more reliable confidence intervals which does not consist of toy MC studies (see section 4.5), the advice to the reader is to give more importance to the difference in χ^2_{marg} between normal and inverted hierarchies in the respective best-fit points, instead of the 1σ ranges of inverted hierarchy, which are smaller by construction. It is also worth to notice that the 1σ ranges for normal hierarchy (best-fit) are set correctly with the constant $\Delta\chi^2$ method in the gaussian approximation.

The best-fit spectra, shown in fig. 28 are computed as described in section 4.3 and are compared to the unoscillated nominal spectra. For e-like samples, where 2-dimensional templates are used, the projection onto the reconstructed neutrino energy axis is shown. All the systematic and oscillation parameters are marginalised following the procedure described in section 4.3, while $\sin^2\theta_{23}$, δ_{CP} and the mass hierarchy were fixed to their best-fit values shown in table 26. Indeed these oscillation parameters have the strongest

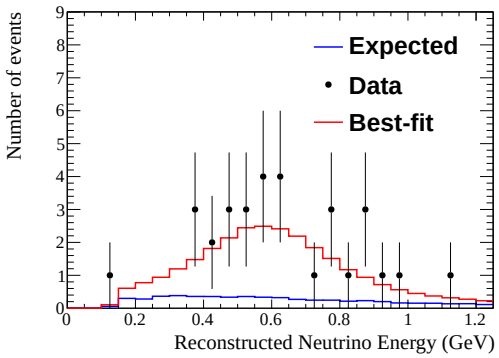
	Reactors	Normal Hierarchy			Inverted Hierarchy		
Parameter		Best-fit	χ^2_{min}	$\pm1\sigma$ interval	Best-fit	χ^2_{min}	$\pm1\sigma$ interval
δ_{CP}	YES	-1.854	296.51	[-2.737; -0.859]	-1.351	298.16	-
δ_{CP}	NO	-2.231	299.49	[-π; -0.640] & [2.793;+π]	1.100	300.20	[-1.735;-0.469]
$\sin^2\theta_{13}$	NO	0.0299	297.09	[0.0224; 0.0387]	0.0339	298.09	-
$\sin^2\theta_{23}$	YES	0.534	295.92	[0.478; 0.577]	0.536	297.98	-
Δm^2_{32} or Δm^2_{13} (10^{-3} eV$^2/c^4$)	YES	2.545	295.36	[2.455; 2.634]	2.510	297.56	-

Table 26: The measured oscillation parameter best-fit and minimum χ^2 values as well as the 1σ CL intervals are shown, for both normal and inverted hierarchy. The $\pm1\sigma$ interval is obtained by using the *constant* $\Delta\chi^2$ method. All the parameters are measured using the reactor constraint $\sin^2 2\theta_{13} = 0.085 \pm 0.005$ [71], except for δ_{CP} and $\sin^2\theta_{13}$. As explained in section 4.5, the gaussian approximation does not work well for discrete parameters like the mass hierarchy however, since there is not a clear way to compute 1σ ranges for both mass hierarchies without performing CPU intensive toy MC studies, the constant $\Delta\chi^2$ method is used.

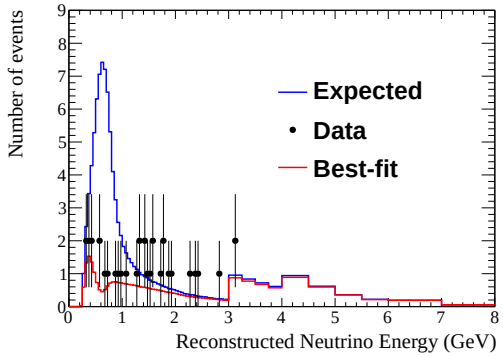
effect on the measurement, since $\sin^2\theta_{13}$ is constrained with the reactor measurement.



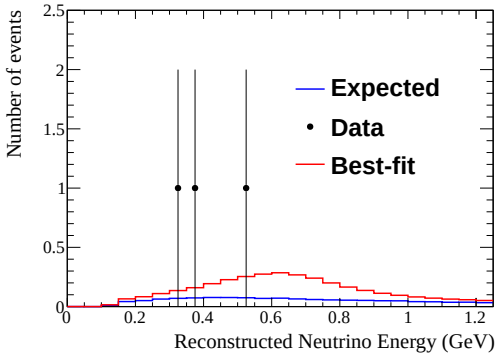
(a) Muon-like, neutrino mode



(b) Electron-like, neutrino mode



(c) Muon-like, antineutrino mode



(d) Electron-like, antineutrino mode

Figure 28: The best-fit spectra as a function of the neutrino reconstructed energy are shown for each selected sample of the Run 1-6 data set. They correspond to the fit of $\sin^2\theta_{23}$, δ_{CP} and mass hierarchy with reactor constraint where all the nuisance parameters are marginalised. The method described in eq. (11) is used. The best-fit distributions are compared with the predicted unoscillated spectrum (all the oscillation parameters are set to 0), generated with the systematic parameters measured with the ND280 data shown in table 6, 7, 8 and the nominal Super-K systematic parameters shown in table 9.

A goodness of fit test is performed by comparing data with the best fit spectra shown in fig. 28. In order to determine how compatible the data is with our model the log likelihood is calculated for each sample as in eq. (3) and the $\chi^2/ndof$ is used to estimate the fit probability using the Poisson statistic, given the number of degrees of freedom equal to $ndof = N_{bins} - N_{free}(3 = 8 - 5)$, where N_{bins} is the number of template bins and N_{free} is the number of free parameters in the fit. The free parameters are $\sin^2\theta_{23}$, δ_{CP} and the

mass hierarchy in this case. In this analysis, a coarse binning scheme is used for the goodness of fit test as in previous analyses. This is used so that there are enough events in each bin to calculate p-values in the gaussian approximation. 5 bins (0-0.4, 0.4-0.7, 0.7-1.0, 1.0-2.0 and 2.0-30.0 GeV) are used for the neutrino mode muon-like spectrum, 1 bin is used for each of the muon-like antineutrino and electron-like spectra in order to enter the gaussian regime, where the χ^2 distribution is well defined. The χ^2 and resulting p-value for each spectrum is shown in table 27. We see for all the spectra, the data and best fit spectra are very compatible.

	χ^2 / ndf	p-value
MH fixed	2.08917 / 6	0.84
MH fitted	2.08917 / 5	0.91

Table 27: Goodness of fit for the 2-dimensional fit of $\sin^2\theta_{23}$, δ_{CP} and mass hierarchy (both fitted and fixed) where all the other parameters are marginalised.

6.2. Feldman-Cousins and significance for the simultaneous fit of δ_{CP} and mass hierarchy

In fig. 29 the confidence intervals obtained with the "Feldman-Cousins" method are shown for δ_{CP} with reactor constraint. The critical values at both 90% and 2σ CL are computed for both the mass hierarchies and the following true values of δ_{CP}: $-\pi$, -2.356, $-\pi/2$, -0.785, 0, +0.785, $+\pi/2$, -2.356, $+\pi$. The procedure described in section 4.4 is used. From the zoom on a small range of $\Delta\chi^2$ it is clear that $\delta_{CP} = 0$ is excluded with a significance of 2σ, while $\delta_{CP} = \pi$ is not excluded at 90% CL. Between 8k and 10k toy experiments were used for the computation of the critical values.

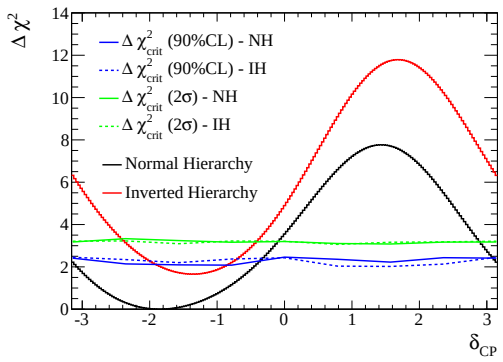
The $\pm1\sigma$ error on the critical values are shown as well. They are computed assuming that the critical value follows a binomial p.d.f., which becomes clear when looking to eq. (19), since it behaves as an efficiency: the 10% of $\Delta\chi^2$ values above the 90% CL critical value are not accounted in the integral. The critical values corresponding to respectively $\Delta\chi^2_{crit} \pm 1\sigma$ are computed for the X% CL using the following equation:

$$\Delta\chi^2_{crit} \pm 1\sigma : \int_{-\infty}^{\Delta\chi^2_{crit}\pm1\sigma} f(\Delta\chi^2)d(\Delta\chi^2) \tag{20}$$

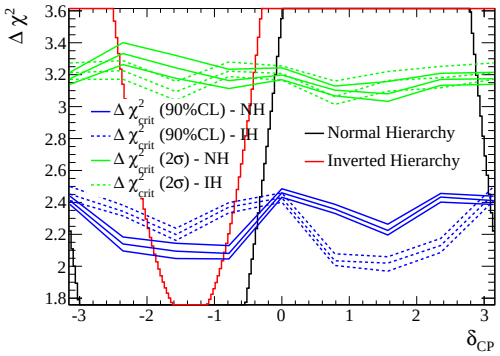
where N is the number of toy experiments, $f(\Delta\chi^2)$ is the p.d.f. of $\Delta\chi^2$, $d(\Delta\chi^2)$ is the differential of the integral and the standard deviation on the critical value is defined as $1\sigma = \sqrt{N \cdot X\% \cdot (1 - X\%)}$. The higher is the confidence level to be computed the more toy experiments are needed. Indeed higher confidence levels corresponds to higher critical values, for which the tail of the $\Delta\chi^2$ distribution, which is less populated, becomes more important. However from fig. 29 it is shown that even 4k toy experiments would be enough for the computation of the critical values, since the $\pm1\sigma$ variation of the critical values does not change significantly the confidence intervals.

The significance to $\delta_{CP} = 0$ and π, shown in fig. 30 and table 28, is computed by performing the p-value using the same $\Delta\chi^2$ distributions obtained with the Feldman-Cousins method and 10k toy experiments, as explained in appendix E. The p-value for $\delta_{CP} = 0$ is 0.032, which corresponds to a significance of Z = 2.14 standard deviations, assuming the errors to be gaussian distributed. Instead the p-value for $\delta_{CP} = \pi$ is 0.122 which corresponds to only Z = 1.55 standard deviations.

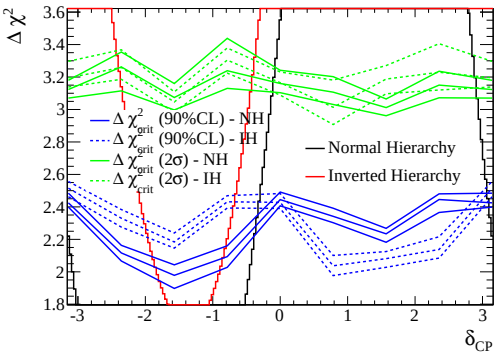
$\delta_{CP} = \pi$ is favored by the data with respect to $\delta_{CP} = 0$. Indeed $\delta_{CP} = \pi$ produces a small event migration toward higher energy, which is favored by the e-like Run 1-6 data, for both neutrino and antineutrino mode. The variation of δ_{CP} from 0 to π in the best-fit spectra, keeping $\sin^2\theta_{23}$ and the mass hierarchy fixed to the global best-fit values, produces an event migration to lower energy values with respect to 0.6 GeV as shown in table 29 and fig. 31. In particular the data set of the e-like antineutrino mode sample shows all the events with reconstructed energy smaller than 0.6 GeV.



(a) Confidence intervals with more than 8k toys MC in each point of δ_{CP}.



(b) Confidence intervals with more than 8k toys MC in each point of δ_{CP} with a zoom on the critical values. The $\pm1\sigma$ errors on the critical values are shown as well.



(c) Confidence intervals with 4k toys MC in each point of δ_{CP} with a zoom on the critical values. The $\pm1\sigma$ errors on the critical values are shown as well.

Figure 29: Measured $\Delta\chi^2$ distributions with the full Run 1-6 data set. Both normal (black) and inverted (red) hierarchies are shown. The "Feldman-Cousins" method is used to set the $\Delta\chi^2$ critical values in order to provide confidence intervals with the proper coverage. The ranges of δ_{CP} which correspond to $\Delta\chi^2$ values larger than the critical values (blue for 90% CL and green for 2σ CL) are excluded at a certain confidence level. Both normal and inverted hierarchy $\Delta\chi^2$ distributions are shifted with respect to the same global minimum. All the oscillation parameters which are not parameters of interest are marginalised using the prior distributions shown in table 22. The $\pm1\sigma$ error on the critical values is shown as well, where specified, and is represented by the lowest and highest set of critical values, with the same color of the central ones.

δ_{CP}	$\Delta\chi^2_{data}$	p-value	# of std. deviations
0	3.552	0.032	2.14
π	2.221	0.122	1.55

Table 28: Summary of best-fit $\Delta\chi^2$ value measured from the Run 1-6 data set, p-value and number of standard deviations with respect to the shown value of δ_{CP}. The best-fit mass hierarchy is normal. All the other systematic parameters are marginalised.

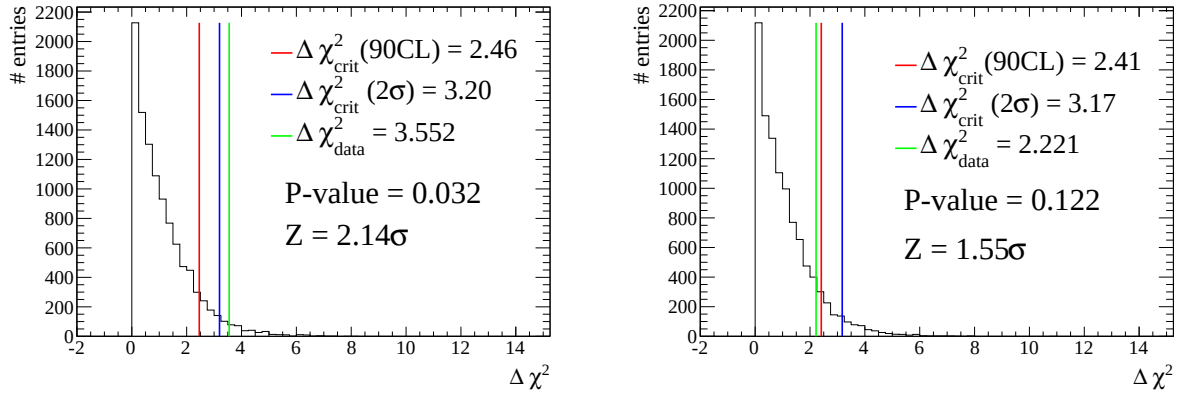
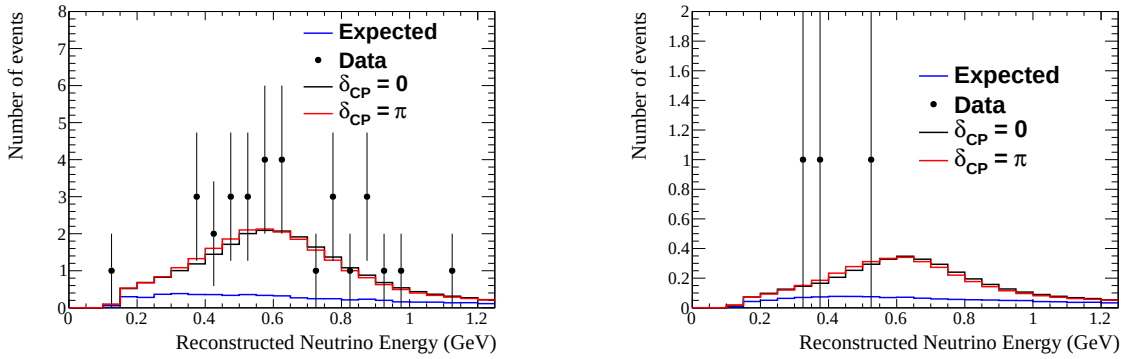


Figure 30: Distributions of $\Delta\chi^2$. $10k$ toy experiments are performed with true values of $\delta_{CP} = 0$ (left) and π (right), assuming normal hierarchy and applying the reactor constraint. Both the mass hierarchy and δ_{CP} are profiled in the fit. All the other oscillation parameters are marginalised. The statistics corresponds to the Run 1-6 data set. The $\Delta\chi^2$ value (green line) measured from the Run 1-6 data set as well as the p-value and the significance as number of standard deviations (Z), are shown as well.

e-like neutrino mode			
	Best-fit, $\delta_{CP} = 0$	Best-fit, $\delta_{CP} = \pi$	Data
Events with $E_{reco} < 0.6$ GeV	58.4%	61.1%	64.5%
$< E_{reco} >$	0.616	0.604	0.627
***e*-like antineutrino mode**			
	Best-fit, $\delta_{CP} = 0$	Best-fit, $\delta_{CP} = \pi$	Data
Events with $E_{reco} < 0.6$ GeV	53.3%	56.3%	100%
$< E_{reco} >$	0.644	0.631	0.408

Table 29: The fraction of events with reconstructed neutrino energy smaller than 0.6 GeV and the mean reconstructed neutrino energy is shown for the best-fit ($\sin^2\theta_{23}$ vs δ_{CP} vs mass hierarchy) spectra where δ_{CP} is kept fixed to either 0 or π, for both neutrino and antineutrino mode *e*-like samples, and for the Run 1-6 data set. The other two free oscillation parameters, $\sin^2\theta_{23}$ and mass hierarchy, as set to the same best-fit values as in fig. 28. The reactor constraint is applied and all the nuisance parameters are marginalised. The method described in section 4.3 is used.



(a) Muon-like, neutrino mode

(b) Electron-like, neutrino mode

Figure 31: The best-fit spectra as a function of the neutrino reconstructed energy where δ_{CP} is fixed to either 0 (red) or π (black) are shown for both neutrino (left) and antineutrino (right) mode *e*-like samples of the Run 1-6 data set. The other two free oscillation parameters, $\sin^2\theta_{23}$ and mass hierarchy, as set to the same best-fit values as in fig. 28. The reactor constraint is applied and all the nuisance parameters are marginalised. The method described in section 4.3 is used as well as the same best-fit values for $\sin^2\theta_{23}$ and mass hierarchy. The best-fit distributions are compared with the predicted unoscillated spectrum (all the oscillation parameters are set to 0), generated with the systematic parameters measured with the ND280 data shown in table 6, 7, 8 and the nominal Super-K systematic parameters shown in table 9.

6.3. 2-dimensional confidence intervals

The same 2-dimensional confidence intervals as in section 5.1 are produced. In fig. 32 the corresponding 2-dimensional contours produced with the Run 1-6 data set are shown drawing the normal and inverted hierarchy contours independently, i.e. each contour is produced with respect to the global minimum under that hierarchy hypothesis. In fig. 33 both normal and inverted hierarchy contours are produced with respect to the same global minimum, i.e. the normal hierarchy hypothesis.

As already explained in section 4.5, "global" and "independent" contours correspond respectively to the least and most conservative approach.

As in 1-dimensional analysis results, looking at contours of $\sin^2\theta_{13}$ vs δ_{CP} with T2K only data, in fig. 32a, it is clear that the reactor measurements are in agreement with the T2K data at 1σ level if $\delta_{CP} \sim -\frac{\pi}{2}$. The tension between the $\sin^2\theta_{13}$ reactor constraint and the T2K data constraint is clear at $\sim \frac{+\pi}{2}$ where the T2K data prefer a lager mixing angle with both hierarchy hypotheses. This is the cause of the closed 90% contour in fig. 32b which is also in good agreement with the 1D δ_{CP} fit in fig. 25a. This effect is much greater for the inverted hierarchy fits and thus leads to a much tighter constraint on δ_{CP}. As a result, we see that all δ_{CP} constraints in fig. 32b, 32c, 32d are tighter for the inverted hierarchy hypothesis than for the normal. fig. 32c also shows a slight anti-correlation between δ_{CP} and Δm^2_{32} in the normal hierarchy fit and a correlation in the inverted most likely coming from the fact that Δm^2_{32} and $cos(\delta_{CP})$ have similar effects on the shape of the appearance spectra. In fig. 32d we see correlation where the constraint on δ_{CP} is narrower for smaller values of $\sin^2\theta_{23}$ as they both appear in the appearance term. In fig. 32e we see that $\sin^2\theta_{23}$ and Δm^2_{32} are not very correlated however the inverted hierarchy contour is shifted down in Δm^2_{32} with respect to the normal hierarchy contour. This effect is due to the correlation between Δm^2_{32} and the mass hierarchy shown in fig. 34. Indeed the inverted hierarchy hypothesis produces an event migration toward higher energy which is balanced in the fit by a lower fit value of Δm^2_{13} with respect to the normal hierarchy case.

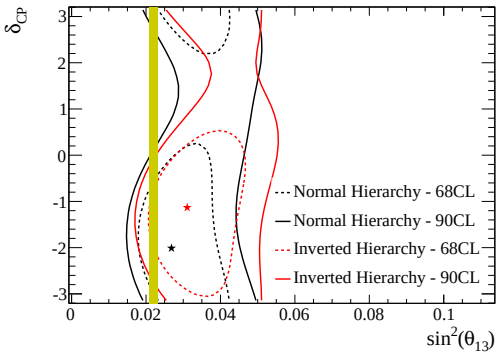
The *constant* $\Delta\chi^2$ contours with respect to the global minimum in fig. 33 show similar trends. As the normal hierarchy was preferred in all fits, the normal hierarchy contours are identical to those in fig. 32. The inverted hierarchy contours are noticeably tighter in all cases. In fig. 33a we see that there is a closed 90% CL contour, and in all others the 90% CL contour is of similar size to the 68% CL contour for the normal hierarchy.

6.4. Discussion of the results

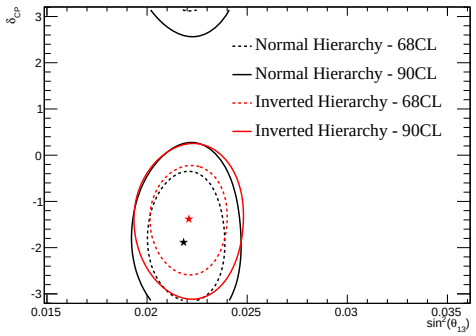
The value of δ_{CP} favored by the joint fit analysis when the reactor constraint is applied is -1.854 and corresponds to normal hierarchy. $\delta_{CP} = 0$ is excluded at 2.14 standard deviations and a large fraction of the range between $\delta_{CP} = 0$ and $+\pi$ is excluded with a significance of 2σ (with the constant $\Delta\chi^2$ method), while $\delta_{CP} = \pi$ is not excluded at 90% CL. Basically with the Run 1-6 data set we cannot exclude the hypothesis of no CP violation in the leptonic sector at 90% CL.

The inverted hierarchy is disfavored at $\sim 1\sigma$ level and it is partially excluded at 3σ for values of δ_{CP} around $+\pi/2$. It has been shown this exclusion, stronger than what is expected from the fit of the Asimov data sets for $\delta_{CP} = -1.601$, is due to the small excess of events in the neutrino mode e-like sample. Indeed the impact of antineutrino mode samples on the result is quite small. If the reactor constraint is not applied, the best-fit value of δ_{CP} is still quite close to $-\pi/2$. However the exclusion is much weaker than when the reactor constraint is applied. The contour of $\sin^2\theta_{13}$ vs δ_{CP} shows an agreement at the level of 1σ with the measurement of the reactors if $\delta_{CP} \sim -\pi/2$.

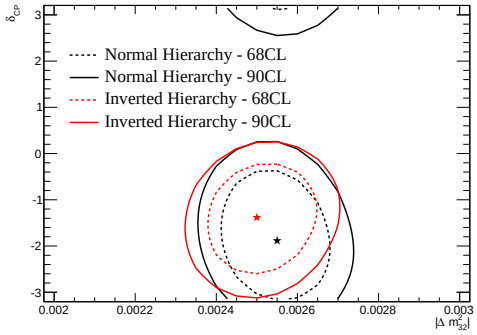
The result shows a stronger exclusion of δ_{CP} compared to the 2014 joint fit analysis [8]. This is mainly due to the new reactor constraint which has lower $\sin^2\theta_{13}$ (0.0217 instead of 0.0250) and a smaller uncertainty: 0.085 ± 0.005 instead of 0.095 ± 0.01.



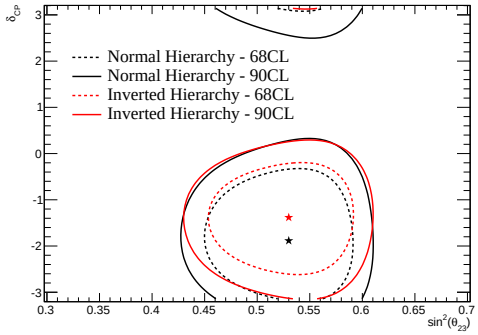
(a) $\sin^2\theta_{13}$ vs δ_{CP} with T2K data only



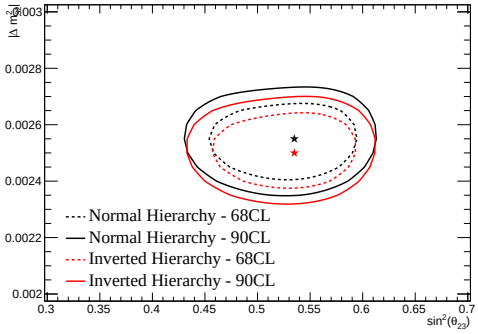
(b) $\sin^2\theta_{13}$ vs δ_{CP} with reactor constraint



(c) Δm^2_{32} (Δm^2_{13}) vs δ_{CP} with reactor constraint

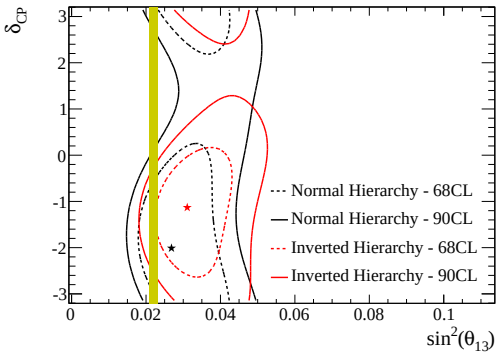


(d) $\sin^2\theta_{23}$ vs δ_{CP} with reactor constraint

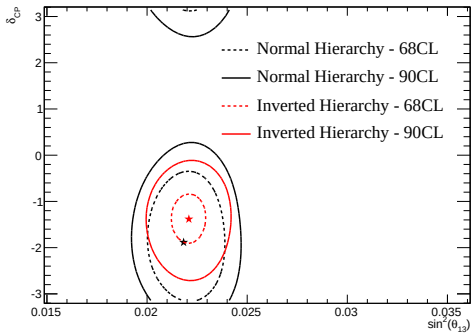


(e) $\sin^2\theta_{23}$ vs Δm^2_{32} (Δm^2_{13}) with reactor constraint

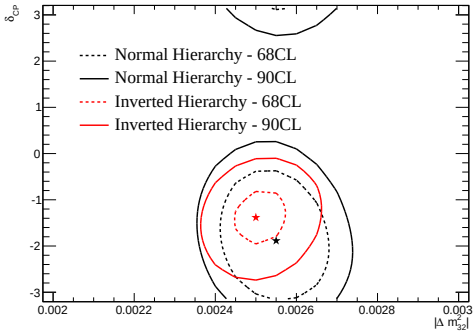
Figure 32: Measured confidence intervals with the full Run 1-6 data set. Both normal (black) and inverted (red) hierarchies are shown. Both 68% (dashed line) and 90% CL (solid line) are shown. Confidence intervals are produced using the *constant* $\Delta\chi^2$ method. Contours for normal (black) and inverted (red) hierarchy are computed independently. All the oscillation parameters which are not parameters of interest are marginalised using the prior distributions shown in table 11. The measured contour of $\sin^2\theta_{13}$ vs δ_{CP} without reactor constraint, in fig. 32a is compared with the measurement by reactor experiments, $\sin^2\theta_{13} = 0.0219 \pm 0.0012$ [71], shown by the yellow bar.



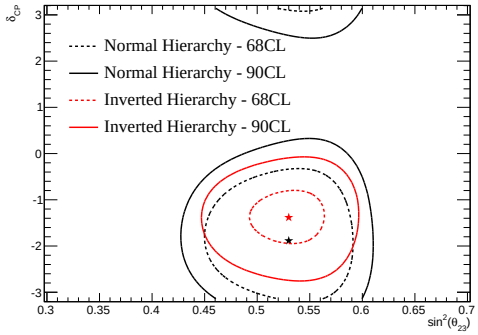
(a) $\sin^2\theta_{13}$ vs δ_{CP} with T2K data only



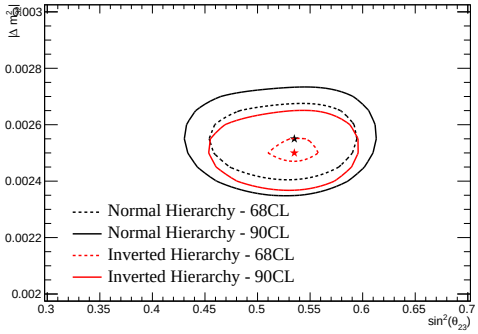
(b) $\sin^2\theta_{13}$ vs δ_{CP} with reactor constraint



(c) Δm^2_{32} (Δm^2_{13}) vs δ_{CP} with reactor constraint

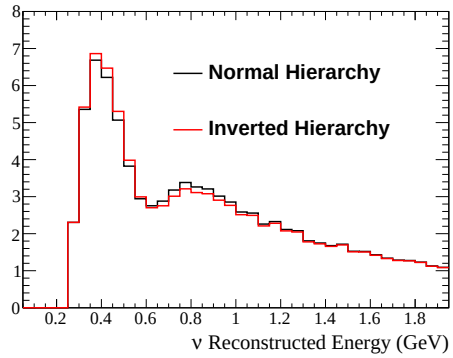


(d) $\sin^2\theta_{23}$ vs δ_{CP} with reactor constraint

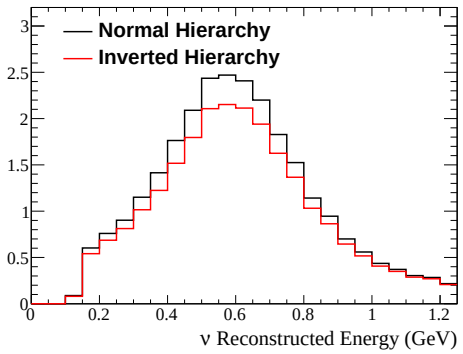


(e) $\sin^2\theta_{23}$ vs Δm^2_{32} (Δm^2_{13}) with reactor constraint

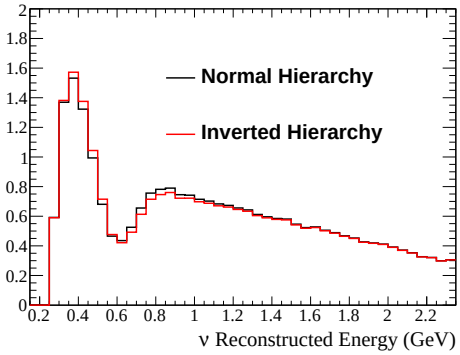
Figure 33: Measured confidence intervals with the full Run 1-6 data set. Both normal (black) and inverted (red) hierarchies are shown. Contours for normal (black) and inverted (red) hierarchy are computed with respect to the minimum of the best-fit hierarchy (star), which corresponds to normal hierarchy. Both 68% (dashed line) and 90% CL (solid line) are shown. Confidence intervals are produced using the *constant* $\Delta\chi^2$ method. All the oscillation parameters which are not parameters of interest are marginalised using the prior distributions shown in table 11. The measured contour of $\sin^2\theta_{13}$ vs δ_{CP} without reactor constraint is compared with the measurement by reactor experiments, $\sin^2\theta_{13} = 0.0219 \pm 0.0012$ [71], shown by the yellow bar.



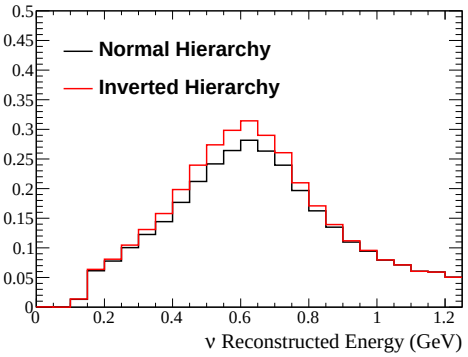
(a) Muon-like, neutrino mode



(b) Electron-like, neutrino mode



(c) Muon-like, antineutrino mode



(d) Electron-like, antineutrino mode

Figure 34: The expected spectra as a function of the neutrino reconstructed energy are shown for each selected sample of the Run 1-6 data set for both normal (black) and inverted (red) hierarchies. They correspond to the Asimov data set A in table 12 and are generated with the systematic parameters measured with the ND280 data shown in table 6, 7, 8 and the nominal Super-K systematic parameters shown in table 9.

7. Summary

In this technical note, we have presented the results of the 3-flavour $\bar{\nu}/\nu$ joint analysis performed by the *VALOR* group, on the combined Run 1-6 data set. The goal of the analysis is to search for CP violation in the leptonic and put limits on δ_{CP} as well as measure all the other oscillation parameters to which T2K is sensitive: $\sin^2\theta_{13}$, $\sin^2\theta_{23}$, Δm^2_{32} (Δm^2_{13}) and partially mass hierarchy, to which T2K has little sensitivity.

This analysis predicts 125.9 ± 6.5 (syst) for neutrino μ-like, 34.3 ± 1.8 (syst) for antineutrino μ-like, 26.6 ± 1.8 (syst) for neutrino e-like, and 3.22 ± 0.24 (syst) for antineutrino e-like single ring events in Super K and observes respectively 124 for neutrino μ-like, 34 for antineutrino μ-like, 31 for neutrino e-like, 3 for antineutrino e-like.

The $\bar{\nu}/\nu$ analysis was performed in a framework of 3-flavour oscillations including matter effects in constant-density matter. The observed reconstructed energy spectrum of single μ-like ring events and reconstructed energy versus lepton angle of single e-like ring events was used to compute the confidence intervals. All 102 systematic parameters as well as the oscillation parameters which were not parameters of interest considered in this analysis were marginalised.

From the fit of the Run 1-6 data we can see that the favored value of δ_{CP} is -1.854, close to $-\pi/2$. The correct coverage is provided by using the "Feldman-Cousins" method with which the significance to excluded $\delta_{CP} = 0$ becomes 2.14 standard deviations, while $\delta_{CP} = \pi$ is excluded at less than 90% CL. The Run 1-6 data excludes no CP violation in the leptonic sector at 90% CL.

In case of inverted hierarchy a not negligible range of δ_{CP} around $+\pi/2$ is excluded with a significance of 3σ.

From the contour of $\sin^2\theta_{13}$ vs δ_{CP} without the reactor constraint it is clear that the T2K data are in agreement with the reactor measurements within 1σ if $\delta_{CP} \sim -\pi/2$.

It has also been shown that the result of the search for CP violation is dominated by the excess of event in the neutrino mode e-like sample, though the antineutrino mode e-like also prefers values of δ_{CP} close to $-\pi/2$. However the significance is not high.

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Appendices

A. Predicted number of events

In table 30, 31 and 32, 33 the total number of events expected for the Asimov data set B (see table 12) is shown.

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	69.747	0.035	4.335	0.002	0.151	0.001	74.270
CC1pi	17.206	0.017	1.572	0.001	0.049	0.000	18.845
CCcoherent	0.330	0.000	0.080	0.000	0.001	0.000	0.412
2p-2h	19.725	0.010	0.852	0.000	0.036	0.000	20.623
CCother	7.742	0.008	0.494	0.001	0.002	0.000	8.247
NC1pi0	0.537	0.016	0.017	0.001	N/A	N/A	0.572
NC1piPM	3.944	0.089	0.142	0.008	N/A	N/A	4.183
NCcoherent	0.016	0.000	0.001	0.000	N/A	N/A	0.018
NCother	2.993	0.121	0.165	0.012	N/A	N/A	3.290
NC1gamma	0.004	0.000	0.000	0.000	N/A	N/A	0.004
Total	122.243	0.297	7.659	0.025	0.238	0.002	130.464

Table 30: Predicted number of events in the neutrino mode μ-like sample obtained after applying the BANFF weight for a total exposure of 6.914×10^{20} POT. The Asimov data set B in table 12 is used.

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	7.088	0.004	13.306	0.005	0.004	0.016	20.422
CC1pi	2.718	0.003	3.066	0.002	0.002	0.003	5.794
CCcoherent	0.054	0.000	0.173	0.000	0.000	0.000	0.228
2p-2h	2.525	0.001	2.118	0.001	0.001	0.002	4.649
CCother	1.282	0.001	0.886	0.000	0.000	0.000	2.170
NC1pi0	0.060	0.002	0.050	0.001	N/A	N/A	0.114
NC1piPM	0.378	0.015	0.380	0.010	N/A	N/A	0.784
NCcoherent	0.001	0.000	0.006	0.000	N/A	N/A	0.008
NCother	0.530	0.025	0.296	0.012	N/A	N/A	0.862
NC1gamma	0.000	0.000	0.000	0.000	N/A	N/A	0.000
Total	14.637	0.051	20.282	0.032	0.008	0.022	35.032

Table 31: Predicted number of events in the antineutrino mode μ-like sample obtained after applying the BANFF weight for a total exposure of 4.011×10^{20} POT. The Asimov data set B in table 12 is used.

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	0.062	2.129	0.001	0.088	10.811	0.121	13.211
CC1pi	0.017	0.348	0.000	0.027	1.236	0.020	1.648
CCcoherent	0.000	0.002	0.000	0.002	0.011	0.002	0.017
2p-2h	0.009	0.710	0.000	0.022	2.750	0.022	3.513
CCother	0.001	0.045	0.000	0.003	0.025	0.001	0.076
NC1pi0	0.524	0.012	0.017	0.001	N/A	N/A	0.553
NC1piPM	0.099	0.002	0.003	0.000	N/A	N/A	0.105
NCcoherent	0.174	0.004	0.018	0.001	N/A	N/A	0.197
NCother	0.126	0.005	0.008	0.001	N/A	N/A	0.139
NC1gamma	0.464	0.007	0.021	0.001	N/A	N/A	0.494
Total	1.476	3.264	0.069	0.145	14.833	0.167	19.954

Table 32: Predicted number of events in the neutrino mode e-like sample obtained after applying the BANFF weight for a total exposure of 6.914×10^{20} POT. The Asimov data set B in table 12 is used.

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	0.002	0.189	0.003	0.339	0.269	1.376	2.178
CC1pi	0.002	0.040	0.001	0.060	0.045	0.163	0.312
CCcoherent	0.000	0.000	0.000	0.006	0.000	0.022	0.029
2p-2h	0.001	0.069	0.000	0.063	0.087	0.219	0.439
CCother	0.001	0.008	0.000	0.004	0.003	0.003	0.019
NC1pi0	0.045	0.002	0.049	0.001	N/A	N/A	0.098
NC1piPM	0.011	0.000	0.008	0.000	N/A	N/A	0.020
NCcoherent	0.021	0.001	0.081	0.001	N/A	N/A	0.103
NCother	0.022	0.001	0.012	0.000	N/A	N/A	0.035
NC1gamma	0.037	0.002	0.081	0.002	N/A	N/A	0.121
Total	0.142	0.313	0.235	0.478	0.404	1.783	3.355

Table 33: Predicted number of events in the antineutrino mode μ-like sample obtained after applying the BANFF weight for a total exposure of 4.011×10^{20} POT. The Asimov data set B in table 12 is used.

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	338.683	0.037	8.672	0.002	0.000	0.000	347.394
CC1pi	39.668	0.018	2.276	0.001	0.000	0.000	41.963
CCcoherent	0.678	0.000	0.136	0.000	0.000	0.000	0.815
2p-2h	74.638	0.011	1.548	0.000	0.000	0.000	76.198
CCother	8.732	0.009	0.560	0.001	0.000	0.000	9.301
NC1pi0	0.537	0.016	0.017	0.001	N/A	N/A	0.572
NC1piPM	3.944	0.089	0.142	0.008	N/A	N/A	4.183
NCcoherent	0.016	0.000	0.001	0.000	N/A	N/A	0.018
NCother	2.993	0.121	0.165	0.012	N/A	N/A	3.290
NC1gamma	0.004	0.000	0.000	0.000	N/A	N/A	0.004
Total	469.893	0.301	13.518	0.025	0.000	0.000	483.737

Table 34: Predicted number of events in the neutrino mode μ-like sample obtained after applying the BANFF weight for a total exposure of 6.914×10^{20} POT. Oscillations are not applied.

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	15.420	0.004	55.525	0.005	0.000	0.000	70.954
CC1pi	4.071	0.003	6.536	0.002	0.000	0.000	10.611
CCcoherent	0.074	0.000	0.612	0.000	0.000	0.000	0.687
2p-2h	4.721	0.001	7.811	0.001	0.000	0.000	12.534
CCother	1.457	0.001	1.008	0.000	0.000	0.000	2.467
NC1pi0	0.060	0.002	0.050	0.001	N/A	N/A	0.114
NC1piPM	0.378	0.015	0.380	0.010	N/A	N/A	0.784
NCcoherent	0.001	0.000	0.006	0.000	N/A	N/A	0.008
NCother	0.530	0.025	0.296	0.012	N/A	N/A	0.862
NC1gamma	0.000	0.000	0.000	0.000	N/A	N/A	0.000
Total	26.713	0.052	72.225	0.032	0.000	0.000	99.021

Table 35: Predicted number of events in the antineutrino mode μ-like sample obtained after applying the BANFF weight for a total exposure of 4.011×10^{20} POT. Oscillations are not applied.

In table 34, 35 and 36, 37 the total number of unoscillated events is shown.

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	0.413	2.294	0.002	0.094	0.000	0.000	2.803
CC1pi	0.069	0.371	0.001	0.028	0.000	0.000	0.469
CCcoherent	0.002	0.003	0.000	0.002	0.000	0.000	0.006
2p-2h	0.095	0.759	0.000	0.023	0.000	0.000	0.878
CCother	0.004	0.046	0.000	0.003	0.000	0.000	0.054
NC1pi0	0.524	0.012	0.017	0.001	N/A	N/A	0.553
NC1piPM	0.099	0.002	0.003	0.000	N/A	N/A	0.105
NCcoherent	0.174	0.004	0.018	0.001	N/A	N/A	0.197
NCother	0.126	0.005	0.008	0.001	N/A	N/A	0.139
NC1gamma	0.464	0.007	0.021	0.001	N/A	N/A	0.494
Total	1.970	3.504	0.071	0.154	0.000	0.000	5.698

Table 36: Predicted number of events in the neutrino mode e-like sample obtained after applying the BANFF weight for a total exposure of 6.914×10^{20} POT. Oscillations are not applied.

	ν_μ	ν_e	$\bar{\nu_\mu}$	$\bar{\nu_e}$	Osc. ν_e	Osc. $\bar{\nu_e}$	Total
CCQE	0.012	0.203	0.018	0.361	0.000	0.000	0.593
CC1pi	0.004	0.043	0.003	0.064	0.000	0.000	0.114
CCcoherent	0.000	0.000	0.000	0.006	0.000	0.000	0.007
2p-2h	0.003	0.074	0.003	0.067	0.000	0.000	0.147
CCother	0.001	0.009	0.000	0.004	0.000	0.000	0.015
NC1pi0	0.045	0.002	0.049	0.001	N/A	N/A	0.098
NC1piPM	0.011	0.000	0.008	0.000	N/A	N/A	0.020
NCcoherent	0.021	0.001	0.081	0.001	N/A	N/A	0.103
NCother	0.022	0.001	0.012	0.000	N/A	N/A	0.035
NC1gamma	0.037	0.002	0.081	0.002	N/A	N/A	0.121
Total	0.156	0.334	0.255	0.507	0.000	0.000	1.253

Table 37: Predicted number of events in the antineutrino mode e-like sample obtained after applying the BANFF weight for a total exposure of 4.011×10^{20} POT. Oscillations are not applied.

B. Predicted *e*-like spectra

In fig. 35 each flavor component of the neutrino and antineutrino mode e-like samples is shown separately in order to compare the different shapes as a function of E_{reco}. The NC spectrum distribution, which is the largest component of the background, is quite flat over all the energy range and is quite difficult to separate. The $\nu_\mu \to \nu_e$ and $\bar{\nu}_\mu \to \bar{\nu}_e$ events populate the same energy region, around 0.6 GeV, with a very similar shape.

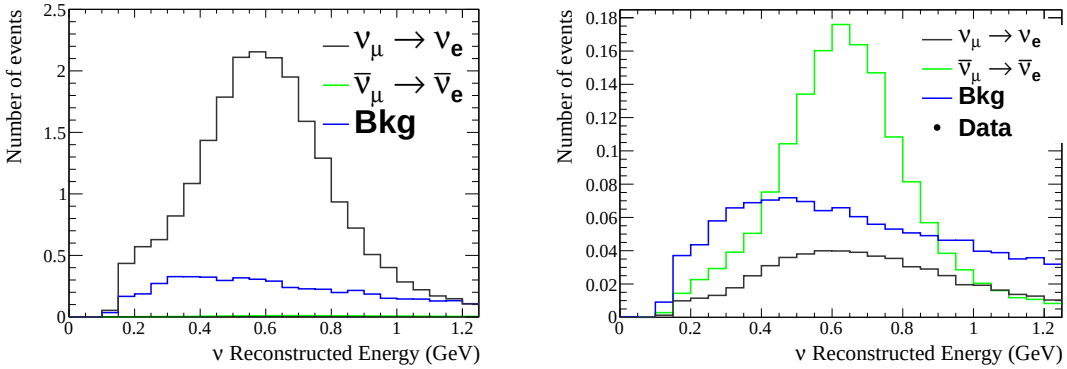
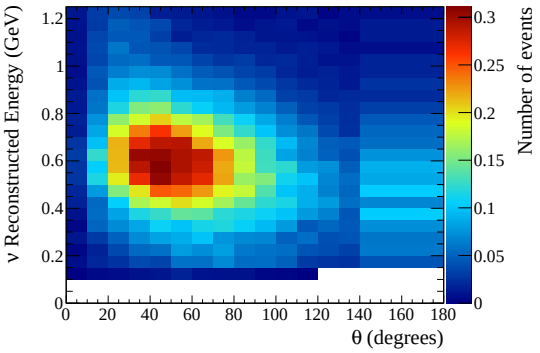


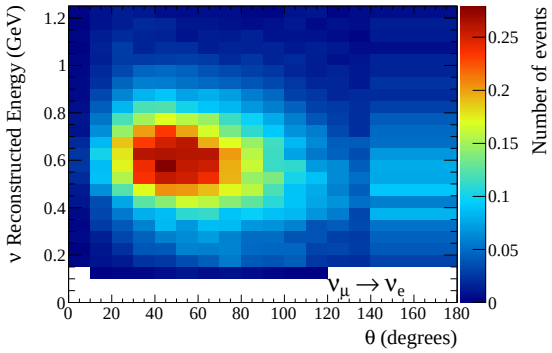
Figure 35: Predicted spectra of reconstructed neutrino energy for electron-like neutrino mode (left) and antineutrino mode (right) events. The distributions correspond to the collected statistics in the full Run 1-6 data set. The distributions are broken down by the following components: $\bar{\nu}_\mu \to \bar{\nu}_e$ (green), $\nu_\mu \to \nu_e$ (black/white), background, which includes NC, $\nu_e + \bar{\nu}_e$ intrinsic beam (yellow), ν_μ intrinsic beam (light blue) and $\bar{\nu}_\mu$ intrinsic beam (blue). The spectra were generated with the systematic parameters measured with the ND280 data shown in table 6, 7, 8, the nominal Super-K systematic parameters shown in table 9 and the oscillation parameters corresponding to the Asimov data set A shown in table 12. It is not a stacked plot.

Another kinematical variable is thus needed in order to better separate different flavour / reaction compoonents. In fig. 36 and 37 the 2-dimensional E_{reco} vs θ spectra are shown for each flavour component and NC for both neutrino and antineutrino mode e-like samples. In particular the shapes of $\nu_\mu \to \nu_e$ and $\bar{\nu}_\mu \to \bar{\nu}_e$ distributions are quite different, thus this templates have a better capability to discriminate signal from background. In general this template parametrization produces quite different shapes for all the main flavour and NC components.

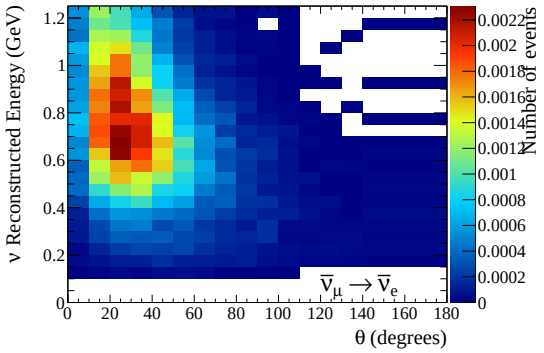
In fig. 38 the projection of the 2-dimensional e-like templates onto the E_{reco} and θ axes are shown.



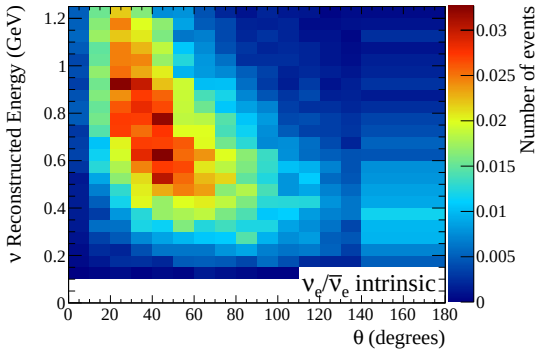
(a) Total



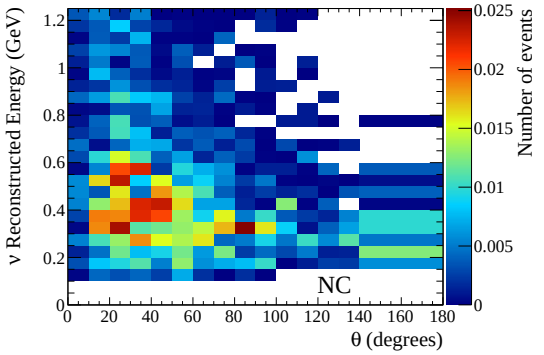
(b) $\nu_\mu \rightarrow \nu_e$.



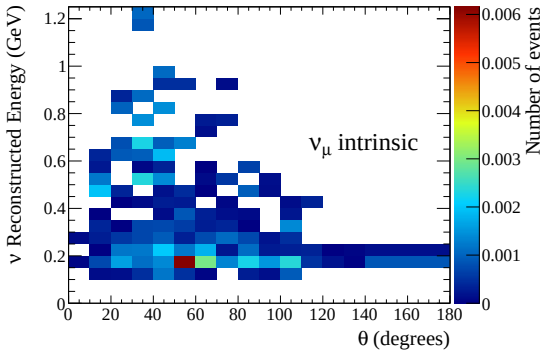
(c) $\bar{\nu}_\mu \rightarrow \bar{\nu}_e$.



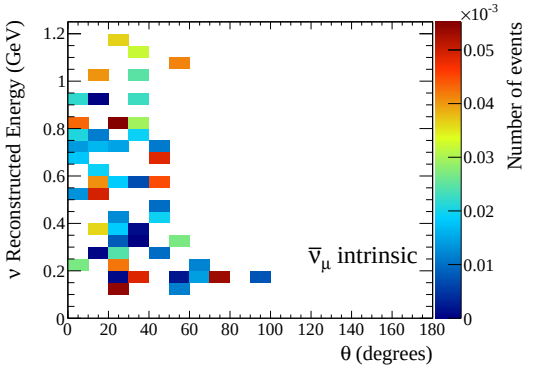
(d) $\nu_e + \bar{\nu}_e$ intrinsic beam.



(e) Neutral current.

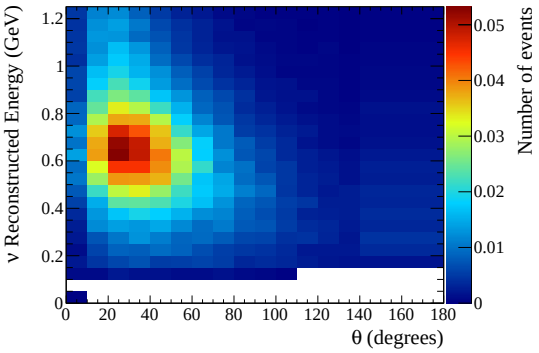


(f) ν_μ intrinsic beam.

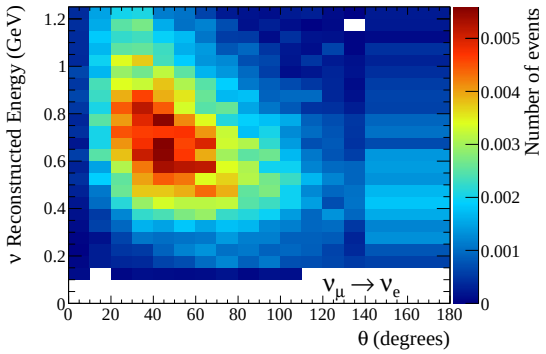


(g) $\bar{\nu}_\mu$ intrinsic beam.

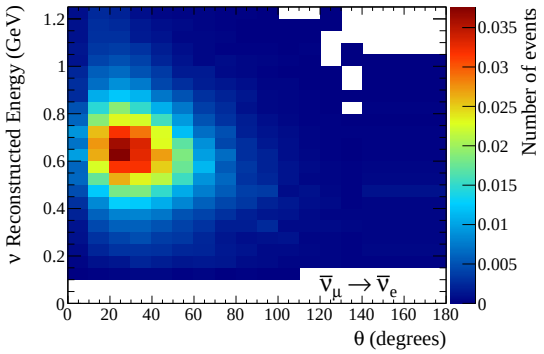
Figure 36: Predicted spectra of electron-like events as a function of the reconstructed neutrino energy and the angle between the outgoing lepton and the neutrino direction. The distributions correspond to the collected statistics in the full Run 1-6 data set. From top left to bottom right the following components are shown: total, $\nu_\mu \rightarrow \nu_e$, $\bar{\nu}_\mu \rightarrow \bar{\nu}_e$, $\nu_e + \bar{\nu}_e$ intrinsic beam, NC, ν_μ intrinsic beam and $\bar{\nu}_\mu$ intrinsic beam. The spectra were generated with the systematic parameters measured with the ND280 data shown in table 6, 7, 8, the nominal Super-K systematic parameters shown in table 9 and the oscillation parameters corresponding to the Asimov data set A shown in table 12.



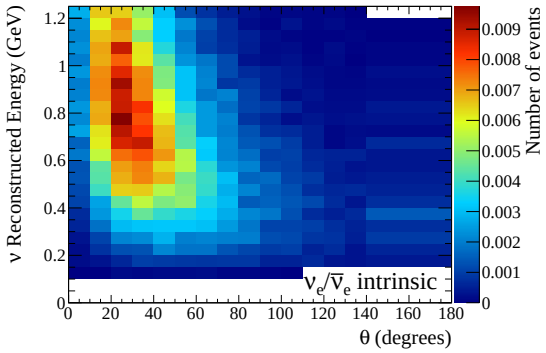
(a) Total



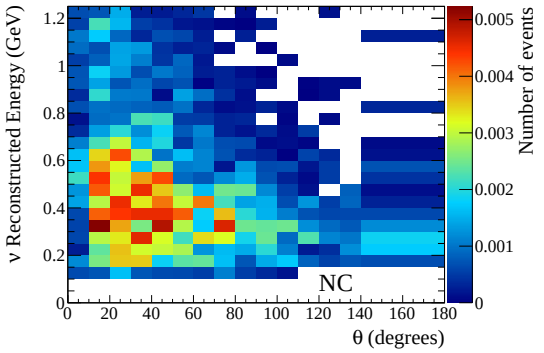
(b) $\nu_\mu \rightarrow \nu_e$.



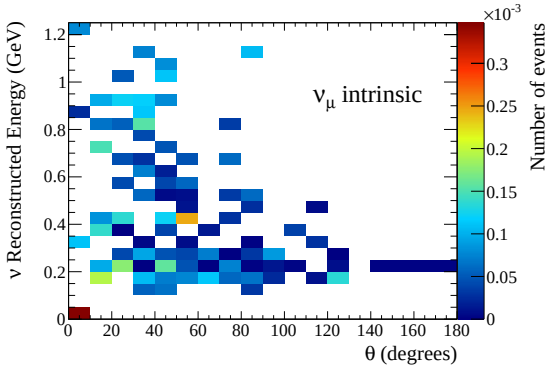
(c) $\bar{\nu}_\mu \rightarrow \bar{\nu}_e$.



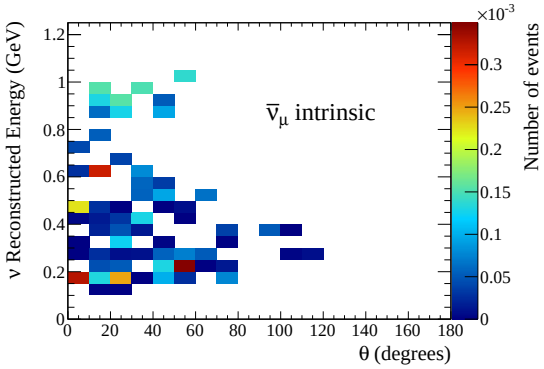
(d) $\nu_e + \bar{\nu}_e$ intrinsic beam.



(e) Neutral current.



(f) ν_μ intrinsic beam.



(g) $\bar{\nu}_\mu$ intrinsic beam.

Figure 37: Predicted spectra of electron-like events as a function of the reconstructed neutrino energy and the angle between the outgoing lepton and the neutrino direction. The distributions correspond to the collected statistics in the full Run 1-6 data set for an exposure. From top left to bottom right the following components are shown: total, $\nu_\mu \rightarrow \nu_e$, $\bar{\nu}_\mu \rightarrow \bar{\nu}_e$, $\nu_e + \bar{\nu}_e$ intrinsic beam, NC, ν_μ intrinsic beam and $\bar{\nu}_\mu$ intrinsic beam. The spectra were generated with the systematic parameters measured with the ND280 data shown in table 6, 7, 8, the nominal Super-K systematic parameters shown in table 9 and the oscillation parameters corresponding to the Asimov data set A shown in table 12.

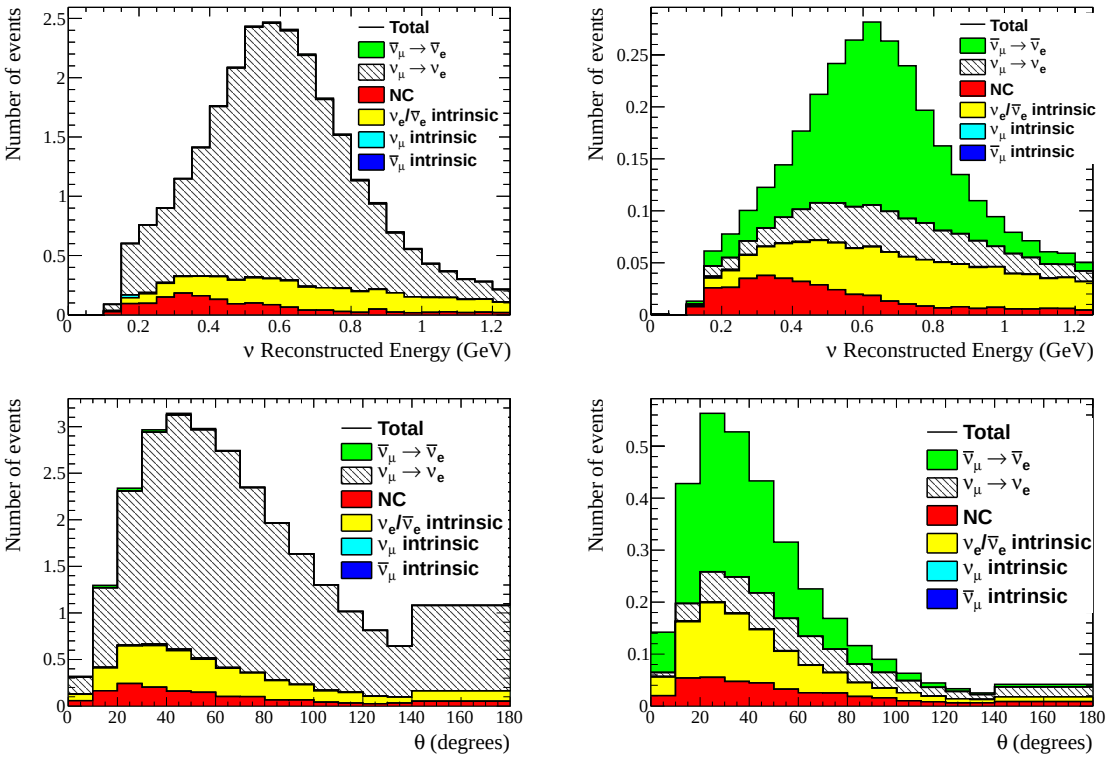
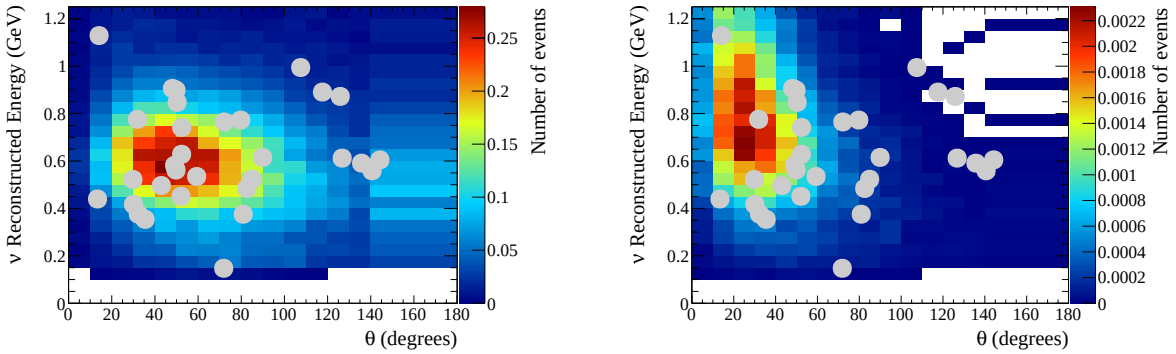


Figure 38: Predicted spectra of the reconstructed neutrino energy (top) and the angle between the neutrino and the outgoing lepton directions (bottom) for electron-like neutrino mode (left) and antineutrino mode (right) events. The distributions correspond to the collected statistics in the full Run 1-6 data set. The distributions are broken down by the following components: $\bar{\nu}_\mu \rightarrow \bar{\nu}_e$ (green), $\nu_\mu \rightarrow \nu_e$ (black/white), NC (red), $\nu_e + \bar{\nu}_e$ intrinsic beam (yellow), ν_μ intrinsic beam (light blue) and $\bar{\nu}_\mu$ intrinsic beam (blue). The spectra were generated with the systematic parameters measured with the ND280 data shown in table 6, 7, 8, the nominal Super-K systematic parameters shown in table 9 and the oscillation parameters corresponding to the Asimov data set A shown in table 12.

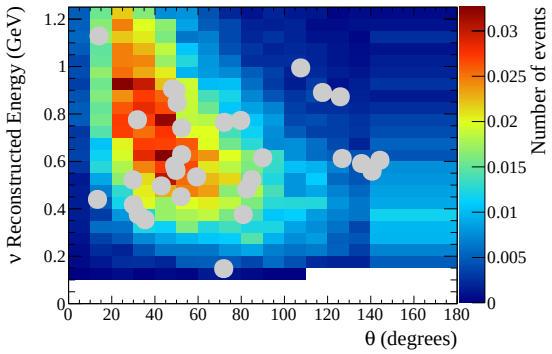
C. Observed *e*-like spectra

In fig. 39 and 40 the predicted breakdown of the *e*-like 2-dimensional spectra are compared to data for both neutrino and antineutrino beam mode.

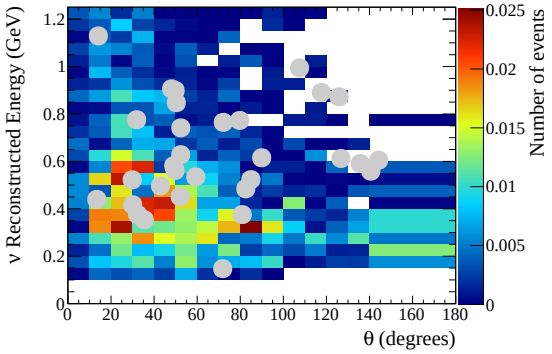


(a) $\nu_\mu \to \nu_e$.

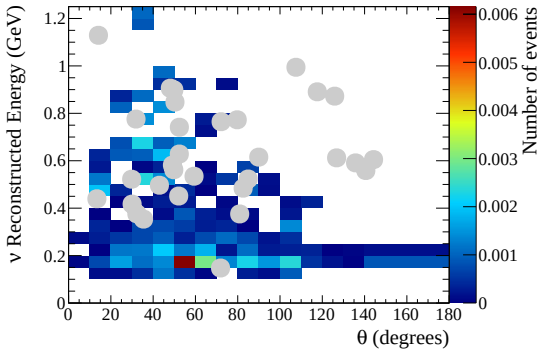
(b) $\bar{\nu}_\mu \to \bar{\nu}_e$.



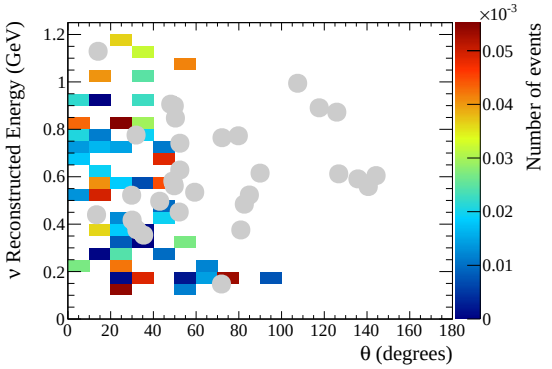
(c) $\nu_e + \bar{\nu}_e$ intrinsic beam.



(d) Neutral current.

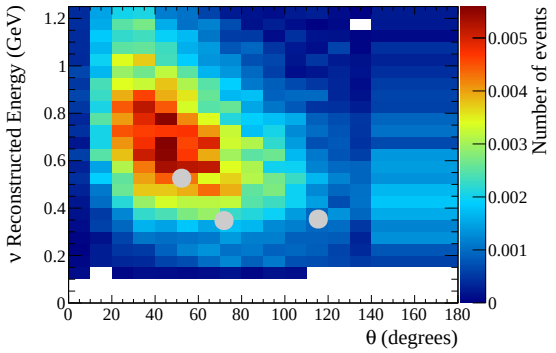


(e) ν_μ intrinsic beam.

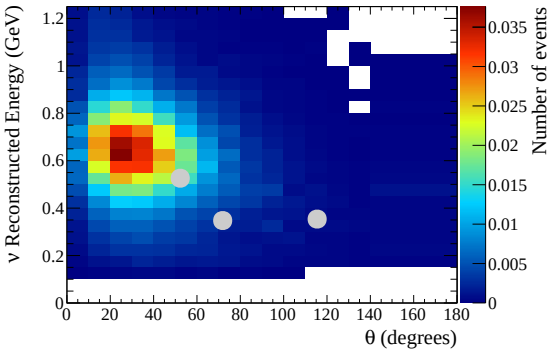


(f) $\bar{\nu}_\mu$ intrinsic beam.

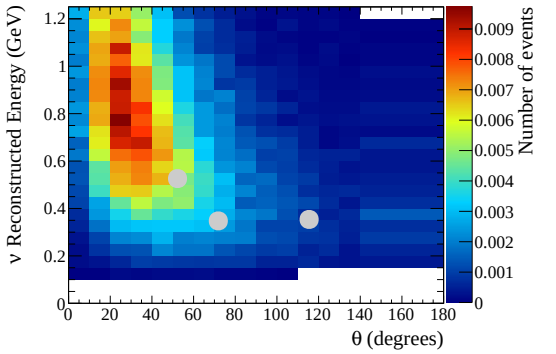
Figure 39: Predicted and observed (grey dots) spectra of electron-like events as a function of the reconstructed neutrino energy and the angle between the outgoing lepton and the neutrino direction. The distributions correspond to the collected statistics in the full Run 1-6 data set. From top left to bottom right the following components are shown: $\nu_\mu \to \nu_e$, $\bar{\nu}_\mu \to \bar{\nu}_e$, $\nu_e + \bar{\nu}_e$ intrinsic beam, NC, ν_μ intrinsic beam and $\bar{\nu}_\mu$ intrinsic beam. The spectra were generated with the systematic parameters measured with the ND280 data shown in table 6, 7, 8, the nominal Super-K systematic parameters shown in table 9 and the oscillation parameters corresponding to the Asimov data set A shown in table 12.



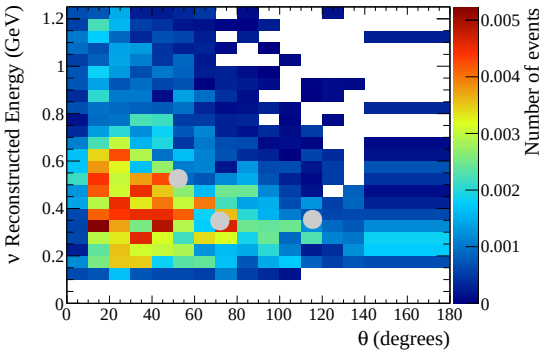
(a) $\nu_\mu \to \nu_e$.



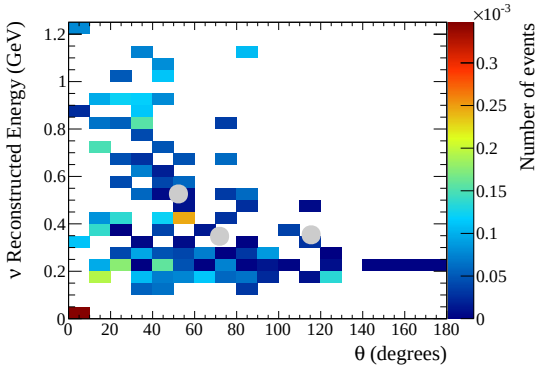
(b) $\bar{\nu}_\mu \to \bar{\nu}_e$.



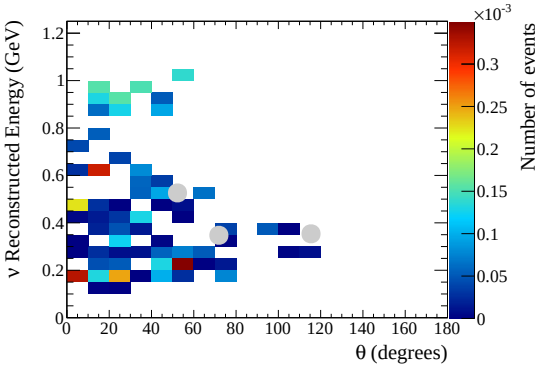
(c) $\nu_e + \bar{\nu}_e$ intrinsic beam.



(d) Neutral current.



(e) ν_μ intrinsic beam.



(f) $\bar{\nu}_\mu$ intrinsic beam.

Figure 40: Predicted and observed (grey dots) spectra of electron-like events as a function of the reconstructed neutrino energy and the angle between the outgoing lepton and the neutrino direction. The distributions correspond to the collected statistics in the full Run 1-6 data set. From top left to bottom right the following components are shown: $\nu_\mu \to \nu_e$, $\bar{\nu}_\mu \to \bar{\nu}_e$, $\nu_e + \bar{\nu}_e$ intrinsic beam, NC, ν_μ intrinsic beam and $\bar{\nu}_\mu$ intrinsic beam. The spectra were generated with the systematic parameters measured with the ND280 data shown in table 6, 7, 8, the nominal Super-K systematic parameters shown in table 9 and the oscillation parameters corresponding to the Asimov data set A shown in table 12.

D. Effect of systematic uncertainties

In table 38, 39, 40 and 41 it is shown which systematic parameter is applied to which reaction and flavour modes for all the Super-K samples.

$f^{SK}_{E;r}$	$f^{SK+FSI}_{0;t,r}$	$f^{SK+FSI}_{1;t,r}$	$f^{SK+FSI}_{2;t,r}$	$f^{SK+FSI}_{3;t,r}$	$f^{SK+FSI}_{4;t,r}$	$f^{SK+FSI}_{5;t,r}$	$f^{banff}_{0;t,r}$	$f^{banff}_{1;t,r}$	$f^{banff}_{2;t,r}$	$f^{banff}_{3;t,r}$	$f^{banff}_{4;t,r}$	$f^{banff}_{5;t,r}$	$f^{banff}_{6;t,r}$	$f^{banff}_{7;t,r}$	$f^{banff}_{8;t,r}$	$f^{banff}_{9;t,r}$	$f^{banff}_{10;t,r}$	$f^{banff}_{11;t,r}$	$f^{banff}_{12;t,r}$	$f^{banff}_{13;t,r}$	$f^{banff}_{14;t,r}$	$f^{banff}_{15;t,r}$	$f^{banff}_{16;t,r}$	$f^{banff}_{17;t,r}$	
✓	✓	✓	✓				✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓							
✓				✓			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓							
✓				✓			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓							
✓	✓	✓	✓				✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓							
✓				✓			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓							
✓						✓

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	$f_{E;r}^{SK}$	$f_{6;t,r}^{SK+FSI}$	$f_{7;t,r}^{SK+FSI}$	$f_{8;t,r}^{SK+FSI}$	$f_{9;t,r}^{SK+FSI}$	$f_{10;t,r}^{SK+FSI}$	$f_{11;t,r}^{SK+FSI}$	$f_{12;t,r}^{SK+FSI}$	$f_{13;t,r}^{SK+FSI}$	$f_{14;t,r}^{SK+FSI}$	$f_{15;t,r}^{SK+FSI}$	$f_{16;t,r}^{SK+FSI}$	$f_{17;t,r}^{SK+FSI}$	$f_{0;t,r}^{banff}$	$f_{1;t,r}^{banff}$	$f_{2;t,r}^{banff}$	$f_{3;t,r}^{banff}$	$f_{4;t,r}^{banff}$	$f_{5;t,r}^{banff}$	$f_{6;t,r}^{banff}$	$f_{7;t,r}^{banff}$	$f_{8;t,r}^{banff}$	$f_{9;t,r}^{banff}$	$f_{10;t,r}^{banff}$	$f_{11;t,r}^{banff}$
ν_μ CCQE	✓				✓	✓	✓							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
ν_μ CC 1π	✓				✓	✓	✓							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
ν_μ CC coherent	✓				✓	✓	✓							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
ν_μ CC 2p-2h	✓				✓	✓	✓							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
ν_μ CC other	✓				✓	✓	✓							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
ν_μ/ν_τ NC $1\pi^{+/-}$	✓										✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
ν_μ/ν_τ NC $1\pi^0$	✓										✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
ν_μ/ν_τ NC coherent	✓										✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
ν_μ/ν_τ NC other	✓										✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
ν_e NC 1γ	✓										✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		
$\bar{\nu}_\mu$ CCQE	✓				✓		✓																			
$\bar{\nu}_\mu$ CC 1π	✓				✓	✓	✓																			
$\bar{\nu}_\mu$ CC coherent	✓				✓	✓	✓																			
$\bar{\nu}_\mu$ CC 2p-2h	✓				✓	✓	✓																			
$\bar{\nu}_\mu$ CC other	✓				✓	✓	✓																			
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^{+/-}$	✓										✓	✓	✓													
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^0$	✓										✓	✓	✓													
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC coherent	✓										✓	✓	✓													
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC other	✓										✓	✓	✓													
$\bar{\nu}_\mu$ NC 1γ	✓										✓	✓	✓													
ν_e CCQE	✓							✓	✓	✓																
ν_e CC 1π	✓							✓	✓	✓																
ν_e CC coherent	✓							✓	✓	✓																
ν_e CC 2p-2h	✓							✓	✓	✓																
ν_e CC other	✓							✓	✓	✓																
ν_e NC $1\pi^{+/-}$	✓									✓	✓	✓	✓													
ν_e NC $1\pi^0$	✓									✓	✓	✓	✓													
ν_e NC coherent	✓									✓	✓	✓	✓													
ν_e NC other	✓									✓	✓	✓	✓													
$\bar{\nu}_\mu$ NC 1γ	✓									✓	✓	✓	✓													
$\bar{\nu}_e$ CCQE	✓				✓		✓							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
$\bar{\nu}_e$ CC 1π	✓				✓	✓	✓							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
$\bar{\nu}_e$ CC coherent	✓				✓	✓	✓							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
$\bar{\nu}_e$ CC 2p-2h	✓				✓	✓	✓							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
$\bar{\nu}_e$ CC other	✓				✓	✓	✓							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
$\bar{\nu}_e$ NC $1\pi^{+/-}$	✓										✓	✓	✓													
$\bar{\nu}_e$ NC $1\pi^0$	✓										✓	✓	✓													
$\bar{\nu}_e$ NC coherent	✓										✓	✓	✓													
$\bar{\nu}_e$ NC other	✓										✓	✓	✓													
$\bar{\nu}_e$ NC 1γ	✓										✓	✓	✓													
Osc. $\bar{\nu}_e$ CCQE	✓	✓	✓	✓																						✓
Osc. $\bar{\nu}_e$ CC 1π	✓	✓	✓	✓																						✓
Osc. $\bar{\nu}_e$ CC coherent	✓	✓	✓	✓																						✓
Osc. $\bar{\nu}_e$ CC 2p-2h	✓	✓	✓	✓																						✓
Osc. $\bar{\nu}_e$ CC other	✓	✓	✓	✓																						✓

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	$f_{E;r}^{SK}$	$f_{0;t,r}^{SK+FSI}RHC$	$f_{1;t,r}^{SK+FSI}RHC$	$f_{2;t,r}^{SK+FSI}RHC$	$f_{3;t,r}^{SK+FSI}RHC$	$f_{4;t,r}^{SK+FSI}RHC$	$f_{5;t,r}^{SK+FSI}RHC$	$f_{0;t,r}^{banff}RHC$	$f_{1;t,r}^{banff}RHC$	$f_{2;t,r}^{banff}RHC$	$f_{3;t,r}^{banff}RHC$	$f_{4;t,r}^{banff}RHC$	$f_{5;t,r}^{banff}RHC$	$f_{6;t,r}^{banff}RHC$	$f_{7;t,r}^{banff}RHC$	$f_{8;t,r}^{banff}RHC$	$f_{9;t,r}^{banff}RHC$	$f_{10;t,r}^{banff}RHC$	$f_{11;t,r}^{banff}RHC$	$f_{12;t,r}^{banff}RHC$	$f_{13;t,r}^{banff}RHC$	$f_{14;t,r}^{banff}RHC$	$f_{15;t,r}^{banff}RHC$	$f_{16;t,r}^{banff}RHC$	$f_{17;t,r}^{banff}RHC$
ν_μ CCQE	✓	✓	✓	✓				✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓							
ν_μ CC 1π	✓							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓							
ν_μ CC coherent	✓				✓			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓							
ν_μ CC 2p-2h	✓	✓	✓	✓				✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓							
ν_μ CC other	✓				✓			✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓							
ν_μ/ν_τ NC $1\pi^{+/-}$	✓						✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓							
ν_μ/ν_τ NC $1\pi^0$	✓						✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓							
ν_μ/ν_τ NC coherent	✓						✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓							
ν_μ/ν_τ NC other	✓						✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓							
ν_e NC 1γ	✓						✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓							
$\bar{\nu}_\mu$ CCQE	✓	✓	✓	✓														✓	✓	✓	✓	✓	✓	✓	✓
$\bar{\nu}_\mu$ CC 1π	✓				✓														✓	✓	✓	✓	✓	✓	✓
$\bar{\nu}_\mu$ CC coherent	✓				✓														✓	✓	✓	✓	✓	✓	✓
$\bar{\nu}_\mu$ CC 2p-2h	✓	✓	✓	✓															✓	✓	✓	✓	✓	✓	✓
$\bar{\nu}_\mu$ CC other	✓				✓														✓	✓	✓	✓	✓	✓	✓
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^{+/-}$	✓						✓												✓	✓	✓	✓	✓	✓	✓
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^0$	✓						✓												✓	✓	✓	✓	✓	✓	✓
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC coherent	✓						✓												✓	✓	✓	✓	✓	✓	✓
$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC other	✓						✓												✓	✓	✓	✓	✓	✓	✓
$\bar{\nu}_\mu$ NC 1γ	✓						✓													✓	✓	✓	✓	✓	✓
ν_μ CCQE	✓							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓							
ν_e CC 1π	✓							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓							
ν_e CC coherent	✓							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓							
ν_e CC 2p-2h	✓							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓							
ν_e CC other	✓							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓							
ν_e NC $1\pi^{+/-}$	✓							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓							
ν_e NC $1\pi^0$	✓							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓							
ν_e NC coherent	✓							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓							
ν_e NC other	✓							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓							
$\bar{\nu}_\mu$ NC 1γ	✓							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓							
Osc. ν_e CCQE	✓							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓							
Osc. ν_e CC 1π	✓							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓							
Osc. ν_e CC coherent	✓							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓							
Osc. ν_e CC 2p-2h	✓							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓							
Osc. ν_e CC other	✓							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓							
$\bar{\nu}_e$ CCQE	✓																								
$\bar{\nu}_e$ CC 1π	✓																								
$\bar{\nu}_e$ CC coherent	✓																								
$\bar{\nu}_e$ CC 2p-2h	✓																								
$\bar{\nu}_e$ CC other	✓																								
$\bar{\nu}_e$ NC $1\pi^{+/-}$	✓																								
$\bar{\nu}_e$ NC $1\pi^0$	✓																								
$\bar{\nu}_e$ NC coherent	✓																								
$\bar{\nu}_e$ NC other	✓																								
$\bar{\nu}_e$ NC 1γ	✓																								
Osc. $\bar{\nu}_e$ CCQE	✓							✓											✓		✓		✓		
Osc. $\bar{\nu}_e$ CC 1π	✓							✓											✓	✓	✓		✓		
Osc. $\bar{\nu}_e$ CC coherent	✓							✓											✓	✓	✓	✓	✓		
Osc. $\bar{\nu}_e$ CC 2p-2h	✓							✓											✓	✓	✓	✓	✓		
Osc. $\bar{\nu}_e$ CC other	✓							✓											✓	✓	✓	✓	✓		

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$f^{SK}_{E;r}$	$f^{SK+FSI}_{6;t,r}$	$f^{SK+FSI}_{7;t,r}$	$f^{SK+FSI}_{8;t,r}$	$f^{SK+FSI}_{9;t,r}$	$f^{SK+FSI}_{10;t,r}$	$f^{SK+FSI}_{11;t,r}$	$f^{SK+FSI}_{12;t,r}$	$f^{SK+FSI}_{13;t,r}$	$f^{SK+FSI}_{14;t,r}$	$f^{SK+FSI}_{15;t,r}$	$f^{SK+FSI}_{16;t,r}$	$f^{SK+FSI}_{17;t,r}$	$f^{banff}_{0;t,r}$	$f^{banff}_{1;t,r}$	$f^{banff}_{2;t,r}$	$f^{banff}_{3;t,r}$	$f^{banff}_{4;t,r}$	$f^{banff}_{5;t,r}$	$f^{banff}_{6;t,r}$	$f^{banff}_{7;t,r}$	$f^{banff}_{8;t,r}$	$f^{banff}_{9;t,r}$	$f^{banff}_{10;t,r}$	$f^{banff}_{11;t,r}$	
✓													✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	ν_μ CCQE
✓						✓							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		ν_μ CC 1π
✓						✓							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		ν_μ CC coherent
✓						✓							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		ν_μ CC 2p-2h
✓						✓							✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		ν_μ CC other
✓										✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		ν_μ/ν_τ NC $1\pi^{+/-}$
✓										✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		ν_μ/ν_τ NC $1\pi^0$
✓										✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		ν_μ/ν_τ NC coherent
✓										✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		ν_μ/ν_τ NC other
✓										✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		ν_e NC 1γ
✓						✓	✓																	✓	$\bar{\nu}_\mu$ CCQE
✓						✓	✓																	✓	$\bar{\nu}_\mu$ CC 1π
✓						✓	✓																	✓	$\bar{\nu}_\mu$ CC coherent
✓						✓	✓																	✓	$\bar{\nu}_\mu$ CC 2p-2h
✓						✓	✓																	✓	$\bar{\nu}_\mu$ CC other
✓										✓	✓	✓												✓	$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^{+/-}$
✓										✓	✓	✓												✓	$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC $1\pi^0$
✓										✓	✓	✓												✓	$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC coherent
✓										✓	✓	✓												✓	$\bar{\nu}_\mu/\bar{\nu}_\tau$ NC other
✓										✓	✓	✓												✓	ν_μ NC 1γ
							✓	✓	✓																ν_e CCQE
							✓	✓	✓																ν_e CC 1π
							✓	✓	✓																ν_e CC coherent
							✓	✓	✓																ν_e CC 2p-2h
							✓	✓	✓																ν_e CC other
										✓	✓	✓												✓	ν_e NC $1\pi^{+/-}$
										✓	✓	✓												✓	ν_e NC $1\pi^0$
										✓	✓	✓												✓	ν_e NC coherent
										✓	✓	✓												✓	ν_e NC other
										✓	✓	✓												✓	$\bar{\nu}_\mu$ NC 1γ
✓	✓	✓	✓	✓	✓								✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		Osc. ν_e CCQE
✓	✓	✓	✓	✓	✓								✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		Osc. ν_e CC 1π
✓	✓	✓	✓	✓	✓								✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		Osc. ν_e CC coherent
✓	✓	✓	✓	✓	✓								✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		Osc. ν_e CC 2p-2h
✓	✓	✓	✓	✓	✓								✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		Osc. ν_e CC other
							✓	✓	✓																$\bar{\nu}_e$ CCQE
							✓	✓	✓																$\bar{\nu}_e$ CC 1π
							✓	✓	✓																$\bar{\nu}_e$ CC coherent
							✓	✓	✓																$\bar{\nu}_e$ CC 2p-2h
							✓	✓	✓																$\bar{\nu}_e$ CC other
										✓	✓	✓													$\bar{\nu}_e$ NC $1\pi^{+/-}$
										✓	✓	✓													$\bar{\nu}_e$ NC $1\pi^0$
										✓	✓	✓													$\bar{\nu}_e$ NC coherent
										✓	✓	✓													$\bar{\nu}_e$ NC other
										✓	✓	✓													$\bar{\nu}_e$ NC 1γ
✓	✓	✓	✓	✓	✓								✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		Osc. $\bar{\nu}_e$ CCQE
✓	✓	✓	✓	✓	✓								✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		Osc. $\bar{\nu}_e$ CC 1π
✓	✓	✓	✓	✓	✓								✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		Osc. $\bar{\nu}_e$ CC coherent
✓	✓	✓	✓	✓	✓								✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		Osc. $\bar{\nu}_e$ CC 2p-2h
✓	✓	✓	✓	✓	✓								✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓		Osc. $\bar{\nu}_e$ CC other

Continued on next page

Table 41 – *Continued from previous page*

| | | |

Table 41 – *Continued from previous page*

Table 41: Each row of this table represents one of the 102systematic parameters (oscillation excluded) considered in this analysis and each column represents one of the 50 MC templates used to construct the single e-like ring reconstructed energy spectrum p.d.f. in antineutrino mode. A ✓symbol denotes that the given MC template is modified when the given systematic parameter is tweaked. Only the systematic parameters that affect this sample are shown.

In table 42, 43, 44, 45 the effect of each group of systematic uncertainties on the total number of events and on the ratio between neutrino and antineutrino mode is shown for the Asimov data set B respectively for neutrino mode μ-like and e-like and antineutrino mode μ-like and e-like. Table 46 shows the corresponding uncertainty on the ratio of neutrino and antineutrino mode events.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ (Asimov B)
SKDet+FSI+SI	4.02%
Flux	3.56%
2p-2h (corr)	3.49%
2p-2h-bar (corr)	0.20%
NC other (uncorr)	0.77%
NC 1gamma (uncorr)	0.00%
XSec nue/numu (uncorr)	0.01%
XSec Tot (corr)	4.03%
XSec Tot	4.10%
Flux+XSec (All)	2.85%
Flux+XSec+SKDet+FSI+SI	4.96%
Oscillations	0.10%
All	4.96%

Table 42: Effect of 1σ variations of the groups of systematics parameters on total number of 1-ring muon-like events for neutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in table 6, 7 8, the nominal Super-K systematic parameters shown in table 9 and the oscillation parameters of the Asimov data set B shown in table 12 are used. Since the parameters δ_{CP}, θ_{23} and Δm^2_{32} (Δm^2_{13}) and mass hierarchy will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ (Asimov B)
SKDet+FSI+SI	3.67%
Flux	3.61%
2p-2h (corr)	3.94%
2p-2h-bar (corr)	0.07%
NC other (uncorr)	0.21%
NC 1gamma (uncorr)	1.94%
XSec nue/numu (uncorr)	2.55%
XSec Tot (corr)	4.16%
XSec Tot	5.26%
Flux+XSec	4.29%
Flux+XSec+SKDet+FSI+SI	5.68%
Oscillations	4.18%
All	7.19%

Table 43: Effect of 1σ variations of the groups of systematics parameters on total number of 1-ring electron-like events for neutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in table 6, 7 8, the nominal Super-K systematic parameters shown in table 9 and the oscillation parameters of the Asimov data set B shown in table 12 are used. Since the parameters δ_{CP}, θ_{23} and Δm^2_{32} (Δm^2_{13}) and mass hierarchy will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.

In fig. 41 the absolute error envelope and the fractional error envelopes expressed as a ratio of event count are shown for the Asimov data set B (see table 12). These plots show the total error envelope for the reconstructed energy spectra; this is calculated as the $\pm1\sigma$ spread with respect to the median of bin contents from 10k toy MC experiments generated with randomized systematic parameters. All systematic

Source of uncertainty	$\delta N_{SK}/N_{SK}$ (Asimov B
SKDet+FSI+SI	3.83%
Flux	3.70%
2p-2h (corr)	2.95%
2p-2h-bar (corr)	1.83%
NC other (uncorr)	0.73%
NC 1gamma (uncorr)	0.00%
XSec nue/numu (uncorr)	0.00%
XSec Tot (corr)	4.10%
XSec Tot	4.17%
Flux+XSec	3.31%
Flux+XSec+SKDet+FSI+SI	5.04%
Oscillations	0.06%
All	5.11%

Table 44: Effect of 1σ variations of the groups of systematics parameters on total number of 1-ring muon-like events for antineutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in table 6, 7 8, the nominal Super-K systematic parameters shown in table 9 and the oscillation parameters of the Asimov data set B shown in table 12 are used. Since the parameters δ_{CP}, θ_{23} and Δm^2_{32} (Δm^2_{13}) and mass hierarchy will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.

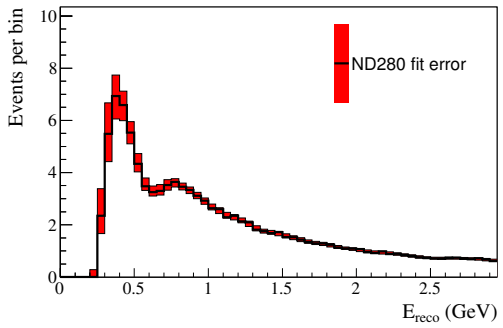
Source of uncertainty	$\delta N_{SK}/N_{SK}$ (Asimov B)
SKDet+FSI+SI	3.88%
Flux	3.81%
2p-2h (corr)	2.92%
2p-2h-bar (corr)	2.56%
NC other (uncorr)	0.32%
NC 1gamma (uncorr)	2.81%
XSec nue/numu (uncorr)	1.68%
XSec Tot (corr)	4.52%
XSec Tot	5.59%
Flux+XSec	4.62%
Flux+XSec+SKDet+FSI+SI	6.18%
Oscillations	3.59%
All	7.09%

Table 45: Effect of 1σ variations of the groups of systematics parameters on total number of 1-ring electron-like events for antineutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in table 6, 7 8, the nominal Super-K systematic parameters shown in table 9 and the oscillation parameters of the Asimov data set B shown in table 12 are used. Since the parameters δ_{CP}, θ_{23} and Δm^2_{32} (Δm^2_{13}) and mass hierarchy will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.

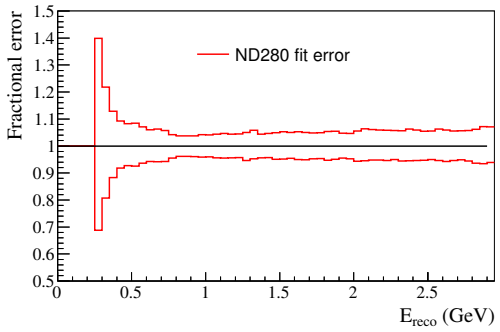
parameters are considered and their correlations are taken into account. The spectra were tuned with the systematic parameters measured with the ND280 data shown in table 6, 7, 8, the nominal Super-K systematic parameters shown in table 9 and the oscillation parameter priors shown in table 11.

Source of uncertainty	μ-like $\delta\left(\frac{\#\nu\text{-mode}}{\#\bar{\nu}\text{-mode}}\right) / \left\langle\frac{\#\nu\text{-mode}}{\#\bar{\nu}\text{-mode}}\right\rangle$	e-like $\delta\left(\frac{\#\nu\text{-mode}}{\#\bar{\nu}\text{-mode}}\right) / \left\langle\frac{\#\nu\text{-mode}}{\#\bar{\nu}\text{-mode}}\right\rangle$
SKDet	0.6%	1.2%
FSI+SI	2.5%	3.6%
SKDet+FSI+SI	2.6%	3.7%
Flux	1.6%	1.7%
XSec NC other (uncorr)	0.0%	0.1%
XSec NC 1γ (uncorr)	0.0%	0.9%
XSec ν_e / ν_μ (uncorr)	0.0%	3.3%
XSec Tot (All)	1.7%	4.4%
Flux+XSec	1.8%	4.1%
Flux+XSec+SDet+FSI+SI	3.1%	5.6%
Oscillations	0.0%	0.6%
All	3.2%	5.6%

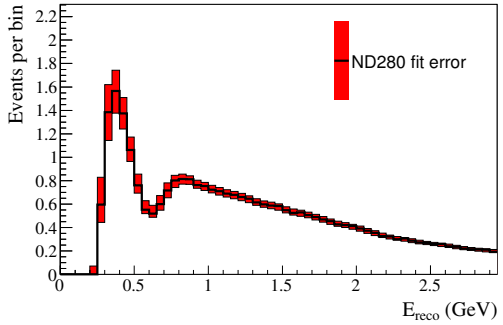
Table 46: Effect of 1σ variations of the several source of systematic uncertainties on the ratio between the total number of events in neutrino and antineutrino modes for 1-ring electron- and muon-like samples obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in table 6, 7 8, the nominal Super-K systematic parameters shown in table 9 and the oscillation parameters of the Asimov data set B shown in table 12 are used. Since the parameters δ_{CP}, θ_{23} and Δm^2_{32} (Δm^2_{13}) and mass hierarchy will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account.



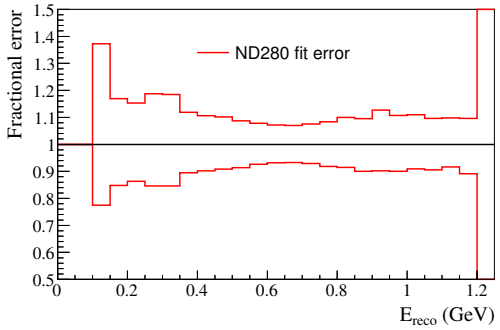
(a) Muon-like, neutrino mode



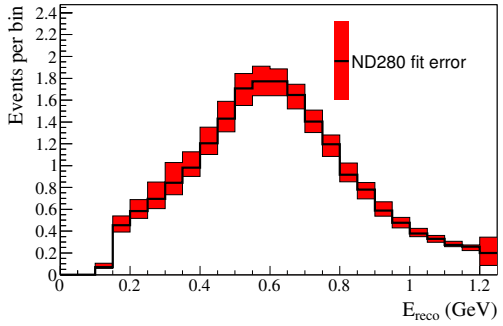
(b) Muon-like, neutrino mode



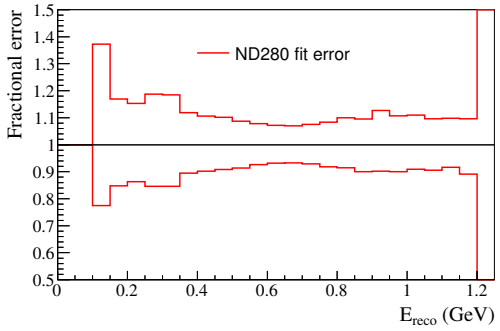
(c) Muon-like, antineutrino mode



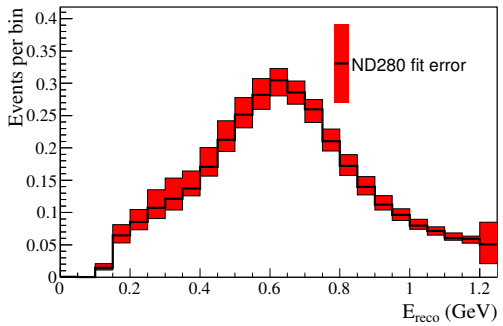
(d) Muon-like, antineutrino mode



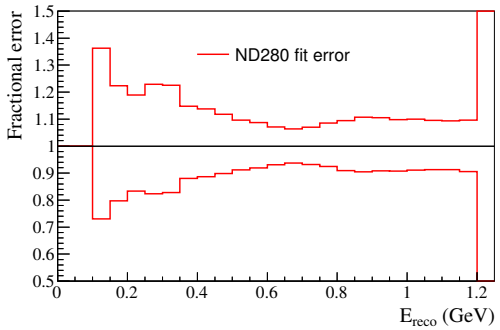
(e) Electron-like, neutrino mode



(f) Electron-like, neutrino mode



(g) Electron-like, antineutrino mode



(h) Electron-like, antineutrino mode

Figure 41: Error envelopes (left) and fractional errors (right) for the reconstructed neutrino energy spectrum as a result of all the systematic parameters. 10k toy MC experiments are generated for the Asimov data set B shown in table 12 with randomized systematic parameters, taking into account their correlations. Since the parameters δ_{CP}, θ_{23} and Δm^2_{32} (Δm^2_{13}) and mass hierarchy will be measured, only the uncertainty on $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. BANFF tuned MC templates are used. The distributions correspond to the collected statistics in the full Run 1-6 data set.

In table 47, 48, 49 and 50 the effect of each group of systematic uncertainties on the total number of events and on the ratio between neutrino and antineutrino mode is shown for the Asimov data set A, but without taking into account the correction due to the bias in the post-fit BANFF systematic parameters as well as the additional covariance matrix that take into account the bias in the flux parameters of the BANFF fit. The impact of the uncertainty due to the bias of the BANFF fit is small and has a small impact compared to the other systematic uncertainties. Indeed the only variation is in the flux uncertainty and the difference is about 1.5% in each sample.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ (Asimov A)
SKDet+FSI+SI	4.11%
Flux	3.26%
2p-2h (corr)	3.53%
2p-2h-bar (corr)	0.20%
NC other (uncorr)	0.79%
NC 1gamma (uncorr)	0.00%
XSec nue/numu (uncorr)	0.01%
XSec Tot (corr)	3.96%
XSec Tot	4.04%
Flux+XSec	2.75%
Flux+XSec+SKDet+FSI+SI	5.03%
Oscillations	0.03%
All	5.00%

Table 47: Effect of 1σ variations of the groups of systematics parameters on total number of 1-ring muon-like events for neutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in table 6, 7 8 and 9 and the oscillation parameters of the Asimov data set A shown in table 12 are used. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. The correction of the post-fit BANFF parameters as well as the additional covariance matrix used to take account for bias in the BANFF fit is not applied.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ (Asimov A)
SKDet+FSI+SI	3.47%
Flux	3.40%
2p-2h (corr)	3.95%
2p-2h-bar (corr)	0.05%
NC other (uncorr)	0.16%
NC 1gamma (uncorr)	1.49%
XSec nue/numu (uncorr)	2.63%
XSec Tot (corr)	4.17%
XSec Tot	5.15%
Flux+XSec	4.08%
Flux+XSec+SKDet+FSI+SI	5.51%
Oscillations	4.19%
All	6.93%

Table 48: Effect of 1σ variations of the groups of systematics parameters on total number of 1-ring electron-like events for neutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in table 6, 7 8 and 9 and the oscillation parameters of the Asimov data set A shown in table 12 are used. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. The correction of the post-fit BANFF parameters as well as the additional covariance matrix used to take account for bias in the BANFF fit is not applied.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ (Asimov A)
SKDet+FSI+SI	3.94%
Flux	3.40%
2p-2h (corr)	3.00%
2p-2h-bar (corr)	1.76%
NC other (uncorr)	0.76%
NC 1gamma (uncorr)	0.00%
XSec nue/numu (uncorr)	0.00%
XSec Tot (corr)	4.20%
XSec Tot	4.27%
Flux+XSec	2.90%
Flux+XSec+SKDet+FSI+SI	4.90%
Oscillations	0.03%
All	4.91%

Table 49: Effect of 1σ variations of the groups of systematics parameters on total number of 1-ring muon-like events for antineutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in table 6, 7 8 and 9 and the oscillation parameters of the Asimov data set A shown in table 12 are used. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. The correction of the post-fit BANFF parameters as well as the additional covariance matrix used to take account for bias in the BANFF fit is not applied.

Source of uncertainty	$\delta N_{SK}/N_{SK}$ (Asimov A)
SKDet+FSI+SI	3.94%
Flux	3.53%
2p-2h (corr)	3.11%
2p-2h-bar (corr)	2.24%
NC other (uncorr)	0.33%
NC 1gamma (uncorr)	2.98%
XSec nue/numu (uncorr)	1.52%
XSec Tot (corr)	4.44%
XSec Tot	5.57%
Flux+XSec	4.39%
Flux+XSec+SKDet+FSI+SI	6.11%
Oscillations	3.98%
All	7.24%

Table 50: Effect of 1σ variations of the groups of systematics parameters on total number of 1-ring electron-like events for antineutrino mode obtained by performing 10k toys MC. The systematic parameter values measured with the ND280 data shown in table 6, 7 8 and 9 and the oscillation parameters of the Asimov data set A shown in table 12 are used. Since the parameters θ_{23} and Δm^2_{32} (Δm^2_{13}) will be measured, only the uncertainty on the parameters $\sin^2\theta_{13}$, $\sin^2\theta_{12}$ and Δm^2_{21} are taken into account. The correction of the post-fit BANFF parameters as well as the additional covariance matrix used to take account for bias in the BANFF fit is not applied.

E. Method for significance calculation

The significance for a certain oscillation hypothesis given by the measured data is provided as following. First the $\Delta\chi^2$ distribution, $f(\Delta\chi^2)$, is obtained by many toy experiments (e.g. $10k$), then the $\Delta\chi^2_{data}$ value is measured with the data set. This quantities are shown in fig. 42.

First the p-value is calculated as:

$$\text{p-value} = \int_{-\infty}^{\Delta\chi^2_{data}} f(\Delta\chi^2) d(\Delta\chi^2) \tag{21}$$

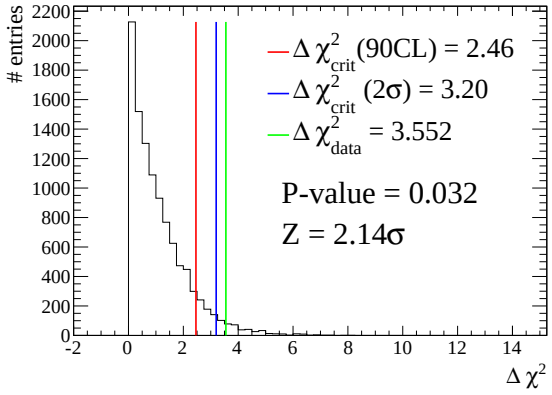


Figure 42: Distributions of $\Delta\chi^2$. $10k$ toy experiments are performed with true values of $\delta_{CP} = 0$ (left) and π (right), assuming normal hierarchy and applying the reactor constraint. Both the mass hierarchy and δ_{CP} are profiled in the fit. All the other oscillation parameters are marginalised. The statistics corresponds to the Run 1-6 data set. The $\Delta\chi^2$ value (green line) measured from the Run 1-6 data set as well as the p-value and the significance as number of standard deviations (Z), are shown as well.

Then the significance can be shown as the number of standard deviations where the measured quantity is assumed to be gaussian distributed. In fig. 43 the gaussian distribution of the measured variable μ is shown as a function of the Z-score, i.e. how many standard deviations a certain measurement x is far from the expected value:

$$Z = \frac{x - \mu}{\sigma} \tag{22}$$

If an experiment measures x, α is the area of the p.d.f. corresponding to a certain CL (90% in the example). The quantity σ is the bias of the measured value with respect to the expectation that contains $1 - \alpha = 68.27\%$ of the total area.

The significance can be reported with the Z-score in the following way:

- the p-value is measured using eq. (21)
- Assuming the p-value to be 1-sided (see fig. 42) and α 2-sided (see fig. 43), we set α = p-value
- Z, shown in eq. (22), is computed using the quantile distribution. Given the cumulative distribution function of the normal standard distribution with Mean=0 and RMS=1 ($\sigma = 1$),

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{Z}^{+\infty} e^{-t^2/2} dt = \alpha/2 \tag{23}$$

the quantile is defined as

$$Z = \Phi^{-1}(1 - \alpha/2). \tag{24}$$

The ROOT function Math::gaussian_quantile, that takes α as input, and calculate the cumulative distribution function of the lower tail of the normal (Gaussian) distribution is defined as following:

```
TF1 q("q","ROOT::Math::gaussian_quantile(1.-(x/2.)),1.)",0.,1.);
double Zscore = q.Eval(pvalue);
```

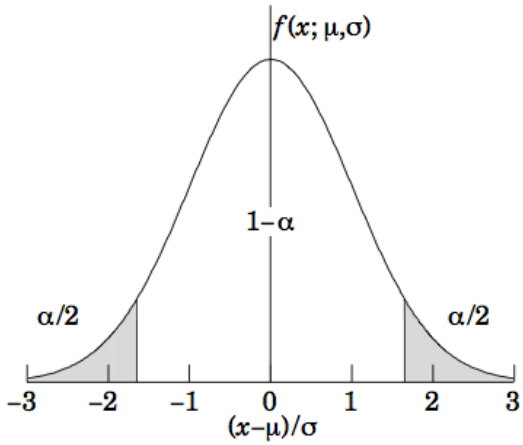


Figure 43: Illustration of a symmetric 90% confidence interval (unshaded) for a measurement of a single quantity with Gaussian errors. Integrated probabilities, defined by $\alpha = 0.1$, are shown.